Differences in performance between logistic regression and machine learning for clinical risk prediction

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Abstract

Supervised machine learning techniques such as random forests, support vector machines and feedforward neural networks are often used in the context of clinical risk prediction. Despite their theoretical benefits over logistic regression in being able to model non-linearities and interactions automatically, the performance of these algorithms is unclear in recent studies. This thesis finds that some supervised machine learning algorithms confer a performance advantage in discrimination when predicting mortality over a ten-year period, although this often comes at the cost of considerable computational expense and poorer calibration of the risk estimates. The benefits of supervised machine learning for differentiating high risk patients from low risk patients are most obvious in the cases of random forest and ensemble learning, and the performance improvement is strongest in the event where the sample size and feature count are high, irrespective of any feature selection.

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submitted by

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for the degree of MSc in Data Science & Statistics of the

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DECLARATION

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1. Introduction

1.1. Clinical Prediction Models

Clinical risk prediction models are explicit, empirical approaches to estimate the probabilities of disease using person level data (Steyerberg, 2019, p. 2). Clinical prediction models fall into two main categories: clinical prediction models for diagnosis and clinical risk prediction models for prognosis. Clinical risk prediction models for diagnosis estimate the probability of a patient having a disease, whereas prognosis models estimate the likely course of a disease. Prognosis models are the focus of this project, where the models estimate the probability of death from any cause over a ten-year period using person level data including but not limited to age, gender, and body mass index (BMI).

Clinical risk prediction models can have large impacts on public health. A famous recent example of this is during the COVID pandemic (Alballa & Al-Turaiki, 2021), where clinical risk prediction models were used for both diagnosis, estimating the probability an individual had COVID-19, and prognosis, estimating the risk of either mortality or severe disease given a patient contracted the disease using individual level data like age and BMI. Such models had significant societal benefits in terms of improving resource allocation, clinical care, and public health. Another famous set of examples are models that estimate the risk of cardiovascular disease such as the Framingham risk score (Wilson et al., 1998), which estimates the ten-year risk of cardiovascular disease using individual level data such as age, diabetes, smoking status, blood pressure and cholesterol. Such models are widely used today in clinical practice (Finnikin et al., 2017) to identify high risk patients and allows for preventative treatment such as statin therapy and lifestyle modifications such as quitting smoking.

Many clinical risk prediction models for prognosis concern themselves with problems with a binary outcome, where the outcome can take one of two values. In the case of the Framingham risk score, the binary outcome was whether the patient was diagnosed with cardiovascular disease or not in the next ten years. In this thesis, the binary outcome is mortality, whether the US individual died from any cause in a ten-year period. The models estimate the risk of mortality conditional on individual level data. It would also be possible to model this mortality data as time-to-event data, modelling how long until each patient dies, however this work is motivated around comparing the methods in the context of binary classification.

As these models are widely used in clinical practice, it is important that they estimate patient risk as accurately as possible. There are two main competing approaches for fitting clinical risk prediction models for prognosis. The more traditional approach is to use logistic regression, however, there is much interest in using supervised machine learning algorithms due to their hypothetical benefits in being able to model more complex relationships automatically (Christodoulou et al., 2019).

1.2. Logistic Regression

Logistic regression is an extension of linear regression for modelling binary outcomes (Osborne, 2015, p.3). The outcome variable **Y** is assumed to follow a Bernoulli distribution. In the mortality prediction setting, **Y** is the binary outcome of the patient dying or not within a ten-year period. Here, p refers to the probability that the patient dies over a ten-year period (p is the probability the dependent variable **Y** is equal to one), and is conditional on a set of explanatory variables or features **X**. In clinical risk prediction models, **X** is a set of individual level characteristics that usually includes as least demographic data such as age, sex, and BMI.

When fitting the model, the probability is transformed into the log odds and the features are modelled with respect to the log odds.

$$\ln\left(\frac{p}{1-p}\right) = \beta \boldsymbol{X}$$

The main advantage of modelling the dependent variable in this way is that the bounds of the odds are (0, ∞) and the bounds of the log odds are (-∞, ∞). As a result, the log of the odds (the logit function) can be modelled as a linear function of a set of predictor variables, and it can never result in impossible predictions for the probability as the predictions for p are bounded in the 0,1 range. The optimal values of the β vector are determined through maximum likelihood estimation. By rearranging this equation, we can model the conditional probability as below:

$$P(Y = 1 \,|\boldsymbol{X}) = \frac{e^{\beta \boldsymbol{X}}}{1 + \, e^{\beta \boldsymbol{X}}}$$

When faced with a new observation, the probability of the patient dying in a ten-year period can be estimated using the optimal values for β and the observed explanatory variables for the new patient. For classification, the patient is deemed to be at high risk if this conditional probability is over a certain threshold, and preventative treatment can be recommended.

The key limitation of logistic regression arises when specifying the model. The decision of which variables to include, the functional form of the explanatory variables and the interactions between variables are left up to the researcher. As with all parametric methods, the prediction performance of logistic regression is contingent on whether the data follows the assumed model (Couronné et al., 2018) and often, researchers only include the explanatory variables additively and without interactions included. These assumptions are referred to as linearity and additivity. Linearity is the assumption that there is a linear relationship between the logit transformation of the dependent variable and the predictor variables, and additivity is the assumption that there are no interactions between variables (Couronné et al., 2018). These assumptions are strong assumptions in the context of mortality prediction, as non-linearities and interactions are intuitively very likely. For instance, including only a linear term for age implies that the effect of a 20-year-old aging to 21 has the same effect on mortality log odds as a 100-year-old aging to 101. The assumption of linearity and additivity are not necessary for logistic regression, but they are frequently the default of researchers and there is no standard practice on how to choose which non-linearities and interactions to include when modelling.

1.3. Machine Learning

Machine learning is a very broad concept that has many competing definitions. In this review, machine learning refers to supervised classification algorithms that learn patterns directly from data (Mitchell, 1997). Related fields such as unsupervised and reinforcement learning are outside the scope of this thesis as they cannot be directly applied for predicting clinical risk. Differences between machine learning and statistical modelling are not clear cut. One distinction between machine learning and statistical modelling is highlighted by Breiman (2001a) who describes two cultures. Firstly, a statistical modelling culture that assumes the outcome variable is generated by a given stochastic data model. In the case of logistic regression, the given stochastic data model is typically that the dependent variable is Bernoulli distributed and the explanatory variables are linearly and additively associated with log odds of disease. In contrast, the machine learning culture focuses on algorithmic models that treat the underlying data mechanism as unknown. The machine learning culture is therefore more flexible, as it does not directly assume a given distribution or functional

form for the data. As a result, machine learning algorithms are therefore generally much more automatic about modelling non-linearities and interactions between variables than logistic regression (Couronné et al., 2018).

Due to the flexible nature of these models, overfitting becomes a more common occurrence (Tu, 1996). Overfitting refers to the problem where an algorithm captures noise in the dataset it was trained on that will not generalise to other datasets (Chandramouli et al., 2018). Preventing overfitting is one of the main challenges of model specification in machine learning and is generally controlled by hyperparameter tuning. Hyperparameters are parameters that cannot be optimised by the algorithm itself and must be chosen before training starts (Yang and Shami, 2020), and hyperparameter tuning refers to choosing hyperparameters that maximise model performance on new data. Performance of a machine learning model depends to a large degree on the ability of the researcher to tune hyperparameters, and there is no standard practice on how to tune hyperparameters for machine learning algorithms.

1.4. Project Aims and Objectives

The main objective of this project is to compare the performance of logistic regression and supervised machine learning algorithms in the clinical risk prediction setting by applying these methods to clinical risk prediction for modelling the probability of death in a ten-year interval. The machine learning algorithms used include random forests, support vector machines and neural networks as these are the most regularly used in supervised machine learning and typically report the best performance (Kotsiantis et al., 2007). An ensemble of these estimators is also used, as ensemble learning is regularly used to improve the performance of machine learning classifiers (Sagi and Rokach, 2018).

This thesis aims to be clear and transparent with all modelling steps, including how features were selected, how hyperparameters are optimised and how performance was measured. The modelling has been carefully conducted in a clear and unbiased manner, as often in medical literature the model building stage is unclear or biased (Christodoulou et al., 2019), preventing a clear comparison between logistic regression and machine learning. The reliability of estimates will also be explored, an aspect of supervised ML algorithms that is rarely reported in recent literature (Van Calster et al., 2016).

Following on from this, the aim is to show which factors have a large influence in determining the relative performances of logistic regression and machine learning. The analysis will focus on the effect of sample size, the effect of the number of explanatory variables, and the effect of feature selection.

The dataset the algorithms are applied to is the National Health and Nutrition Examination Survey (NHANES) (CDC, 2008) merged with the 2019 publicly available linked mortality file (CDC, 2022). NHANES is a United States cross sectional study running every two years, collecting data through a multitude of methods including interviews, questionnaires, health examinations and laboratory analysis. This dataset gives a rich set of variables on an individual level including data on demographics, diets, medical examinations, and laboratory test results. This cross-sectional data is then linked to a mortality file that specifies the individual's mortality status, cause of death and how long they survived after the interview. The aim is to apply logistic regression and supervised machine learning algorithms to estimate the probability of death over a ten-year period on this dataset. Estimating the probability of death removes the issue of having to address competing risks. Applying these methods to the same dataset will facilitate a direct comparison of the performance of each method. Following on from this, the performance and reliability of these models will be measured to

explore which models show the highest level of performance, and which methods estimates are the most reliable. After fitting the models, the models are again fitted with different levels of training data points and different amounts of explanatory variables to verify if these are important factors in determining the final performance of these classification algorithms.

This project thoroughly explores the relative performance and reliability of supervised machine learning and logistic regression in the clinical risk prediction space for binary classification on a single dataset. The thesis also contributes to understanding what the key drivers behind these differing performance levels are. This will be useful for researchers when determining which technique to use for clinical risk prediction.

The next section is a thorough methods review of how logistic regression and supervised machine learning have been applied in the clinical risk prediction space with a binary outcome. The material in the methods review and the previous introduction is largely inspired by the proposal for this thesis (Husbands, 2022). This is then followed by a more thorough overview of the data I will be using and precisely how the methods have been applied. The section afterwards is a demonstration of results and limitations, and the final section contains concluding remarks and guidance for where future work could focus.

2. Methods Review

2.1. Logistic Regression Approaches

The traditional approach of estimating a clinical risk prediction model is logistic regression. A typical example of this is North et al.'s (2011) model of pre-eclampsia (high blood pressure) in pregnant women. This model estimates the probability of developing pre-eclampsia using individual level data including age, BMI, alcohol intake and family history of pre-eclampsia.

The main challenge of logistic regression arises in model development, which broadly refers to the identification and modelling of the predictor variables (Shipe et al., 2019). In logistic regression there is no agreed upon method for determining the best multivariate diagnostic model from a set of candidate predictors. It is regarded as impractical and poor practice to include all the candidate predictors when the number of candidate predictors is large relative to the number of data points, as this can result in model overfitting (Pavlou et al., 2015), with the fitted model then capturing random noise present in the development dataset. As a rule of thumb, it has been commonly found that there should be at least ten events per feature to mitigate the effect of overfitting (Steyerberg et al., 1999), as the reliability of the model estimates greatly decreases when this is not the case. If the number of features exceeds the number of data points it becomes impossible to estimate a logistic regression (Strobl et al., 2009) and therefore including interaction effects and high order terms often leads to difficulties with reliable parameter estimation. The issue of model development in logistic regression, including which variables to include, and which interactions and non-linearities to model is the central challenge with fitting a logistic regression that achieves good performance.

2.1.1. Feature Selection

There are two main approaches to deciding which features to use for clinical risk prediction for logistic regression, which are often used together. The first is using specialist knowledge to refine the variables to be used in model fitting, whilst the other is to use software-controlled methods. Software-controlled methods are typically iterative methods, such as stepwise regression that either use backward elimination or forward stepwise to iteratively drop the worst predictor or include the next best predictor, based on p-values or the Akaike Information Criterion (Royston et al., 2009). Univariate analysis can also often be used, where each candidate explanatory variable is measured for its association with the outcome and only statistically significant candidate variables are used going forwards. All these methods have limitations. Univariate analysis can result in useful candidate predictors being rejected, as can using specialist knowledge. Stepwise regression is unstable, and the resultant estimated coefficients tend to be too high, resulting in overfitting and therefore an inaccurate evaluation of model performance (Steyerberg and Vergouwe, 2014). An alternative much more rarely used approach in the clinical risk prediction space is to use dimensionality reducing algorithms such as principal component analysis (PCA) (Thottakkara et al., 2016). PCA reduces the dimensionality of the data and extracts the most important information from the data (Abdi and Williams, 2010) and is widely used across many scientific disciplines. It is very rarely used for clinical risk prediction, despite Thottakkara et al. (2016) finding that this means of feature selection improved the performance of logistic regression as well as supervised machine learning. It is important to note here that the decision of what features to include and their functional form also impacts the performance of supervised machine learning, though to a lesser extent as these models, in theory at least, automatically identify non-linearities and interactions present in data.

2.1.2. Modelling Continuous Variables and Interactions

A related issue to feature selection is how to model continuous predictors and interaction terms. The literature is clear that continuous variables should not be dichotomised as this results in information loss (Steyerberg and Vergouwe, 2014) and implicitly assumes there is no effect within levels of the variable. Nevertheless, dichotomisation is relatively common in the clinical risk prediction literature.

The other way that researchers typically model continuous variables is as a linear term. This necessarily assumes that the effect is the same at each part of the range of the predictor (Steyerberg, 2019), for instance, growing a year older has the same effect on the log odds of developing pre-eclampsia whether the patient is 20 or 35. A more flexible approach would be to use restricted cubic splines. Restricted cubic splines are a flexible way of modelling non-linearities and can fit a wide range of curves (Harrell, 2015, p. 23). Despite this, non-linearities are rarely modelled in logistic regressions for clinical risk prediction, and the decision of which non-linearities to model was often unclear (Christodoulou et al., 2019). Similarly, interaction terms are rarely modelled. Interaction terms capture the idea that the value of one predictor can have an impact on the effect of another predictor. As most logistic regressions in the clinical risk prediction space do not model non-linearities or interactions, the models are likely to oversimplify the true relationship between features and the outcome variable and thus underfit, oversimplifying the true relationship making model performance artificially lower than it would have been had interactions and non-linearities been properly addressed.

In the motivating example of pre-eclampsia, a multitude of methods were used for feature selection. The researchers excluded variables based on domain knowledge or where there was a high frequency of missing data. They then performed univariate regression analysis on each of the explanatory variables and excluded those variables which were statistically insignificant at the 10% level. After doing this, they performed a forward stepwise regression. The handling of non-linearities and interactions was unclear, though as neither appeared in the final equation, it is probable the researchers tacitly assumed linearity and additivity. This example highlights a general trend, in that the implementation of feature selection and model building is inconsistent, ad-hoc, and often unclear in clinical risk literature.

After deciding on the number of features and their functional form, it can be recommended to apply shrinkage to the regression coefficients, shrinking them toward zero (Van Calster et al., 2020). Shrinkage is used to prevent overfitting in logistic regression and aims to prevent predicted risks that are too extreme. However, it has been shown that shrinkage and other penalised logistic regression methods often work poorly in individual clinical datasets and do not resolve the problems associated with either small sample size or small numbers of events per variable.

In summary, the main challenge of fitting a logistic regression for clinical risk prediction is the issue of model development. Typically, in the literature, the methodology for doing this either involves directly assuming only linear direct effects, or discretization, transforming the continuous variables into categorical variables and assuming the variable to have a relation only across levels. It is rare in this field for non-linearities and interactions to be considered, despite the fact these relationships in nature are unlikely to be linear. There are other challenges in logistic regression, including model estimation, the encoding of variables, the handling of missing data and the measurement of performance, although these issues are often ones that supervised machine learning algorithms share.

2.2. Machine Learning

2.2.1. Hypothetical Advantages of Machine Learning

Overall, the literature around logistic regression shows that there are clear limitations when it comes to applying this method to real world data. Methods of feature selection and the typical implicit assumption that the explanatory variables are linearly and additively related to the log odds of the outcome variable in modern literature suggest that there are ways to improve model performance with machine learning techniques.

In the clinical risk prediction setting, machine learning generally describes a set of computationally intense and highly flexible algorithms that identify patterns in complex data structures (Jiang et al., 2020). Unsupervised and reinforcement learning are not within the scope of this project, as these machine learning techniques are not used for prediction when the outcome is known. Machine learning contrasts with logistic regression and other more rigid statistical modelling techniques in that it does not require the same statistical assumptions around the functional form of the relationship between the explanatory variables and the outcome. There are at least seventeen different regularly used machine learning classifiers (Fernández-Delgado et al., 2014) and in the clinical risk prediction space, these are most often random forests, support vector machines and artificial neural networks, which are widely used machine learning algorithms across many domains. It is plausible they confer an advantage in this area due to the flexible way they model non-linearities and interactions between variables. It is regularly claimed in the literature that machine learning algorithms are better able to handle both a larger number of predictors and their interactions, and the flexible nature of these algorithms prevents researchers from having to pre-specify the important predictor variables as these algorithms directly learn the important predictors from data (Rajkomar et al., 2018).

However, despite the theoretical improved performance, the real-world studies have resulted in mixed conclusions. A general survey on all classifiers by Fernández-Delgado et al. (2014) on a wide range of datasets found that random forests tended to yield the highest accuracy in classification. Accuracy here is the percentage of correctly classified units, which in the context of this project is the weighted average of the proportion of US individuals who were correctly predicted as deceased and the number of US individuals who were correctly predicted as alive. However, in the clinical risk setting, the theoretical improved performance of machine learning has not been clearly established (Christodoulou et al., 2019).

2.2.2. Random Forests

There are many variants of the random forest algorithm, however, the one implemented here is the original introduced by Breiman (2001b). The random forest is an ensemble learning technique, aggregating the estimate of multiple classification decision trees (Couronné et al., 2018). Each tree in the forest is built based on a recursive partitioning, where the features are iterated over and split such that the resultant splits of data contain observations that are more like one another.

An example of a constituent tree is shown in **Figure 2-1**, which seeks to predict myocardial infarctions over the next ten years based on age and smoking status. To evaluate a constituent tree, the data starts at the root node and at each split, goes left if the data satisfies the criteria and right if not. In this tree, only over 50s who smoke would be classified as at risk of a heart attack.

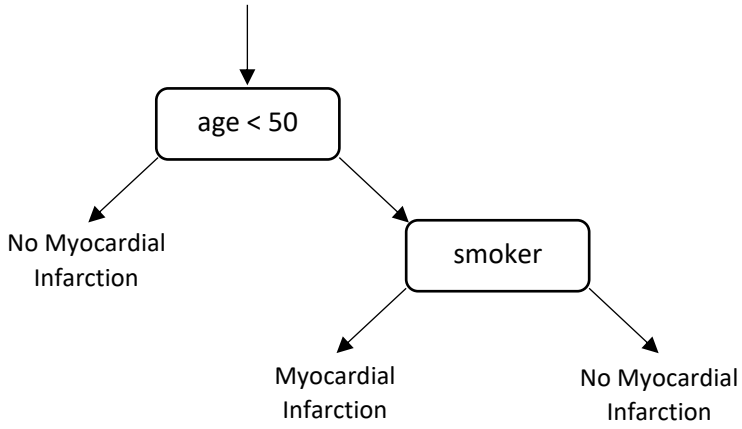

Figure 2-1: Example constituent decision tree. Loosely based on Karaolis et al. (2010)

Any given decision tree has parameters to optimise, and these parameters are which feature to use when splitting the data and what threshold to use. For instance, in the above decision tree, the decision to split upon age first and the decision to use 50 as the threshold for splitting age are parameters that were learned by the model. The original method to choose the optimal split at each decision node was to minimise the Gini impurity. The Gini impurity is the probability when selecting two patients from the dataset randomly with replacement, they will have a different outcome. In the case of binary clinical risk prediction, this is the probability that when two individuals are sampled, one individual has the disease, and one does not. The formula for the Gini impurity for 2 classes, as is the case in binary clinical risk prediction, can be shown below (Raileanu and Stoffel, 2004):

$$G(p) = 2p(1-p)$$

Where p is the proportion of individuals in the dataset who experienced the event outcome, such as the probability of having a myocardial infarction (heart attack) or dying in a ten-year period. The Gini impurity is minimised when there is only one class in a population. Splits are chosen to minimise the overall Gini impurity as below.

$$\min_{split} L(split) = \frac{n_l}{n} G(p_l) + \frac{n_r}{n} G(p_r)$$

Where n_l is the number of data points on the left hand of the split, n_r is the number of data points on the right hand of the split, n is the total number of data points, and G(p) is the Gini impurity. This formula represents the weighted Gini impurity of a split, with the weights being the proportion of data points either side of the split. This loss function minimises the Gini impurity on both the left and right sides of the split (weighted by the number of data points on each side of the split), selecting a split that maximises homogeneity on either side of the split.

The alternative method of selecting the optimal split is to maximise the information gain, which maximises the reduction in entropy after splitting the data. However, the difference between the loss functions in practice is minimal, and as information gain is more computationally expensive, Gini impurity is most often used (Raileanu and Stoffel, 2004).

To prevent each tree being the same, each tree experiences randomisation during training. The first way to force diversity in the random forest is by bagging, where each tree is built on a bootstrap sample of the original dataset (Chandramouli et al., 2018). The second way to force diversity is that, at each split, only a given number of features are given to the tree as candidates to split upon.

Natively, random forests do not output probabilities. There are several ways to infer probabilities from a random forest. Most often, the class probability of a single tree is the proportion of samples of the same class in a terminal leaf node. For instance, if in a terminal leaf node there are 5 dead and 3 alive US individuals, the probability of death for a new individual who was evaluated at this terminal leaf node would be 0.625. The probability is then inferred from the random forest by taking a grand mean across all decision trees of the probability of belonging to a given class (Olson and Wyner, 2018).

This method has potential advantages over logistic regression in that it captures dependency and non-linearity in its features by design, whereas any non-linearities and interaction in logistic regression must be explicitly included by the researcher. Any subsequent split is contingent on the previous splits, and so in classification the features are not modelled additively. It has been shown in simulated data that random forests are able to better classify dependent, non-linear structures in simulated data (Strobl et al., 2009).

A downside of random forests is that the performance is impacted by hyperparameters which are set before the algorithm starts. Examples of these are the number of candidate predictors at each split, the minimum number of data points at terminal nodes and the number of trees in the random forest (Couronné et al., 2018). The performance of random forest is contingent upon these hyperparameters to some degree, and methods to identify the best hyperparameters are not consistent in literature (Probst et al., 2019b). **Section 2.4** covers hyperparameter tuning in more detail.

A recent example of random forests applied to the clinical risk prediction space is implemented by Su et al. (2020). They develop both a logistic regression and random forest to predict whether the patient will experience a fatal or non-fatal cardiovascular event based on individual level data such as age and BMI. Despite the specification for the logistic regression being unclear, seemingly assumingly linearity and additivity in their logistic regression, the random forest had worse discrimination performance. This is a result that broadly generalises across clinical risk prediction literature, despite evidence of random forest generally being one of the top classifiers outside clinical risk prediction (Fernández-Delgado et al, 2014). Ultimately, the best classifier is contingent on the innate characteristics of the dataset such as the number of features, the class imbalance, the number of data points and the distribution of the data (Khan et al., 2020), and it appears from current research that the innate characteristics of clinical risk prediction datasets favour logistic regression (Christodoulou et al., 2019)

2.2.3. Support Vector Machines

Support Vector Machines (SVMs) estimate a high dimensional surface called a hyperplane, defined over the features that best separate groups in the data (Orru et al., 2012). Such groups in the clinical risk prediction context would be individuals who die in the following ten years and those who do not, or those who experience myocardial infarction and those whose who do not. The hyperplane is estimated by maximising the margin of separation between two distinct groups.

A simplified, hypothetical model of this is an algorithm that predicts ten-year risk of myocardial infarction using only age and cholesterol as features. Note that in this example, those that experience a myocardial infarction are encoded as +1 and those that do not are encoded as -1.

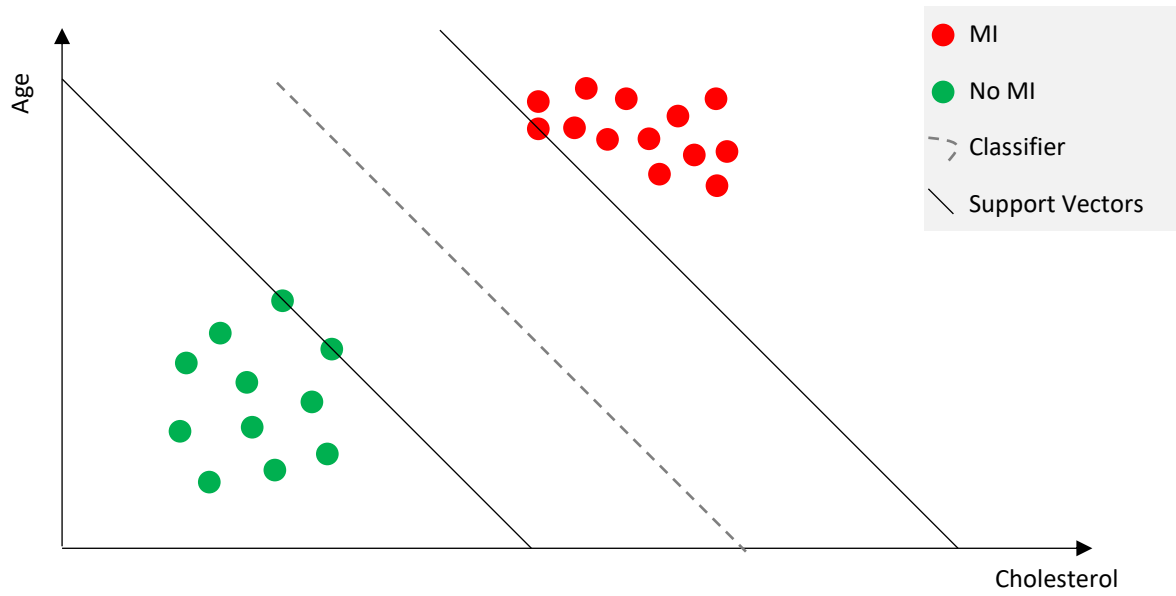

Figure 2-2: Hypothetical support vector machine classifier (hard margin)

Figure 2-2 represents synthetic data. In two dimensions, this toy data is perfectly linearly separable. The goal of a hard margin support vector machine is to estimate a decision boundary that best separates the data. This decision boundary is a hyperplane. A 1-dimensional hyperplane that separates this data can be expressed by:

$$w_1 age + w_2 cholesterol + b = 0.$$

The objective of a support vector machine is to choose values of w_1, w_2 and b to maximise the margin (Jakkula, 2006). The margin is the distance between the decision boundary and the closest observation that experiences a myocardial infarction, plus the distance between the decision boundary and the closest observation who do not experience a myocardial infarction (Nefedov, 2016). The closest observations to the decision boundary are support vectors. These support vectors determine the location of the decision boundary, and these support vectors are defined as:

$$w_1 age + w_2 cholesterol + b = +1.$$

$$w_1 age + w_2 cholesterol + b = -1.$$

The support vector classifies the data using the decision boundary, with data above the boundary classified as having a myocardial infarction and data below the boundary classified as not having a myocardial infarction.

The support vector classifier identifies the optimal decision boundary by maximising the margin, the distance between the decision boundary and the two support vectors. It can be shown that the margin is equal to:

$$\frac{2}{\sqrt{{w_1}^2 + {w_2}^2}}$$

This margin is then maximised subject to the constraint that:

$$y_i(w_1 age_i + w_2 cholesterol_i + b) \geq 1 \; \forall \; i \in (1, \dots, n)$$

This constraint requires that every observation is correctly classified. This is the constraint for the hard margin support vector machine. This constraint therefore allows zero misclassifications in the training data and is hence very sensitive to outliers. However, real data is not perfectly linearly separable. Outliers often exist, meaning that the hard margin support vector machine overfits and poorly generalises to new observations. For overlapping classes, the problem becomes infeasible as no weights can satisfy all these constraints (Nefedov, 2016). As a result, soft margin support vector machines that allow for misclassifications are more commonly used, with constraints that introducing error terms or slack variables into the constraints as below:

$$y_i(w_1 age_i + w_2 cholesterol_i + b) \geq 1 - \varepsilon_i \; \forall \; i \in (1, \dots, n)$$

The soft margin support vector machine maximises an amended objective function, whereby the objective function is as below:

$$\frac{2}{\sqrt{{w_1}^2 + {w_2}^2}} + C \sum_{i=1}^{n} \varepsilon_i$$

Here, C is the complexity parameter. C is a hyperparameter that represents a trade-off between two objectives: maximising the margin and minimising the number of misclassifications.

The last element of a support vector machine is a kernel that facilitates non-linear boundaries. This entails passing the data through a non-linear map resulting in higher dimensional data. Common choices include polynomial and radial basis function kernels. The radial basis function kernel is used for the support vector machine in this work, and its formula is given below (Nefedov, 2016):

$$K(x_i, x) = e^{-\gamma ||x_i - x||}$$

Where gamma is a hyperparameter to be tuned, x_i is one explanatory variable and x represents all other explanatory variables. Kernel functions such as the radial basis function kernel result in non-linearities being automatically modelled by the data, such as quadratic terms and interactions between variables. By using non-linear surfaces, performance typically improves as classes tend to be better separated in higher dimensional spaces (Nefedov, 2016). The choice of kernel is another hyperparameter that can be tuned. **Figure 2-3** captures why kernel functions and soft margins are likely to improve performance, as they result in decision boundaries that can both handle misclassifications and capture non-linearities and interactions that are present in data. A linear classifier in this instance would be unable to construct a decision boundary that separates this data.

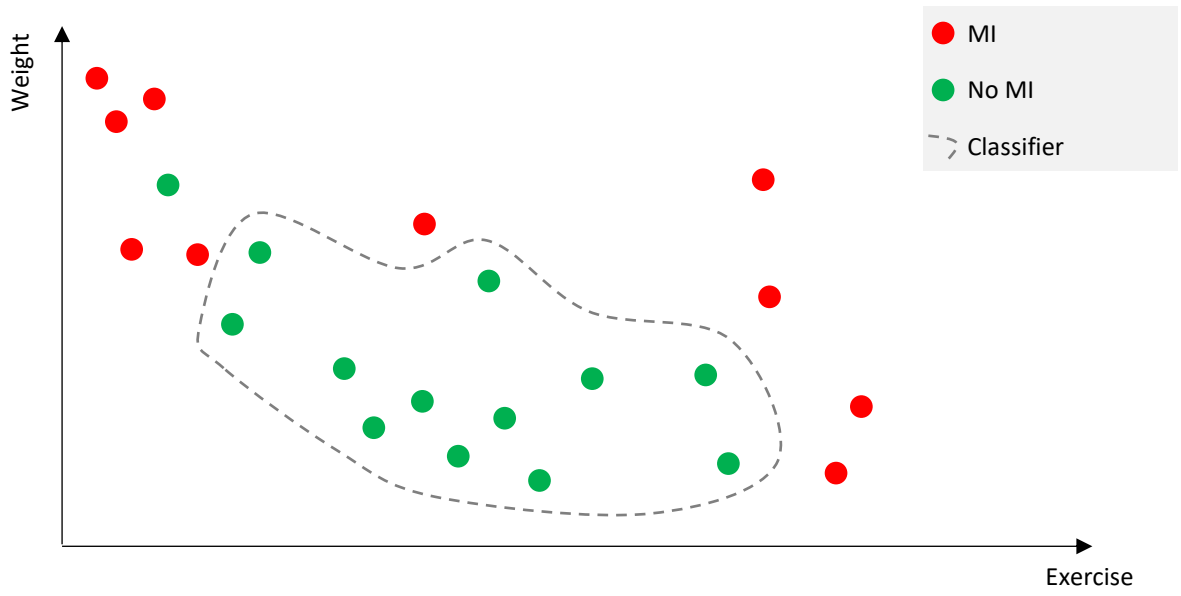

Figure 2-3: Hypothetical soft margin support vector machine classifier

SVMs typically are estimated with more than two features, and thus cannot be easily shown visually. However, the principle of a soft-margin maximum classifier using features passed through a kernel function to make a non-linear classifying boundary is the same regardless of the number of features used. As with random forests, the theoretical advantage of this method is that the algorithm can identify non-linearities and interactions present in the data. However, the model again requires the selection of hyperparameters, this time including the regularisation parameter that determines the soft margin and the choice of kernel function (Duan et al., 2003).

Natively, support vector machines do not output probabilities, just the predicted classification. To obtain probabilities from a support vector machine, Platt scaling is used most often (Platt, 1999). This method fits a logistic regression to the values of the scoring function for each patient. The scoring function for a given observation is given by the below formula for a support vector machine with kernel function k and is analogous to the distance from the decision boundary, with data points further away from the decision boundary getting higher scores.

$$f(x) = \sum_{i=1}^{m} \alpha_i K(x_i, x) + C$$

In the above scoring function of m features, α_i represents the weights to be optimised by the SVM, K represents the given kernel function and C represents the complexity parameter (Gillani et al., 2014). This scoring function is used to predict classes for new observations and can be used to predict class probabilities using Platt scaling. Support vector machines also require feature scaling, as the algorithm is scale variant.

A recent example of SVM applied to clinical risk prediction is implemented by Cui et al. (2018). They develop a support vector machine to measure risk of readmission for patients with diabetes, alongside a logistic regression model and many other supervised machine learning algorithms. This paper engages in multiple issues with data preparation, such as class imbalance correction, which distorts risk predictions and can worsen model performance (Goorbergh et al., 2022). However, this

paper does find considerable gains for support vector machines modelling risk of readmission for patients with diabetes.

2.2.4. Neural Networks

Artificial Neural Networks (ANNs) constitute a multitude of approaches that are inspired by human neural architecture (Amato et al., 2013). The main advantage of these algorithms compared to parametric models such as logistic regression is in their ability to capture non-linearities and interactions automatically. The most widely used neural network model, and the one used in this analysis, is the feedforward neural network. In this network, the neurons are arranged in layers. There is an input layer, that takes in individual level patient data and represents individual patient characteristics, and multiplies them by weights and connections, a hidden layer, that further multiplies these inputs, and an output layer, that typically reflects patient risk of having a particular disease. It is the hidden layer that distinguishes the neural network from a straightforward logistic regression, and the hidden layer or layers allow for neural networks to model many different non-linear and interactive relationships between variables more automatically than linear regression (Steyerberg, 2019, p. 71).

A simple, hypothetical example of this is shown in **Figure 2-4** and predicts ten-year mortality in patients using the variables age and BMI.

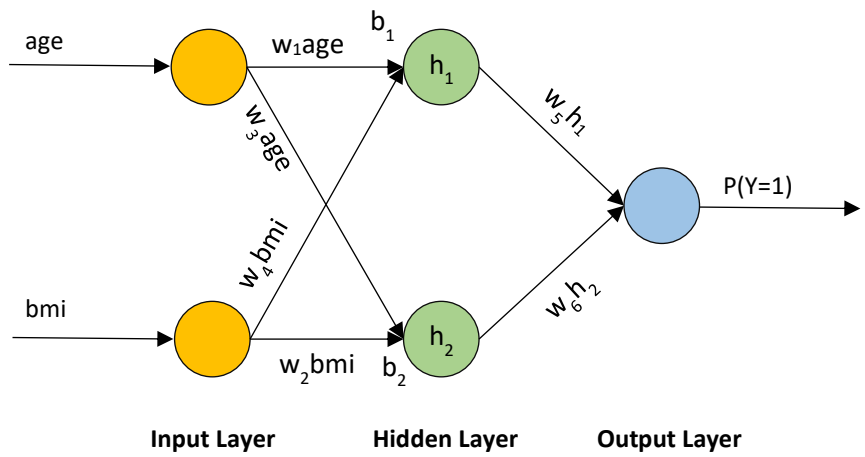

Figure 2-4: Hypothetical Artificial Neural Network (figure loosely based on Fei et al., 2018)

The neural network consists of nodes and connections between the nodes (Chandromouli et al., 2018). The input layer simply acts as a means of inputting data into the algorithm. The algorithm then multiplies the data by weights (w) in each connection and adds a bias term (b) before passing the data to the nodes in the hidden layer. The data is therefore fed through into the first hidden layer through a weighted link function. For instance, the input (x) for the first hidden node in the above network is given by:

$$x_1 = w_1 age + w_4 bmi + b_1$$

The input to a node in a neural network is the sum of all its connections and a bias. In the hidden layer, the input is then passed through an activation function. A default choice for this is the rectified linear unit (ReLU) activation function given below:

$$h_1 = f(x_1) = \max(0, x_1)$$

However, there are many activation functions, and the choice of activation function is also a hypeparameter that can be tuned. After the inputs have been passed through the activation function, they are again passed through connections and biases reaching an output layer. The final output is an estimated probability resulting from the output layer, which usually applies a sigmoid activation function in the case of binary classification to obtain probability of death in a ten-year period. The sigmoid activation function is shown below:

$$p = f(\mathrm{x}) = \frac{1}{1 + e^{-x}}$$

Whilst this is a simple example, typical fitted neural networks contain the same constituent parts, though they regularly contain many more input variables, connections, and hidden layers. The values of the weights and biases of the neural network are parameters to be estimated. These parameters are determined by backpropagation, which trains the model sequentially and updates the weights and biases using gradient descent to determine in which direction the weights should be adjusted, and a learning rule that determines the size of the step. The typical objective function used in backpropagation is the sum of squared residuals. Each training cycle is referred to as an epoch, and results in the weights and biases being adjusted (Basheer and Hajmeer, 2000). Like with random forests and support vector machines, the neural network has hyperparameters that dictate its final performance, including the number of hidden layers, the number of neurons in each layer, the activation function, the learning rate, and the number of epochs.

The algorithm laid out above is a feedforward neural network, in that there are no feedback loops. They contrast with recurrent neural networks, which have a feedback loop where data from later layers is fed back into previous layers (Kostadinov, 2018). This work considers only feedforward neural networks for binary classification, as these are by far the most used in the clinical risk prediction space and recurrent neural networks are mostly used when dealing with sequential data.

There are many different variants and additions to the basic feedforward neural network, and one widely used addition to the feedforward neural network that will be used later is the concept of dropout. Dropout randomly drops out internal nodes during training (along with their connections), having the effect of shrinking weights and reducing overfitting in a neural network. The concept of dropout is captured below:

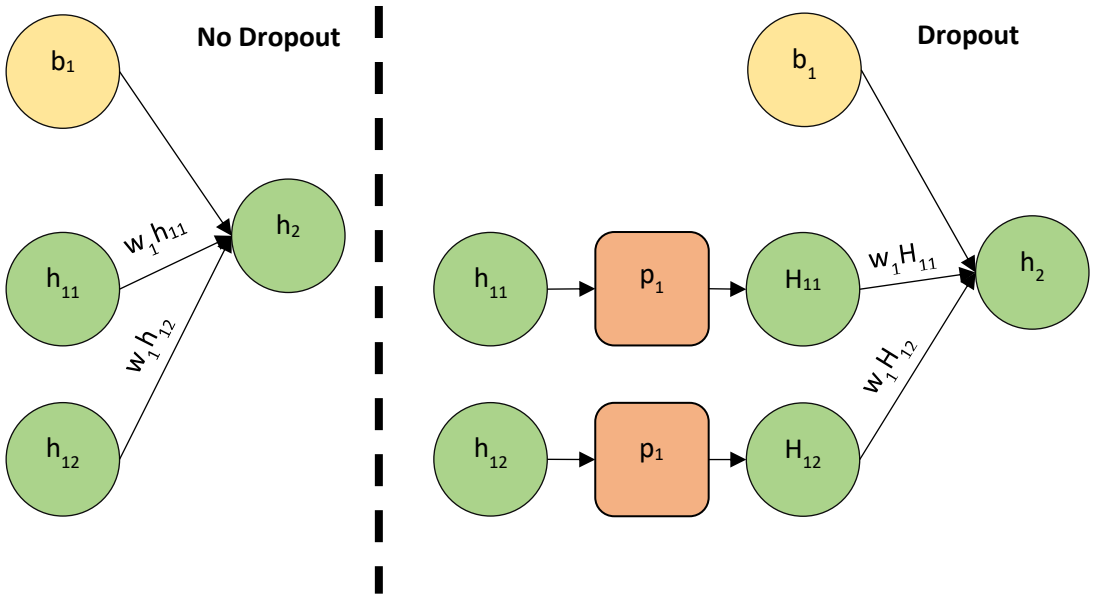

Figure 2-5: Dropout

When there is a dropout layer, the input from a node in a hidden layer has a fixed probability of being removed from the network, giving no input into the next layer, as shown in **Figure 2-5**. In complex neural networks with many hidden layers and internal nodes per layer, this has the effect of shrinking the weights and significantly reducing overfitting in a neural network, resulting in better performance (Srivastava, 2013). Whether or not to include a dropout set and the probability of dropout are additional hyperparameters to be tuned in model validation.

Another widely used addition to the feedforward neural network that will be used later is early stopping. As the model is trained, at each epoch the model is more prone to overfitting, resulting in performance increasing in the training set but worsening on the test set. This is typically prevented by early stopping (Prechelt, 1998). Early stopping refers to when training is stopped in the training set after performance worsens in a validation set. A validation set in this context is a subsection of the training data used to tune hyperparameters, and distinct from the test set that is used to measure performance. Early stopping is captured in **Figure 2-6**, where early stopping would take place when the loss of the validation set is minimised, as any further training is likely to result in overfitting.

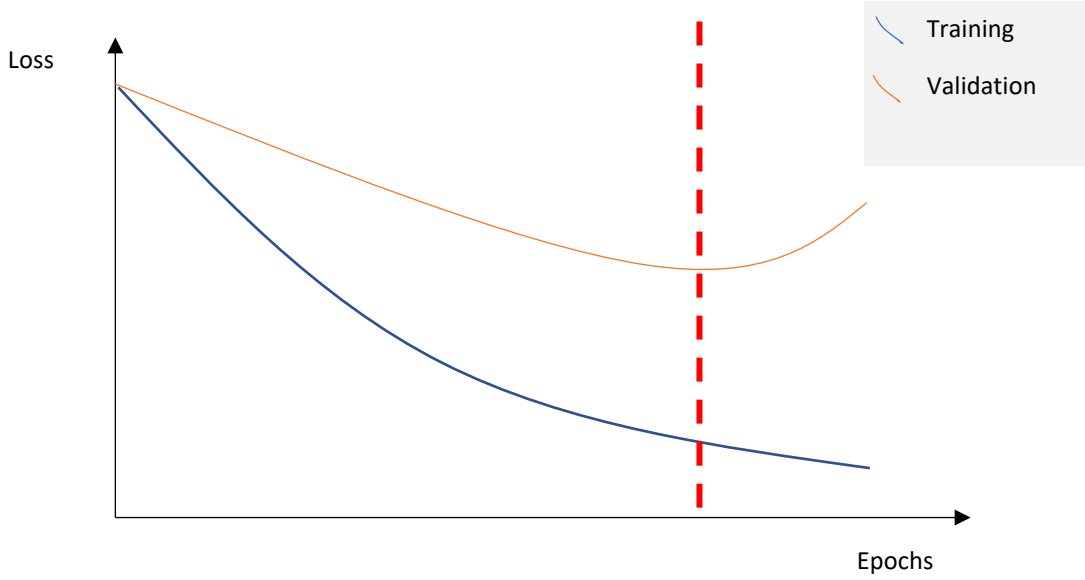

Figure 2-6: Hypothetical training and validation loss over epochs

2.2.5. Ensemble Approaches

An ensemble approach is the approach of combining multiple estimators to improved performance. An example of this would be combining the estimators of a random forest, a support vector machine, and a neural network together and letting them vote on a classification. This tends to improve performance, especially when diversity is high in the estimators and the correlation between errors is small (Sewell, 2008).

Estimators can be combined in multiple ways (Zhou, 2021), however, averaging and plurality voting are the most common. Averaging would entail using the mean of the probabilities from each of the different estimators, and plurality voting entails classifying the outcome of a patient based on the outcome the plurality of classifiers predict. A probability could then be obtained as the proportion of classifiers predicting the outcome. This is exactly how random forests output probabilities.

A more advanced algorithm leverages stacking (Zhou, 2021). Stacking uses the fitted probabilities of other multiple machine learning algorithms as features. For instance, a random forest, support vector machine and neural network's fitted probabilities can be used as features. A stacking algorithm then uses these probabilities and fits logistic regression with these as features, then outputs probabilities in the same way as logistic regression. Ensemble learning tends to improve performance over a single very good classifier as the generalisability of the learner is improved, no longer having a significant risk of training a learner that is poor or in a bad local optimum.

The success of any given ensemble learning classifier largely depends on two key factors, the accuracy and diversity of the individual learners (Zhou, 2021, p. 198). There is no generally accepted definition of diversity (Tang et al., 2006), and measuring diversity empirically is challenging. One definition given by Brown et al. (2005) is that it is the degree of disagreement among individual learners, and is regularly measured by individual classifier correlation, though other measures are valid.

One example of a hybrid ensemble being applied in clinical risk prediction is Nikookar and Naderi (2018), who used an ensemble of five different learners including a random forest and a support vector machine to estimate individual patient risk of heart disease and showed that ensembles of many heterogeneous estimators outperform individual ML algorithms.

2.3. Model Performance

There are a wide range of ways that clinical risk prediction models report performance. There are two main ways through which clinical prediction models should be assessed: discrimination and calibration (Alba et al., 2017). Discrimination refers to the ability of a model to differentiate high risk patients from low risk patients, and calibration refers to the accuracy of the risk estimates.

2.3.1. Discrimination

There are many ways that model performance is evaluated (Steyerberg, 2019). One of the most intuitive is accuracy, which most often uses a threshold of 0.5 in binary classification. This means that any patients with a probability above 0.5 are classified as positive or having the disease, and any patients below are classified as negative or not having the disease. The accuracy is then the total proportion of true positives and true negatives. However, only reporting accuracy is rare in recent literature. The reason for this is that using a 0.5 cut off implicitly assumes that false positives and false negatives are equally important, which is often absurd in the clinical risk prediction context (Greenland, 2008). As such, the most ubiquitous form of measuring discriminatory performance reported is the c-statistic or area under the curve (AUC), which measures the area under the receiver operator curve (ROC) (Steyerberg et al., 2010). The receiver operator curve plots the true positive rate against the false positive rate for every threshold. In the context of clinical risk prediction for mortality prognosis, the true positive rate is the probability that someone who dies in the next ten years will be predicted to die in the next ten years, and the false positive rate is the probability that someone who will be predicted to die in the next ten years survives (Fawcett, 2006). A threshold is the value above which an estimated probability must be to classify as positive and is 0.5 in the case of accuracy. An example receiver operator curve is shown in **Figure 2-7** in the case of logistic regression for predicting ten-year mortality.

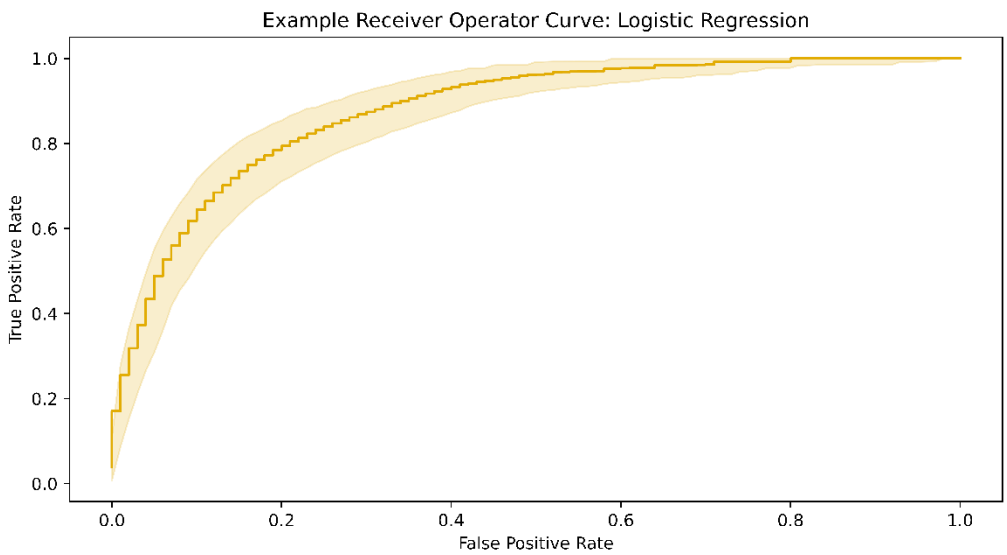

Figure 2-7: Example ROC curve with bootstrapped confidence interval

This receiver operator curve captures the trade-off in measuring performance, in that to increase the true positive rate, the false positive rate necessarily increases. The c-statistic or AUC can be interpreted as the probability that given two patients, one who has the disease and one who does not, the model successfully prescribes a higher probability to the individual who had the disease. As above, most ROC and AUC estimates are accompanied with a confidence interval, which is most often achieved by bootstrapping (Bertail et al., 2008).

Bootstrapping entails fitting the model to bootstrapped samples of the original dataset drawn with replacement, then calculating the ROC and AUC for each sample. Measuring differences in discrimination can then be measured by measuring the difference in AUC between two methods and performing a statistical test on whether a difference in discrimination performance is significant at a given significance level is done by creating confidence intervals for the difference in AUC. Confidence intervals are then calculated using a wide variety of methods (Carpenter and Bithell, 2000), most typically percentile methods, whereby to get a 90% confidence interval, the lower bound is the 5th percentile of the bootstrapped AUCs and the upper bound the 95th percentile of the AUCs. This method requires a symmetrical distribution of the AUCs to be unbiased (Carpenter and Bithelll, 2000), and researchers recommend bootstrap confidence intervals use at least 999 bootstrap samples to estimate reliable confidence intervals.

2.3.2. Calibration

Another important measure of model performance is calibration. Calibration captures the reliability of model estimates, and a model is said to be properly calibrated if, for those individuals with an estimated probability of $\hat{p}$, a proportion $\hat{p}$ develop the predicted outcome. There are many means of assessing a model's calibration (Van Calster et al., 2019), however, this is typically explored with a calibration plot, which plots the predicted probability from the patients against the observed proportions in the data. As the true probability of mortality is not directly observable, this is often done using a LOESS smoother (Steyerberg et al., 2010) as in **Figure 2-8**, or by binning observations with similar probabilities:

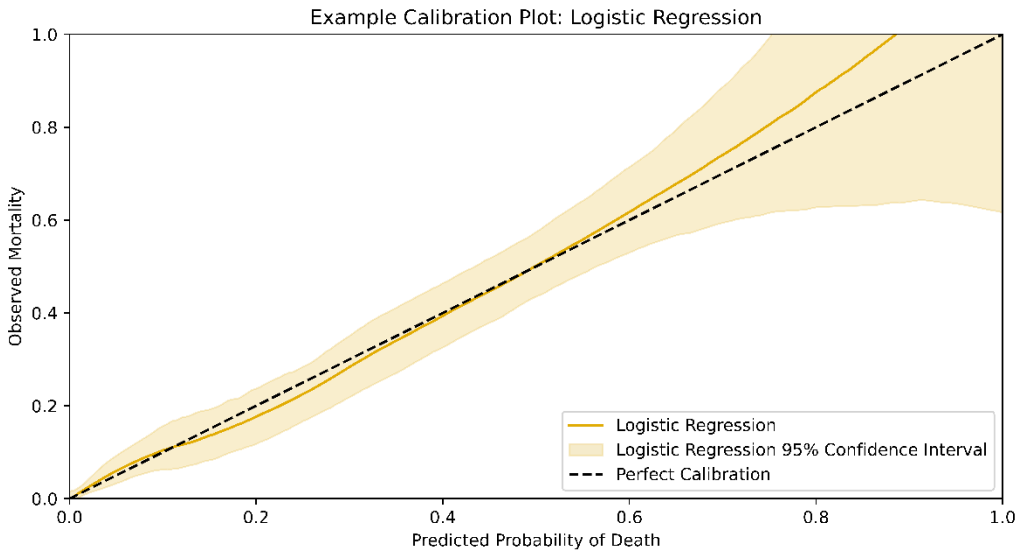

Figure 2-8: Example calibration plot with LOESS smoother. The predicted probabilities were taken from a logistic regression, and extrapolation of the LOESS smoother was used to fit over the 0-1 interval. Confidence intervals were derived using bootstrapping (as described in **Section 4.3**)

For instance, in the above calibration plot that assesses one of the later implementations of logistic regression, it can be observed that the logistic regression is well calibrated. This is because the 95% confidence interval of the predicted probability always lies within the observed probability of the LOESS smoother. Again, this confidence interval is obtained through bootstrapping.

However, using calibration plots is not sufficient to measure calibration performance, and is especially limited in the case of comparing calibration performance across models. This is because calibration curves are qualitative and therefore different people comparing calibration curves may lead to different assessments in performance. Furthermore, a calibration curve can be poorly calibrated only in certain regions where there are very few observations, and well calibrated where most observations lie. This is partially true in **Figure 2-8**, where the calibration of the logistic regression is poorest at the highest predicted probabilities of death where there were very few observations.

There are many means of measuring calibration performance (Huang et al., 2020), but one that will be used throughout this work and is based on calibration curves is the integrated calibration index (ICI) introduced by Austin and Steyerberg (2019). The ICI is equivalent to the mean of the absolute difference between the predicted probabilities from a given model and the observed probabilities that are derived from a smooth calibration curve (Austin et al., 2022), and can be shown in the below formula:

$$ICI = \frac{1}{n}\sum_{i=1}^{n} |\widehat{p_i} - \widehat{o_i}|$$

Where $\widehat{p_i}$ is the estimated probability from a given model that a given patient experiences the outcome, $\widehat{o_i}$ is the observed probability of mortality estimated by the LOESS smoother. Larger differences between observed and predicted probabilities imply a poorly calibrated model, and as such larger ICI values indicate poorer calibration. This measure of calibration has advantages in that it is quantitative and gives higher weight to regions of the calibration curve where there are more observations. However, the ICI is sensitive to the specification of the LOESS smoother.

The calibration of a model's risk estimates can be improved by implementing calibration models (Huang et al., 2020) such as Platt scaling, isotonic regression, and Bayesian binning. However, this further increases computational expense, and this work is aimed largely at comparing the typical methods that are used in practice for clinical risk prediction. As a result, whilst promising, the implementation of calibration models is outside the scope of this work.

Calibration is very rarely discussed in recent literature (Van Calster et al., 2019) despite the fact poorly calibrated models make predictions misleading. This is especially problematic when these models are used to support decision making in clinical practice. One hypothetical example could be treating a patient who has a relatively favourable prognosis, which may result in costly side effects and a poor use of scarce healthcare resources.

2.4. Model Validation

2.4.1. Cross Validation and Bootstrapping

Model validation refers to ensuring the model predictions generalise to new observations. This is typically done by partitioning the dataset into a training set, to fit the model, and a testing set to measure performance. The main reason to do this is to ensure model predictions generalise to data the model was not trained on (Chandramouli et al., 2018). Machine learning algorithms are highly flexible and can overfit, capturing noise in the dataset they were trained on. The most common method of model validation is k-fold cross validation, where the data is randomly split into k-equally sized portions (Saeb et al., 2017). The model is trained on the training set (k-2 portions), the model's hyperparameters are tuned on a portion (the validation set), and the last portion is used to evaluate performance (the test set). This is repeated k-times, and each time a different subsection is used as test data. The average AUC and calibration across the k test sets are then reported as model performance (Chandramouli et al., 2018). Cross-validation ensures a fair assessment of model performance as the reported test set accuracy is independent of the validation and training sets. This method of cross validation also mitigates overfitting as the hyperparameters chosen to maximise the validation set is used to tune hyperparameters, maximising performance on data the model was not directly trained on.

In addition to k-fold cross-validation, bootstrapping is another common way to partition the data into test-validate-train subsections by using simple random sampling with replacement for each section. This allows for the same data points to be in both the training and test sections but is useful for small datasets as the number of training-validation-test data samples are unlimited (Chandramouli et al., 2018).

2.4.2. Hyperparameter Tuning

Hyperparameters are model parameters that are not optimised directly during training (Probst et al., 2019a) and are set by the researcher before model training. They contrast with parameters, which are learned by the model during model training. An example hyperparameter would be the minimum data points per leaf node in the constituent decision trees of a random forest, whereas an example parameter would be which feature to split upon in the constituent decision tree (Luo, 2016). When building ML algorithms, it is possible to use the default values of these parameters from software packages, however, manual, or automatic optimisation of hyperparameters using a validation set typically leads to improved performance (Luo, 2016). It is important to distinguish the difference between the validation and test sets here, as literature often uses these terms interchangeably. In this work, the validation set is a subset of the training set used to optimise hyperparameters, whereas the test set is a set of data not used in model training to measure performance and see how the model generalises to new observations (Chandramouli et al., 2018)

Cross validation is used most often to optimise hyperparameters (Duarte and Wainer, 2017) and results in hyperparameters with improved performance on new data. This typically entails partitioning the dataset into two parts, train, and test. Within the training set, k-fold cross validation is used, with k-1 parts of the data used to train the model and the last part as a validation set to evaluate performance. To tune hyperparameters, typically different combinations of hyperparameters are tried, and the optimal hyperparameters chosen are those that maximise average k-fold validation performance (such as AUC). Final performance is then reported on the test set with the optimal hyperparameters found in the training and validation sets. This method of optimising hyperparameters is used to give an unbiased estimate of model performance (Zhou, 2021), as using the test data to optimise hyperparameters could result in overfitting, capturing

idiosyncrasies in the test set that would not generalise to new observations (Christodoulou et al., 2019).

There are a multitude of ways of determining how to choose which subset of hyperparameters to use in model training, including manual search, grid search, random search, and Bayesian optimisation (Wu et al., 2019). Manual search leverages the knowledge of the researcher and tries out hyperparameter sets by hand. Grid search entails attempting every possible combination of hyperparameters. Random search tries random combinations with a range of values. Bayesian optimisation uses Bayes theorem to direct the search to find the hyperparameters that maximise the objective function of the machine learning algorithm. Each method has their advantages, with exhaustive grid search being guaranteed to find the best hyperparameters from a given range, though it is not feasible in the case of continuous hyperparameters and is the most computationally expensive. Random search is more efficient (Bergstra and Bengio, 2012) but unstable for training complex models. Bayesian optimisation requires a prior distribution of hyperparameters which must be chosen by the researcher. Overall, the automatic tuning of a machine learning algorithm is a problem that is yet to be fully solved in machine learning (Wu et al., 2019).

Hyperparameter tuning is also something that is regularly done in a biased or unclear manner in recent studies. Christodoulou et al. (2019) show that the model validation procedure was biased or unclear in 68% of studies. Typically, this arises when hyperparameters are tuned on the entire dataset (test and train). The results in overfitting as it uses the test data to select hyperparameters, therefore allowing the model to capture noise in the test dataset and result in artificially improved performance of machine learning as the overfitting results in improved performance in the test set that would not generalise to new observations.

3. Data

This section gives an overview of the dataset used and details of how the data was pre-processed. The code used to produce all results and graphs displayed in the following sections is available at https://github.com/SamHusbands21/thesis and is written largely in Python 3.9.5. The LOESS smoothing for the calibration curves was written in R 4.1.2.

3.1. About NHANES

To compare logistic regression and the different supervised machine learning algorithms respective performance, these approaches were each applied in turn to the National Health and Examination Survey (NHANES). The 2007/08 (CDC, 2008) wave was selected and merged with the 2019 Public-Use Linked Mortality File (LMF) to create a variable that reflects ten-year mortality. The NHANES programme has been carried out by the National Center for Health Statistics in the USA since the 1960s as part of the Centers for Disease Control and Prevention, with the main aim of producing high quality health statistics to determine disease prevalence and risk factors for major diseases that help develop public health policy (CDC, 2017).

This survey was selected due to the wide range of variables collected on a large sample of approximately 10,000 Americans, which will allow for comparison of how the different methods fare with different levels of training data and different levels of explanatory variables. The dataset consists of demographic, dietary, examination, laboratory, and questionnaire data, which amounts to over 1,000 candidate explanatory variables. The 2019 Linked Mortality File uses death certificate records to determine the mortality status of participants (CDC, 2022).

3.2. Variables

The aim of initial analysis was to compare how well logistic regression and supervised machine learning estimated the risk of mortality for each patient in the context of a subset of known, powerful predictors of mortality with a large sample size, as this is the typical structure of other clinical risk prediction models. As a result, nine covariates were chosen from the 2007-2008 NHANES wave. These were age, sex, BMI, systolic and diastolic blood pressures, total cholesterol level, smoking status, alcohol status and whether the subject had diabetes. These variables were chosen due to their frequency in other clinical risk prediction models, and it was largely expected that these variables are amongst the strongest predictors of mortality in a ten-year period. This covariate data was then linked to the 2019 linked mortality file.

All analysis in this work is complete case, meaning any individuals with missing data for one or more of the covariates or explanatory variables were removed. This can be problematic as it may introduce bias into the results as often the reason the data is missing can be related to the outcome variable (Sterne et al, 2009), and dropping a large portion of the dataset results in worse precision and power. However, this bias and loss of precision is common to all models and should be sufficient for comparing logistic regression and supervised machine learning. The alternative solution to complete case analysis would be using an imputation-based method such as multiple imputation for handling missing data. This entails creating multiple datasets, then using a predictive distribution based on the observed data to sample the missing data. Multiple imputation also has limitations in that it is computationally more expensive (Sterne et al, 2009) and makes strong assumption such as the 'missing at random' assumption, which means assumes that any systematic differences between observed and missing data can be entirely explained by other observed variables (Bhaskaran and Smeeth, 2014). For simplicity and to reduce computational expense, the analysis was therefore restricted to complete case. After the complete case restriction was applied, the dataset consisted of 4,756 participants.

Table 3-1 gives a description of each of the variables and summary statistics. It is worth noting that age, a key predictor of mortality, was truncated by the CDC at higher values and rounded down to 80. No imputation was used to correct this, and the same data was given to all models.

Variable Name	Description	Summary
RIDAGEYR	Age in years of the participant. Note that any subjects over the age of 80 were rounded down to age 80.	**Mean**: 50.56 **Standard Deviation**: 17.73 **Min**: 20, **Max**: 80
RIAGENDR	Gender of the participant	**Female:** 2,407 **Male:** 2,349
BMXBMI	Body Mass Index of the Participant (kg / m^2)	**Mean**: 28.94 **Standard Deviation**: 6.55 **Min**: 14.20 **Max**: 73.43
SMQ020	Have you smoked at least 100 cigarettes over your lifetime?	**Yes**: 2,279 **No:** 2,477
ALQ101	Have you had at least 12 alcohol drinks in the past year?	**Yes:** 3,376 **No:** 1,380
BPXSYN	Mean of the four readings of systolic blood pressure (BPXSY1, BPXSY2, BPXSY3 and BPXSY4), measured in mm Hg.	**Mean**: 124.13 **Standard Deviation**: 18.63 **Min**: 80.67 **Max**: 212.67
BPXDIN	Mean of the four readings of diastolic blood pressure (BPXDI1, BPXDI2, BPXDI3 and BPXDI4), measured in mm Hg.	**Mean**: 69.94 **Standard Deviation**: 13.02 **Min**: 0 **Max**: 118.67
LBDTSCI	Total Cholesterol (mmol/L)	**Mean**: 5.10 **Standard Deviation**: 1.08 **Min**: 2.35 **Max**: 11.90
DIQ010	Have you ever been told by a doctor you have diabetes or sugar diabetes, other than during pregnancy?	**Yes:** 599 **No:** 4,157
Mortality	Whether or not the participant is deceased 120 months (10 years) from the interview date. This variable is derived from the linked mortality file.	**Deceased**: 658 **Alive**: 4,098

Table 3-1: *Description of variables and summary statistics for the 4,756 participants used in model building. Data comes from the 2007-2008 NHANES wave linked with the 2019 Linked Mortality File.*

Variable Name	Description	Missing Details
RIDAGEYR	Age in years of the participant. Note that any subjects over the age of 80 were rounded down to age 80.	**Count:** 10,149 **Missing**: 0
RIAGENDR	Gender of the participant	**Count:** 10,149 **Missing**: 0
BMXBMI	Body Mass Index of the Participant (kg / m^2)	**Count:** 8,861 **Missing**: 1,288
SMQ020	Have you smoked at least 100 cigarettes over your lifetime?	**Count:** 5,929 **Missing**: 4,220
ALQ101	Have you had at least 12 alcohol drinks in the past year?	**Yes:** 5,227 **No:** 4,922
BPXSYN	Mean of the four readings of systolic blood pressure (BPXSY1, BPXSY2, BPXSY3 and BPXSY4), measured in mm Hg.	**Count:** 7,350 **Missing:** 2,799
BPXDIN	Mean of the four readings of diastolic blood pressure (BPXDI1, BPXDI2, BPXDI3 and BPXDI4), measured in mm Hg.	**Count:** 7,350 **Missing:** 2,799
LBDTSCI	Total Cholesterol (mmol/L)	**Count:** 7,387 **Missing**: 2,762
DIQ010	Have you ever been told by a doctor you have diabetes or sugar diabetes, other than during pregnancy?	**Count:** 9,551 **Missing**: 598
Mortality	Whether or not the participant is deceased 120 months (10 years) from the interview date. This variable is derived from the linked mortality file.	**Count:** 6,219 **Missing**: 3,930

Table 3-2: *Missing Values by Variable*

Table 3-2 demonstrates most of the ineligible data was dropped due to ineligibility of subjects to be part of questionnaires or the linked mortality file. This data was most often missing by design, because the participants were under eighteen, giving rise to data protection concerns.

4. Methods

This section gives a detailed overview of model development for logistic regression and each machine learning algorithm in the context of a subset of known, powerful predictors of mortality with a sample size of 4,756 Americans.

4.1. Logistic Regression

Logistic regression is simpler to train than most supervised machine learning algorithms owing to its lack of hyperparameters. The dataset was split into 80% training data and 20% testing data, a default proportion of train and test data (Chandramouli et al., 2018). The train and test data were also stratified by the dependent variable to prevent differentials in class imbalance between test and train. The logistic regression assumed linearity and additivity, not including any cubic splines or interactions. This was to see if supervised machine learning could outperform default logistic regression, often the default of researchers and the choice of which interactions and non-linearities to include often requires domain knowledge. The logistic regression was fit on the training data and performance was measured on the testing data. The model performance was evaluated on the holdout test set to get a sense of how well logistic regression generalised to new observations.

Performance of the models was measured in two ways. Discriminatory performance was measured by the Receiver Operator Curve and the Area Under the Receiver Operator Curve (AUC), the most ubiquitous way of measuring discriminatory performance, and calibration was assessed using calibration plots and the Integrated Calibration Index (ICI) as recommended by Austin and Steyerberg (2019).

Bootstrapping was used to construct 95% confidence intervals for the ROC, AUC, calibration plots and ICI, in a method that was based on recommendations from Austin and Steyerberg (2019). Bootstrapping was then used as to conduct formal statistical comparison of differences in performance across models. The dataset was bootstrapped 999 times in accordance with Carpenter and Bithell's (2000) recommendation that this was the minimum number of bootstrap samples to obtain reliable bootstrap confidence intervals for a symmetric distribution. The model was trained on 80% of each bootstrapped dataset and the ROC and AUC were measured on the remaining 20% of each bootstrapped dataset. The ROC shown is the median ROC, in that for each false positive rate the median true positive rate across bootstraps is plotted. The confidence interval was again generated using bootstrapping, with the lower bound being the 2.5th percentile of the true positive rate for each false positive rate, and the upper bound being the 97.5th percentile.

Calibration was first assessed using calibration plots. The model was trained on 80% of each bootstrapped dataset and the plot was produced using the remaining 20% of each bootstrapped dataset. The calibration plot was produced using predictions generated on the test data to show how the model generalised to new observations. The calibration curve was fitted using the LOESS function in R with default parameters, following guidance from Austin and Steyerberg (2014) that this tended to result in good performance. To produce the calibration plots, the LOESS function in R was extrapolated to produce estimates over the entire 0-1 range of predicted probabilities for each bootstrap.

The calibration curve shown is the median calibration curve, in that for each predicted probability it is the median observed probability across bootstraps that is generated from the LOESS smoother. The 95% confidence interval was generated using percentiles, with the lower bound of the confidence interval being the 2.5th percentile of the observed probabilities and the upper bound being the 97.5th percentile. However, calibration plots are qualitative and are not well equipped to

capture differences in performance across models. To capture differences in calibration performance, the integrated calibration index (ICI) was used. The ICI was measured for each bootstrap to capture dispersion in calibration performance.

Both the AUC and ICI are then used in hypothesis tests to assess differences in both discrimination and calibration performance. Details of how these hypothesis tests were conducted is found in **Section 4.3**.

Figure 4-1 demonstrates the general methodology for fitting the logistic regression, with the logistic regression fitted assuming linearity and additivity and the same bootstrap datasets were used to fit the later supervised machine learning algorithms.

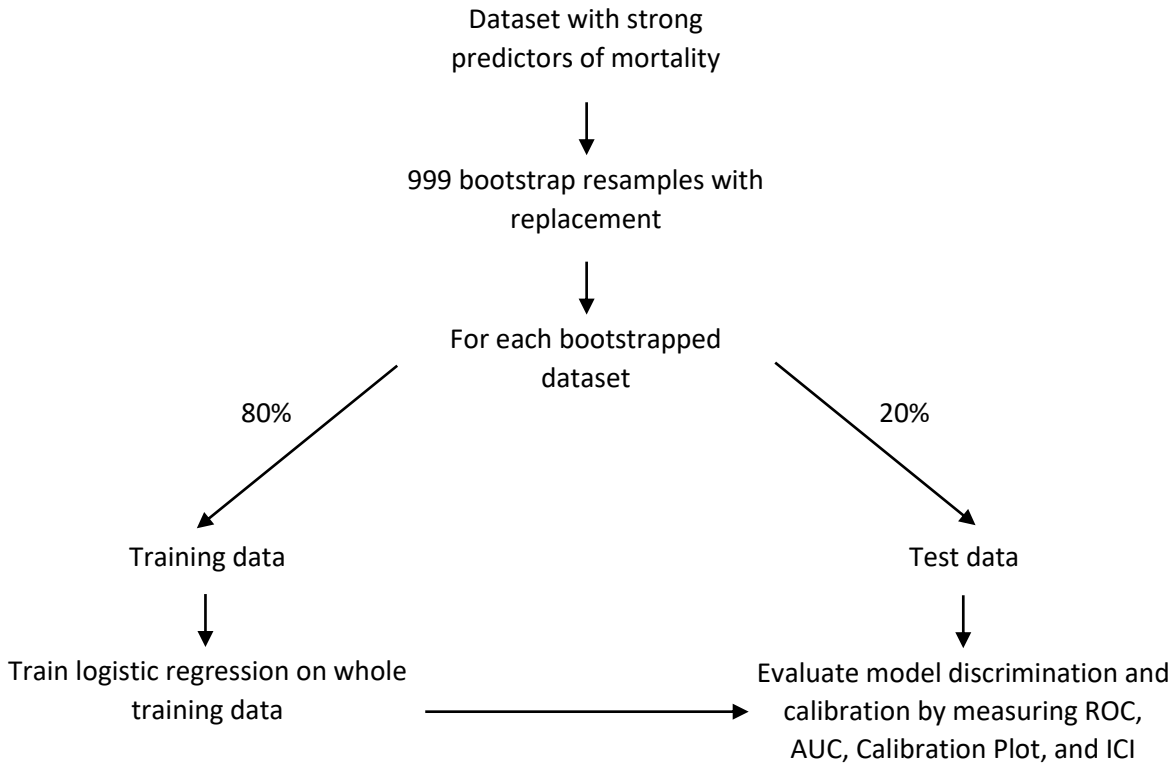

Figure 4-1: Methodology for logistic regression

4.2. Supervised Machine Learning

The model development of all supervised machine learning algorithms is more involved owing to the optimisation of hyperparameters. Insofar as this was possible, the model development was kept consistent between methods to facilitate a fair comparison. As such, for all supervised ML, the dataset was split into 80% training data and 20% testing data, again stratified by the dependent variable. Within this training data, three-fold cross validation was used to validate the model performance with different choices of hyperparameters.

The model was trained on two-thirds of the training data and the validation AUC calculated on the remaining third of the training data. The optimal hyperparameters were those that maximised average validation AUC. It is important to note here that this validation set is part of model training, and distinct from the test set that is not used in model training, but to measure performance. This entailed fitting the model to the training data three times, each with a different third of the training data used as the validation set. The hyperparameters that maximised the average of the three

validation AUCs were chosen as the optimal hyperparameters, and final performance was measured using the AUC and ROC of the test set. Three-fold cross validation was used as opposed to the more conventional choices of five-fold or ten-fold due as to minimise computational cost, though this comes at a cost of potentially suboptimal hyperparameters. Hyperparameters were optimised this way as it is generally found in recent studies that cross validation results in hyperparameters with better performance on new data (Duarte and Wainer, 2017), and hyperparameters were optimised only on training data to prevent hyperparameters being chosen that optimised performance on the test set, something that is likely to result in overfitting and therefore an overestimation of the performance of supervised machine learning (Christodoulou et al., 2019).

The hyperparameters optimised are necessarily bespoke for each algorithm, however, as a general principle random search was used to search for candidate hyperparameters in given ranges as this approach is more efficient and generally yields better hyperparameters than grid search (Bergstra and Bengio, 2012). The hyperparameters were drawn from either a uniform or log-uniform distribution, with the log-uniform distribution preferred when the hyperparameter ranges spanned orders of magnitude. The ranges were chosen by hand and based on guidance for each algorithm. The hyperparameters were then optimised by searching the range, randomly drawing hyperparameters and fitting the model to the training data. Average validation AUC was then measured using three-fold cross validation as above. This process was repeated ten times, performing ten searches of the hyperparameter space, and then training the algorithm with each set of candidate hyperparameters. Ten random searches were chosen again due to computational constraints. The final hyperparameters chosen maximised the average validation AUC using three-fold cross validation, and these hyperparameters were used to train the model on the entire training data.

Final performance was measured in the same way as with logistic regression, using the ROC and AUC of the test set to measure discriminatory performance and the calibration plot and ICI of the test set to measure calibration performance. The ROC, AUC calibration plot, and ICI were measured for each algorithm, using a holdout test set to get a sense of how well each supervised machine learning algorithm generalised to new observations. The hyperparameters were optimised using cross validation instead of on the holdout test set, as using the test set to optimise hyperparameters would be biased model development. This is because using the test set to optimise hyperparameters amounts to overfiring as the hyperparameters could be chosen in a way that captures idiosyncrasies in the test set that would not generalise to new observations.

As before, the dataset was bootstrapped 999 times and the algorithm was run 999 times, to each bootstrap draw of the original dataset. This entailed for each of the 999 bootstrap draws:

1. Splitting the data into 80% train and 20% test.
2. Using random search to get candidate hyperparameters.
3. Fitting the model with these hyperparameters three times to the training set (two-thirds train, one-third validation), each with a different partition of the training set as the validation set.
4. Repeating steps 2 and 3 ten times, performing ten different random searches.
5. Selecting the optimal hyperparameters that maximised validation AUC.
6. Training the model with optimal hyperparameters on the training set.
7. Reporting final performance on the test set (ROC curve, AUC, calibration plot and ICI)

Bootstrapping is used to capture the dispersion in the ROC, AUC, calibration plot, and ICI which is important when comparing the methods discrimination and calibration performance. Again, the AUC

and ICI are used in hypothesis tests to assess differences in both discrimination and calibration performance. Details of how these hypothesis tests were conducted is found in **Section 4.3**.

Figure 4-2 demonstrates the general methodology for the fitting of each of the machine learning algorithms. The fitting of each machine learning model required hyperparameter optimisation, and the details of the hyperparameter optimisation are covered in the subsequent sections.

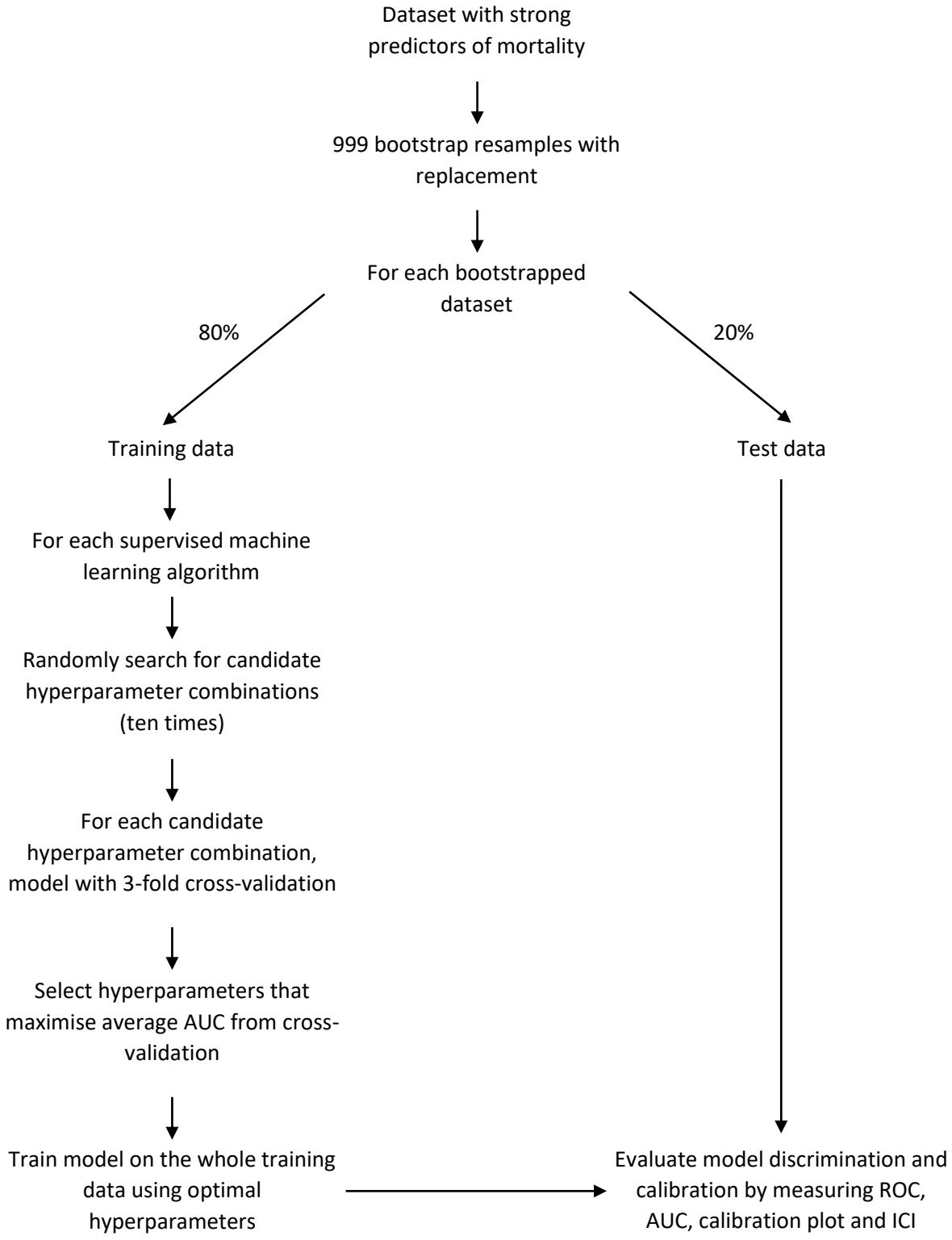

Figure 4-2: Methodology for supervised machine learning

4.2.1. Random Forest

For random forest, the hyperparameters optimised were the number of features considered at each split, the max depth of each constituent decision tree, the minimum data point per leaf node and the number of constituent trees in the random forest. Each decision tree was trained on all the training samples, and the bootstrapping was drawn with replacement. The hyperparameters were optimised using a random search method, randomly choosing hyperparameters from given ranges and choosing hyperparameters that maximised average validation AUC using three-fold cross validation as above.

The random search for the random forest algorithm used a uniform distribution to choose hyperparameters, and the hyperparameters were chosen from the following distributions:

- Max Depth: Discrete Uniform [3, 8][1]
- Max Features: Discrete Uniform [1, 9]
- Minimum Data Points per Leaf: Discrete Uniform [1, 5]
- Number of Trees: Discrete Uniform [100, 1000]

These ranges were chosen by hand, although they are chosen to be broadly in line with typical default values outlined in Probst et al. (2019b). The optimal split was chosen by minimising the Gini impurity as this is less computationally expensive than maximising information gain (Raileanu and Stoffel, 2004) and generally results in similar performance.

The random search algorithm drew from the hyperparameter sample space ten times, drawing ten different combinations of hyperparameters. The final hyperparameters chosen maximised average validation AUC using three-fold cross validation.

As with logistic regression, the main package used for fitting the random forests was scikit-learn (Pedregosa et al., 2011), and all plotting used Matplotlib (Hunter, 2007). SciPy. Stats (Virtanen et al., 2020) was also used to draw the hyperparameters from their respective uniform distributions.

4.2.2. Support Vector Machine

For support vector machines, the hyperparameters optimised in SVM were C and gamma, with gamma being the hyperparameter associated with the radial basis function kernel and C the complexity parameter that represents a trade-off between maximising the margin and minimising the number of misclassifications. Note that feature scaling was used on all features prior to model training using min-max normalisation, as SVM is scale variant. The below feature scaling ensured all features lied in the 0-1 range:

$$x = \frac{x - \min(x)}{\max(x) - \min(x)}$$

Again, a random search algorithm was used to tune hyperparameters. The random search used a log-uniform distribution from SciPy.Stats (Virtanen et al., 2020) to choose hyperparameters as the hyperparameter ranges spanned various orders of magnitude:

- C: Log-Uniform [0.001, 1000]
- Gamma: Log-Uniform [0.001, 1000]

[1] All distributions use [min, max] notation, and the maximum is inclusive.

These ranges were chosen in accordance with guidance from scikit-learn, although random search was used instead of the recommended grid search to speed up computation and to conform with the other supervised ML methods in this thesis.

As before, the random search algorithm drew from the hyperparameter sample space ten times, drawing ten different combinations of hyperparameters. The final hyperparameters chosen maximised average validation AUC using three-fold cross validation, and final performance was then measured using the ROC curve and AUC of the test set. The method was applied to 999 bootstraps to get a sense of dispersion in the ROC and AUC. As previously, the main package used for fitting the SVM was scikit-learn (Pedregosa et al., 2011), and all plotting used Matplotlib (Hunter, 2007).

4.2.3. Neural Network

The model development of the feedforward neural network was again similar to the other supervised machine learning algorithms. As with SVMs, feature scaling was applied prior to model training to increase training speed and performance. The hyperparameters optimised in the neural network were:

- Number of hidden layers: Discrete Uniform [1, 5]
- Internal nodes per hidden layer: Discrete Uniform [36, 360]
- The probability of dropout per node in hidden layer: Discrete Uniform [0, 0.3][2]
- The learning rate: Log Uniform [0.0001, 0.1]

These ranges were chosen manually, as there was limited reliable academic guidance on default ranges for these hyperparameters. It is important to note that in this model development, there are some hyperparameters that were not tuned including:

- The maximum number of epochs (set to 30)
- The batch size (set to 16)
- The activation function (ReLU for hidden layers, sigmoid for last layer to output probabilities)

This was to save on computational efficiency, though by not tuning these hyperparameters it is possible the model performance of the Neural Network is somewhat worsened. It is also important to note that this algorithm utilised early stopping. When the validation set successively increased its loss function, training was stopped early. When fitting the model with optimum hyperparameters, the number of epochs was set to the number of epochs when training was stopped.

As before, the random search algorithm searched the hyperparameter sample space ten times, drawing ten different combinations of hyperparameters. The final hyperparameters chosen maximised average validation AUC using three-fold cross validation. The main package used for fitting the neural networks was TensorFlow (Abadi et al., 2016), as this has more extensive support for neural networks than scikit-learn. As before all plotting utilised Matplotlib (Hunter, 2007).

[2] Equal probability of drawing 0 (no dropout), 0.1 , 0.2 and 0.3

4.3. Measuring Performance

To properly assess the differences in discrimination and calibration performance, hypothesis tests were conducted. When measuring differences in discrimination performance, the null hypothesis was that there is no difference in discriminatory performance between logistic regression and supervised machine learning. The alternate hypothesis was that there was a difference in discriminatory performance between logistic regression and supervised machine learning.

The test statistic for differences in discriminatory performance was the difference in AUC between each supervised machine learning and algorithm and logistic regression. As higher AUC values imply improved discrimination performance, a positive difference in AUC between supervised machine learning and logistic regression implies better discrimination performance for supervised machine learning. This test statistic was calculated for each bootstrap, using the 999 bootstrapped AUCs for logistic regression and each supervised machine learning algorithm. Following on from this, a 95% confidence interval for the difference in AUC was calculated using percentiles. The null hypothesis of there being no difference in discriminatory performance was rejected if this confidence interval did not contain zero.

Similarly, differences in calibration performance were measured using differences in the ICI. The null hypothesis was that there is no difference in calibration performance between logistic regression and supervised machine learning. The alternate hypothesis was that there was a difference in calibration performance between logistic regression and supervised machine learning.

The test statistic for this hypothesis test was differences in the ICI between logistic regression and each supervised machine learning algorithm. As a lower ICI indicates better calibration performance (Austin and Steyerberg, 2019), a positive difference in the ICI between logistic regression and supervised machine learning implies that supervised machine learning algorithm is better calibrated than logistic regression, and a negative difference in the ICI between logistic regression and supervised machine learning implies that logistic regression is better calibrated. The test statistic was measured for each bootstrap. Using the 999 bootstraps, a 95% confidence interval of difference in ICI between logistic regression and each supervised machine learning algorithm was again generated using percentiles, and the null hypothesis of no difference in calibration performance was rejected if the percentile confidence interval does not contain zero.

5. Results

5.1. Discrimination

Discrimination refers to the ability of a model to differentiate high risk patients from low risk patients (Steyerberg et al., 2010) and is most commonly assessed by the Receiver Operator Curve (ROC) **Figures 5-1, 5-2 and 5-3** compare the ROC for logistic regression to each of the supervised machine learning algorithms. These figures demonstrate that the median ROC curve is higher and to the left for both random forests and support vector machines, indicating superior discrimination performance for these algorithms. This is because for any given number of false positives, these algorithms identify more true positive cases. However, these plots do not make clear that these improvements in discriminatory performance are statistically significant, as the confidence intervals of the receiver operator curve intersect for all algorithms.

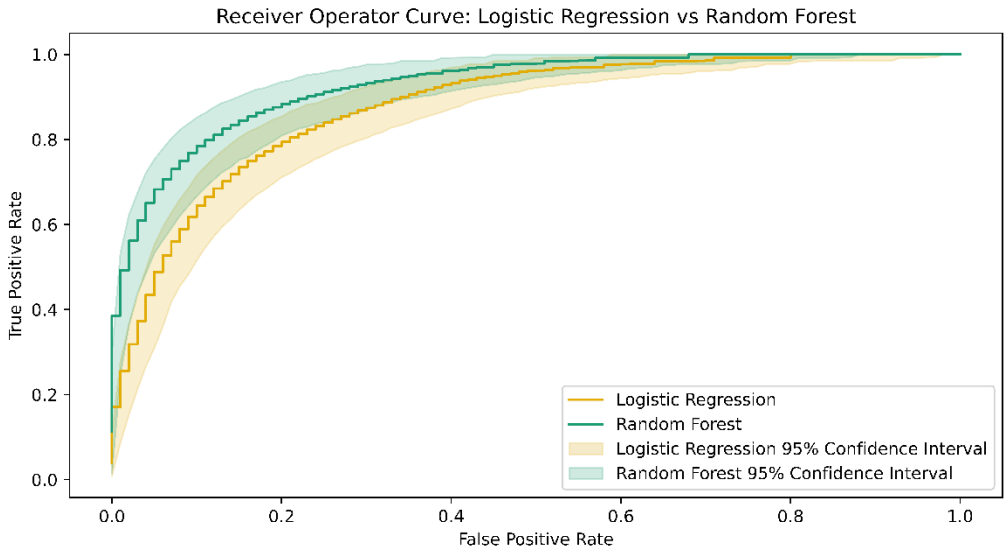

Figure 5-1: Receiver operator curve for logistic regression and random forest. The ROC was calculated for each bootstrap, and the median true positive rate for each false positive rate is plotted as the median ROC. Confidence intervals were derived using bootstrapped percentiles.

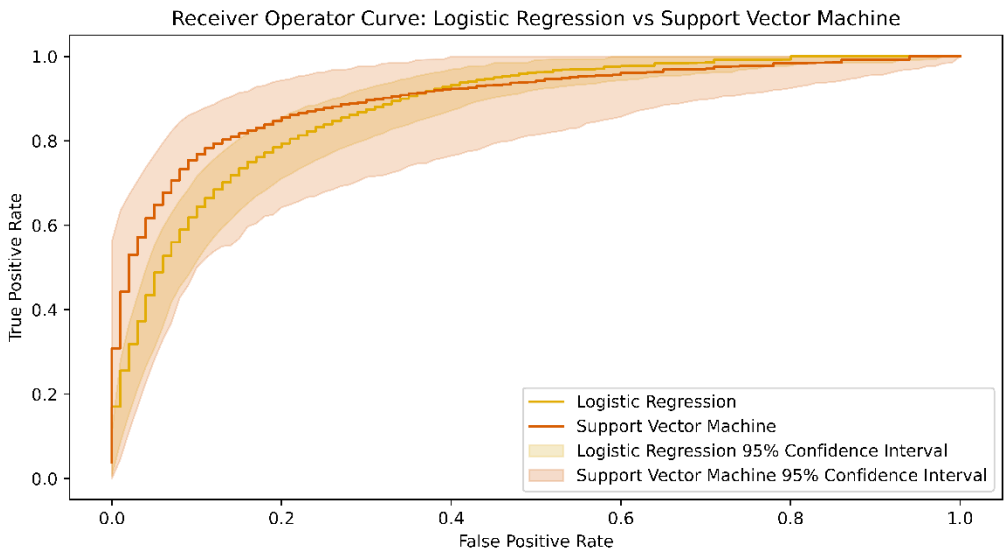

Figure 5-2: Receiver operator curve for logistic regression and support vector machine. The ROC was calculated for each bootstrap, and the median true positive rate for each false positive rate is plotted as the median ROC. Confidence intervals were derived using bootstrapped percentiles.

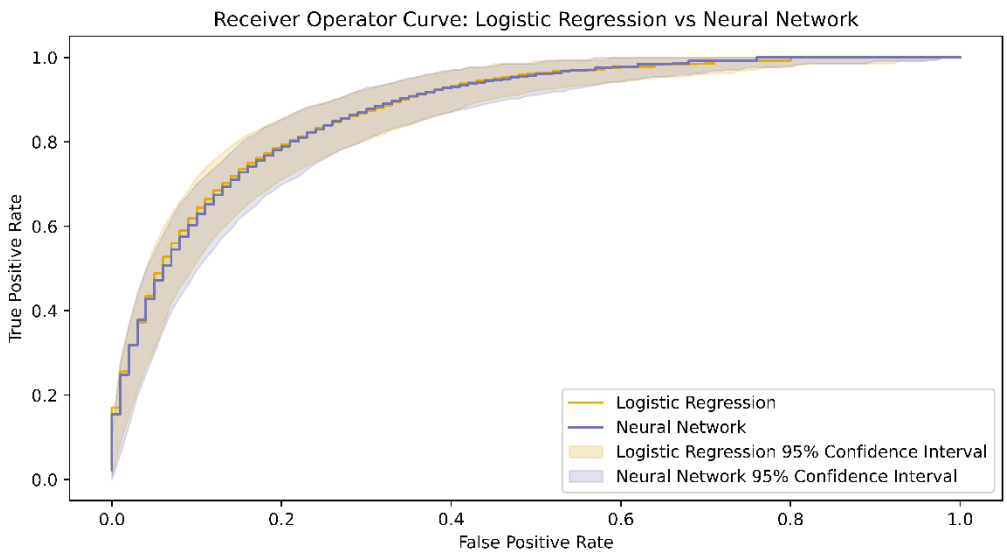

Figure 5-3: Receiver operator curve for logistic regression and neural network. The ROC was calculated for each bootstrap, and the median true positive rate for each false positive rate is plotted as the median ROC. Confidence intervals were derived using bootstrapped percentiles.

Figure 5-4 shows boxplots of the difference in AUC between each supervised machine learning algorithm and logistic regression, with the whiskers showing 95% confidence intervals. This is measuring the difference in discrimination performance between logistic regression and each supervised machine learning algorithm, with positive differences in AUC indicating superior performance for machine learning in discriminating between high and low risk individuals. It is clear from this plot that only random forests have a statistically significant improvement in discriminatory performance over logistic regression at the 5% level. Neither support vector machines nor neural

networks demonstrate a statistically significantly higher AUC. For random forests, the mean AUC across the 999 bootstraps was 0.922, and the mean AUC of logistic regression was 0.872. AUC can be interpreted as the probability that given two patients, one who died and one who did not, the model successfully prescribes a higher probability to the individual who died. Random forests therefore successfully prescribed the higher probability to individuals who died 5% percentage points more often than logistic regression in expectation, and this difference is statistically significant at the 5% significance level. However, neither support vector machines nor neural networks achieved statistically superior discriminatory performance at the 5% significance level.

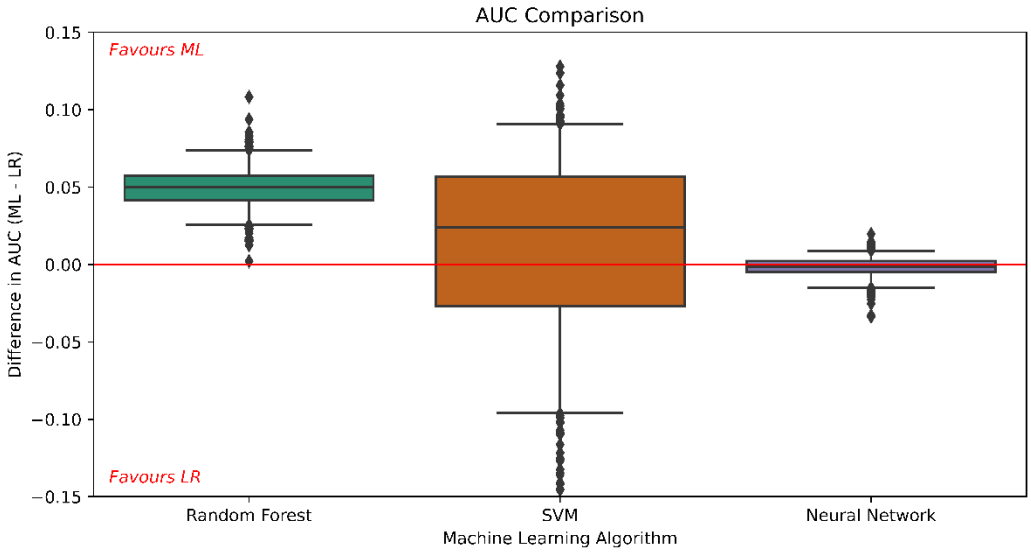

Figure 5-4: Comparing the AUC of each supervised ML algorithm to logistic regression. AUC differences were calculated across 999 bootstraps, with whiskers showing 95% percentile confidence interval

This result is somewhat surprising, as it might be expected that all machine learning approaches would outperform a logistic regression that assumes linearity and additivity, as these advanced approaches are able to automatically learn any non-linearities and interactions present in data. It also does not conform with previous literature such as Christodoulou et al. (2019), which demonstrated that there was no performance benefit from supervised machine learning over logistic regression in the clinical risk prediction space, even for random forests.

5.2. Calibration

Calibration measures the reliability of risk estimates. Calibration is most often assessed visually by means of calibration plots (Steyerberg, 2019, p.297). The below calibration plots (**Figures 5-5: 5-7**) are plotted using the median calibration curve across the test set of the 999 bootstraps. These plots demonstrate qualitatively that logistic regression has the best calibration performance, as logistic regression has predicted probabilities that are always close to the observed probabilities measured by the LOESS smoother. It is notable that the 95% confidence interval is narrowest for logistic regression, implying that there is less dispersion of probability estimates across bootstraps. However, it is not clear quantitatively that logistic regression has an advantage in its calibration performance over any of the machine learning methods.

Using calibration plots to measure calibration performance is qualitative and does not facilitate a formal hypothesis test. **Figure 5-8** demonstrates the difference in ICI between logistic regression and each supervised machine learning algorithm across the 999 bootstraps. Higher ICIs indicate a poorly calibrated model, and as such positive differences between logistic regression and supervised machine learning would suggest machine learning is better calibrated.

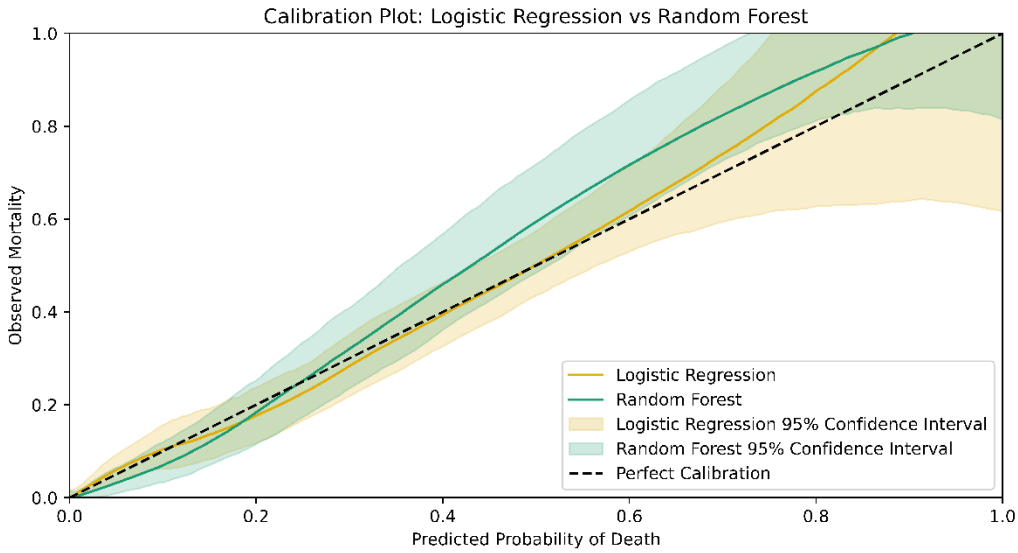

Figure 5-5: Calibration plot for logistic regression and random forest. The calibration curve was calculated for each bootstrap using the default LOESS smoother, and extrapolation of the LOESS smoother was used to fit over the 0-1 interval. Confidence intervals were derived using bootstrapped percentiles.

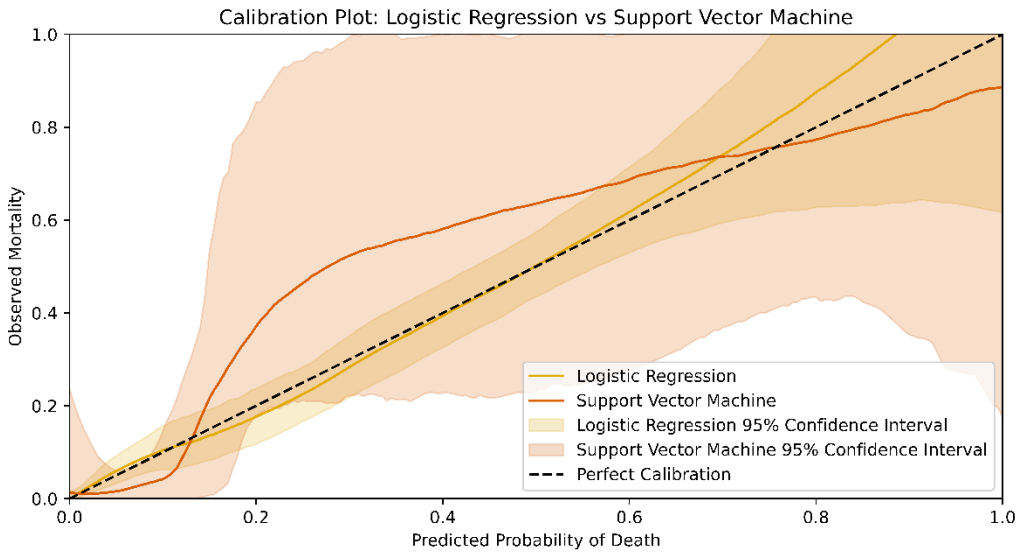

Figure 5-6: Calibration plot for logistic regression and support vector machine. The calibration curve was calculated for each bootstrap using the default LOESS smoother, and extrapolation of the LOESS smoother was used to fit over the 0-1 interval. Confidence intervals were derived using bootstrapped percentiles.

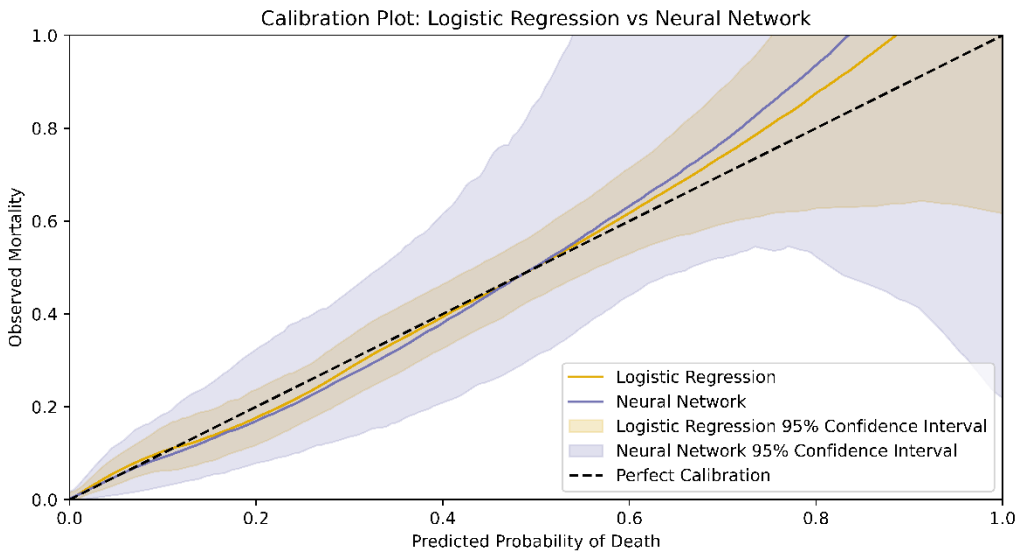

Figure 5-6: Calibration plot for logistic regression and neural network. The calibration curve was calculated for each bootstrap using the default LOESS smoother, and extrapolation of the LOESS smoother was used to fit over the 0-1 interval. Confidence intervals were derived using bootstrapped percentiles.

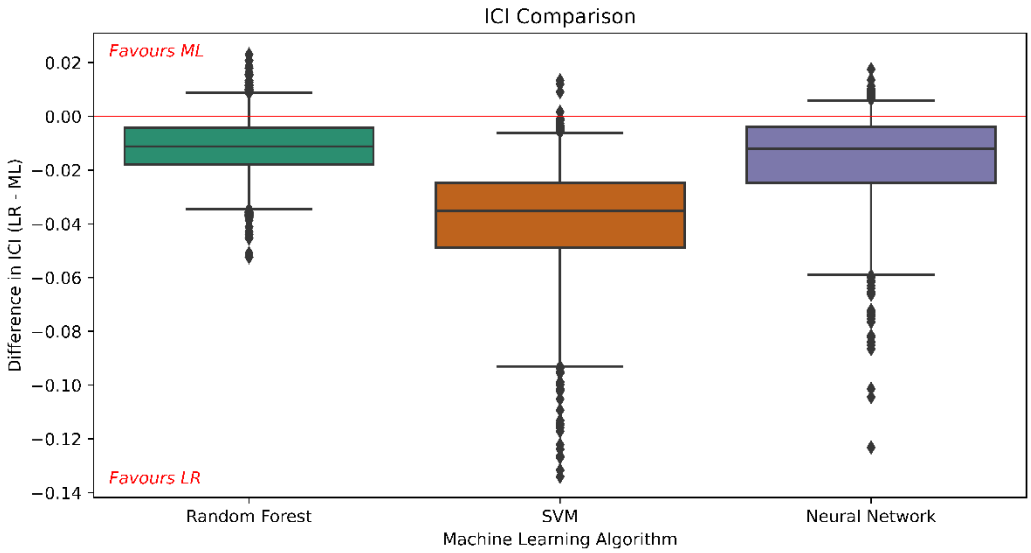

Figure 5-8: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm. ICI differences were calculated across 999 bootstraps, with whiskers showing 95% percentile confidence interval.

Figure 5-8 shows that the median ICI across the 999 bootstraps was smaller for logistic regression than for any of the supervised machine learning algorithms. This suggests that logistic regression is better calibrated and produces risk estimates that are more reliable.

However, only for the support vector machine can the null hypothesis of equal calibration performance be rejected at the 5% significance level, as only for support vector machine does the confidence interval not contain zero. As a result, it cannot be concluded that logistic regression has superior calibration performance to either random forests or neural networks at the 5% significance level, although the median difference in the ICI across bootstraps suggests better calibration performance for logistic regression.

Overall, this initial modelling suggests that there are some, albeit limited, gains in performance for supervised machine learning over logistic regression, though it depends on the specific algorithm used. A simple logistic regression that assumes linearity and additivity was able to have similar discriminatory performance relative to support vector machines and neural networks, when predicting ten-year mortality with a small number of strong predictors. However, the discriminatory performance of random forests measured by the AUC was larger than that of logistic regression and this was statistically significant at the 5% significance level. There was no evidence of gains in calibration performance of any supervised machine learning algorithm over logistic regression, and the difference in the ICI showed suggests logistic regression has superior calibration performance, though the superior calibration performance for logistic regression was only statistically significant compared to the support vector machine, which does not natively support probability estimates.

5.3. Discussion

As shown, initial analysis suggests that in the clinical risk prediction space with a binary outcome, there is a significant improvement in discriminatory performance of random forests compared to logistic regression. However, logistic regression had similar discriminatory performance to both support vector machines and neural networks and there is some evidence that logistic regression had more reliable risk estimates relative to all supervised machine learning algorithms, though this result was only statistically significant compared to support vector machines. However, there are numerous reasons why these initial results may not be representative.

It is possible that hyperparameters have not been tuned to a sufficient degree. Selecting the best hyperparameters has a direct effect on the model's performance (Yang and Shami, 2020) and therefore the performance of supervised machine learning depends to some degree on the tuning of hyperparameters. It is plausible the ranges that were searched did not contain the globally optimum hyperparameters, or that there were insufficient draws in the random search to find the global optimum hyperparameters. There was scope to increase the amount of hyperparameter tuning, although this requires further computational resources on an already computationally costly process. Per bootstrap, fitting the logistic regression took 1.044 seconds, fitting the random forest (including hyperparameter tuning) took 30.983 seconds, fitting the support vector machine took 26.139 seconds and fitting the neural network took 48.391 seconds on an Intel i7-10700k with 3.8 GHz processor speed with 8 cores and 16 threads**.** Fitting to the 999 bootstraps took approximately 30 hours overall, of which only 17 minutes was spent training the logistic regressions, showing that logistic regression is much less computationally expensive. Whilst it is likely further hyperparameter tuning would improve the performance of SVM and Neural Networks relative to logistic regression, this would have added further computational cost, and after a certain point the level of hyperparameter optimisation required to achieved better discriminatory performance becomes impractical.

It is also possible that the choice of features affected the result. The features were deliberately chosen as they were anticipated to be strong predictors of mortality, being often found in other models used in clinical practice. This may have made the problem too simple, and as such a more complex machine learning solution is unlikely to add much value over a logistic regression assuming linearity and additivity.

It is also possible that the lack of an ensemble classifier favoured logistic regression when comparing logistic regression to supervised machine learning. Most state-of-the-art supervised machine learning solutions leverage ensemble learning to improve discriminatory performance. This is

because ensemble classifiers improve generalisability and reduce the risk of poor local optima (Zhou, 2021). However, ensemble classifiers further increase complexity and computational expense.

It is also plausible that there may be an insufficient number of data points or features for supervised machine learning to show its full potential. It is regularly postulated in literature than machine learning, particularly neural networks, require abundant training data to give good discriminatory performance (Van Der Ploeg et al., 2014) and it is also speculated that random forests especially fare better with larger feature counts (Couronné et al., 2018).

Additionally, it was shown that feature selection and engineering such as principal component analysis can improve the performance of all models (Thottakkara et al., 2016). It is plausible that machine learning techniques may benefit more from feature selection and feature engineering, and the lack of any feature engineering and the ad-hoc nature of the initial feature selection may have affected the result in logistic regression favour.

It is also worth emphasising that these are the results of just one dataset, and as such the results may be conditional on meta features associated with this specific dataset such as its signal to noise ratio (Christodoulou et al, 2019). It is also worth considering that this thesis is only comparing the problem of binary classification, and there may be larger performance gains for supervised machine learning techniques over multinomial logistic regression, where the assumptions of multinomial logistic regression are more severe.

However, the lack of any feature selection, or modelling of non-linearities and interactions is likely to have artificially worsened the performance of logistic regression relative to supervised machine learning. This result is even more striking in this context, as the simplest logistic regression model can achieve similar discriminatory performance to both support vector machines and neural networks which automatically model these non-linearities and interactions.

6. Ensemble Learning

One potential reason why the gains from supervised machine learning were not clear cut previously was the lack of an ensemble classifier. Ensemble classifiers tend to improve performance over any single supervised machine learning algorithm as the generalisability of the learner is improved, reducing the risk of training a learner that is poor or in a bad local optimum (Zhou, 2021). Reducing the risk of a poor local optimum may be particularly pertinent in the previous analysis due to the limitations around hyperparameter optimisation. As such, introducing an ensemble classifier may result in a model that further improves performance and shows more substantial gains in performance for supervised machine learning.

6.1. Method

The ensemble classifier used in this analysis leverages stacking. The three previous supervised machine learning models are used as individual learners. The stacking algorithm takes the predictions learned from the 80% training data and fits a logistic regression as a meta learner, with each predictor included as a linear term as recommended by Zhou (2021). It then uses the 20% test data to measure performance in the exact same way to facilitate a comparison with logistic regression.

Model development of the individual learners was done in the exact same way, with 80% training data and 20% testing data, stratified by the dependent variable. Hyperparameter tuning was done with ten random searches of the same hyperparameter space, using three-fold stratified cross validation to determine optimal hyperparameters. The model with optimal hyperparameters was then fitted to the training data and used to make predictions on the test data. This was repeated 999 times for each of the bootstraps, to capture dispersion in the ROC, AUC, calibration plot and ICI.

Figure 6-1 captures the model development of the ensemble learner. It is largely the same model development as described in **Section 4.2**, using the exact same algorithms as individual learners. The key distinction is the use of the predictions on the training data to construct an ensemble classifier using stacking that distinguishes the model development from previous analysis.

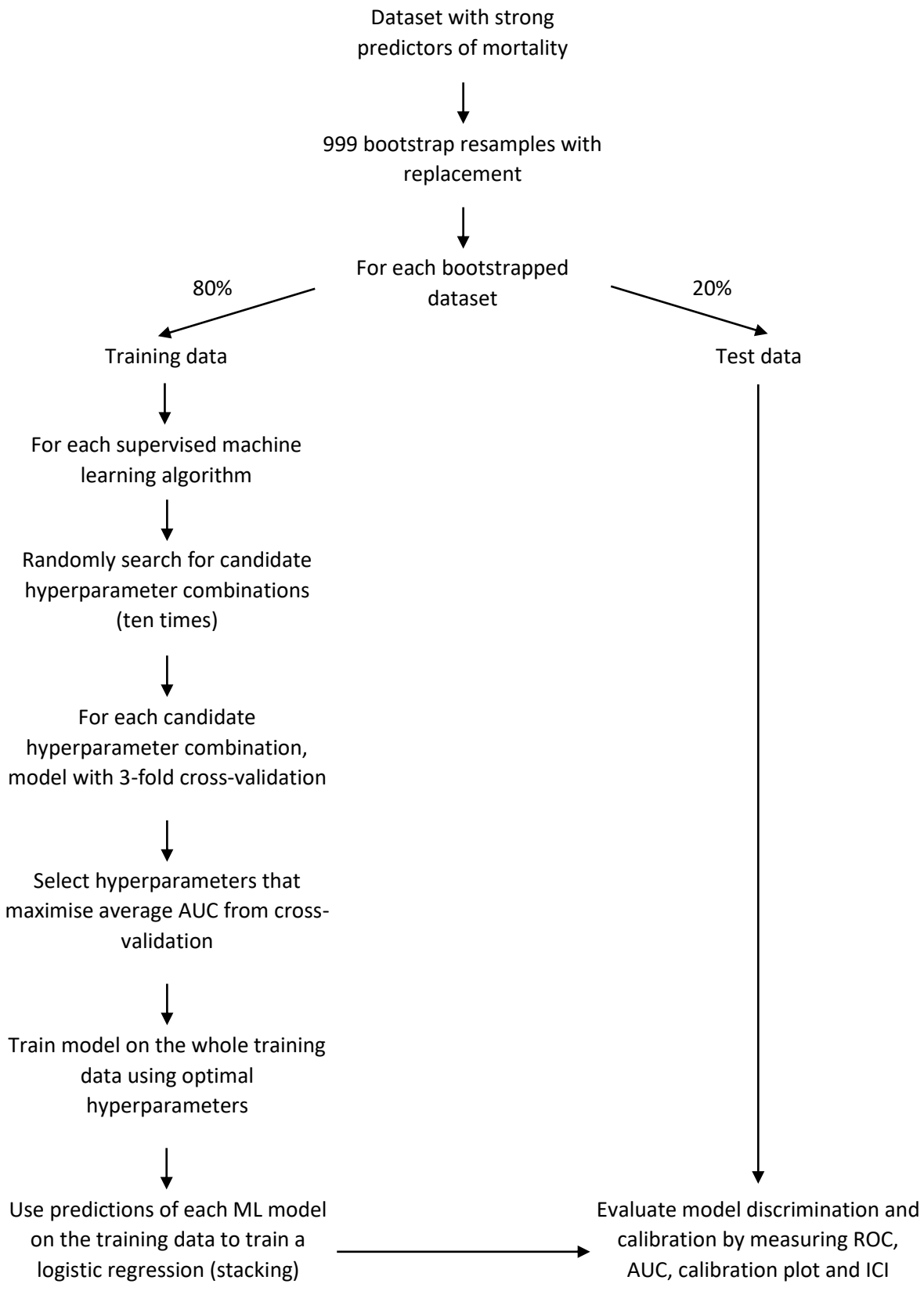

Figure 6-1: Methodology for the ensemble classifier

6.2. Results

Figure 6-2 shows the median ROC curve is higher and to the left for ensemble learning compared to a logistic regression that assumes linearity and additivity, suggesting improved discrimination performance over logistic regression. **Figure 6-3** further supports this conclusion, demonstrating that the ensemble classifier outperforms logistic regression with respect to its discrimination performance as measured by the AUC, and this result is significant at the 5% significance level, showing that the ensemble classifier is better than logistic regression in differentiating higher risk patients from lower risk patients. However, the median difference in AUC between the ensemble and logistic regression is very similar to the difference in AUC between the random forest and logistic regression, suggesting that discriminatory performance is not further improved by the ensemble classifier despite considerable increases in computational expense.

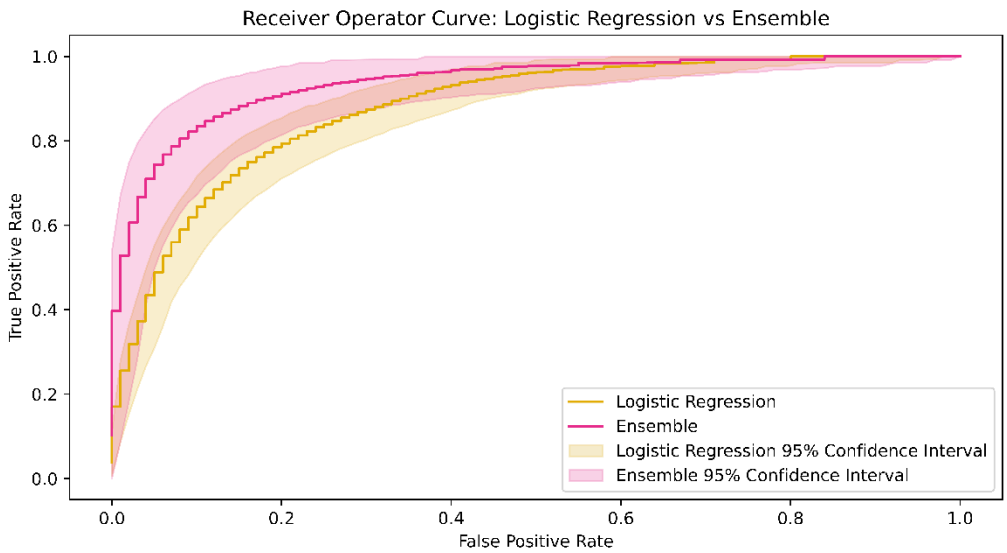

Figure 6-2: Receiver operator curve for logisitic regression and the ensemble classifier. The ROC was calculated for each bootstrap, and the median true positive rate for each false positive rate is plotted as the median ROC. Confidence intervals were derived using bootstrapped percentiles.

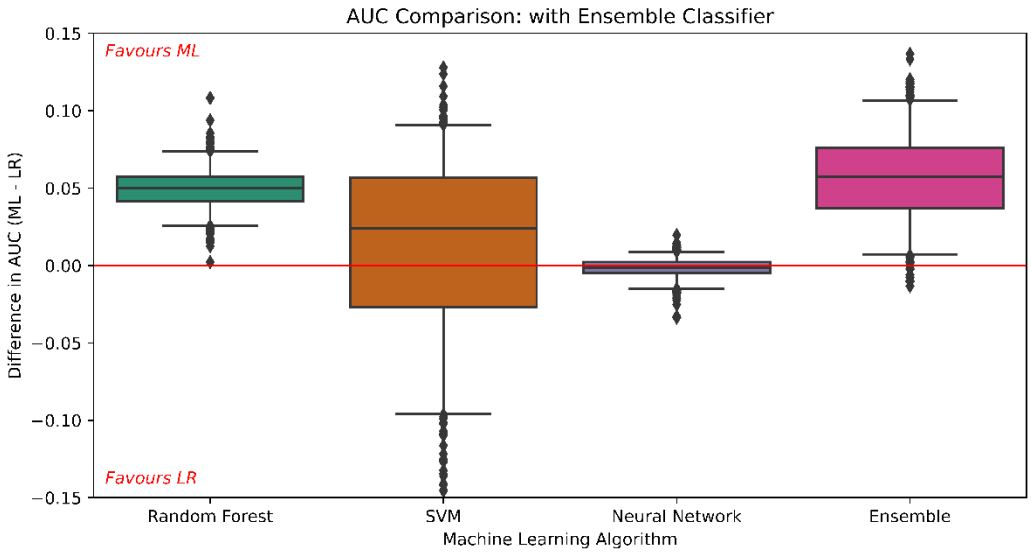

Figure 6-3: Comparing the AUC of each supervised ML algorithm to logistic regression. AUC differences were calculated across 999 bootstraps, with whiskers showing 95% percentile confidence interval

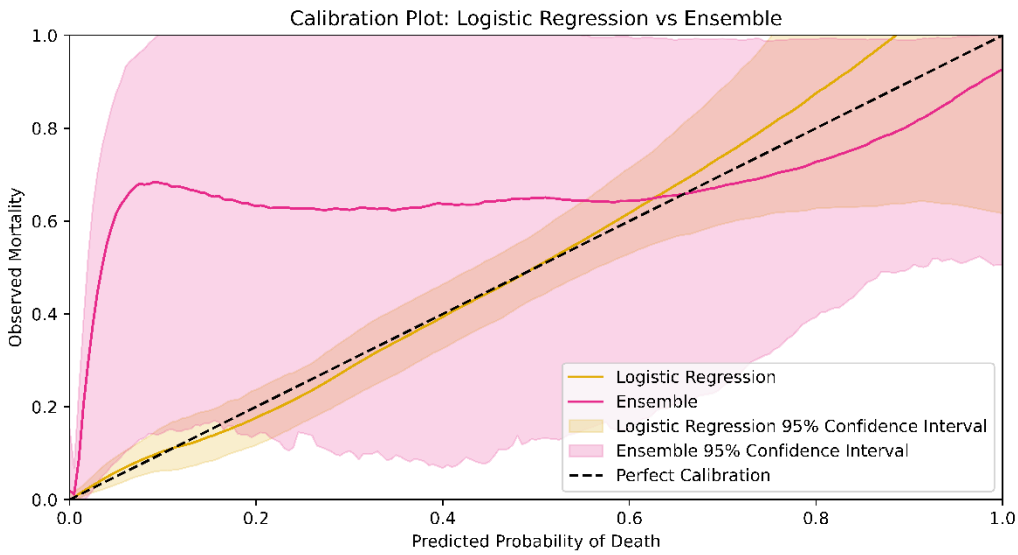

Figure 6-4: Calibration plot for logistic regression and the ensemble classifier. The calibration curve was calculated for each bootstrap using the default LOESS smoother, and extrapolation of the LOESS smoother was used to fit over the 0-1 interval. Confidence intervals were derived using bootstrapped percentiles.

However, the ensemble classifier is worse relative to logistic regression with respect to the reliability of its risk estimates when assessed visually as in **Figure 6-4**. The risk estimates are not reliable, systematically deviating from the perfect calibration line and systematically underestimating the risk of mortality whereas logistic regression is generally well calibrated.

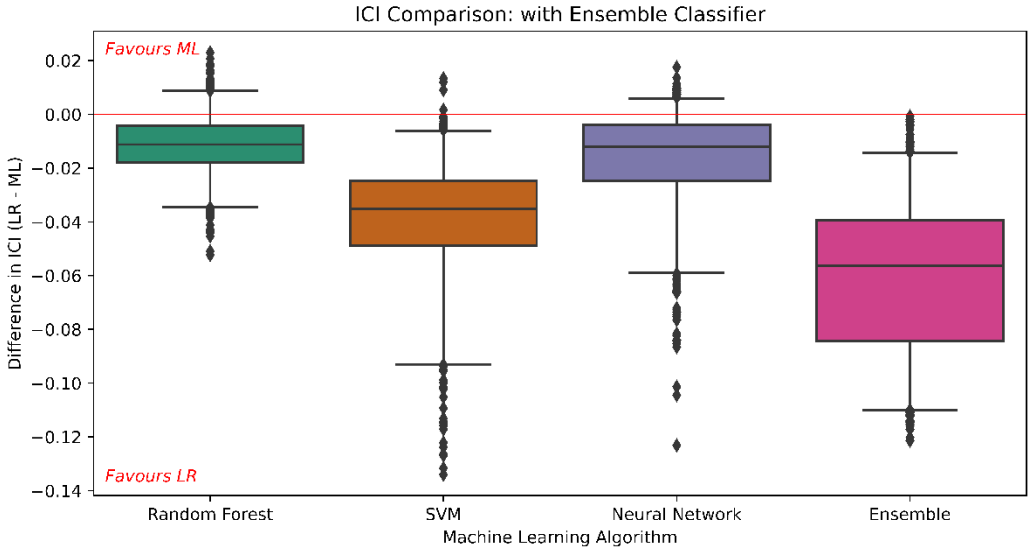

Figure 6-5: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm (including the Ensemble Classifier). ICI differences were calculated across 999 bootstraps, with whiskers showing 95% percentile confidence interval.

Figure 6-5 further demonstrates that the risk estimates are not reliable from this ensemble, with the null hypothesis of the same calibration performance of the ensemble classifier and logistic regression being rejected. As measured by the ICI, logistic regression has better calibration performance, and this is statistically significant at the 5% level.

6.3. Discussion

These results demonstrate that an ensemble classifier can outperform a simple logistic regression that assumed linearity and additivity with respect to discrimination performance. This result lends credence to the conclusion that supervised machine learning has a discrimination performance benefit compared to logistic regression in the clinical risk prediction space, likely due to its capability to automatically learn non-linearities and interactions present in data. However, the reliability of the predictions from this ensemble is worse than those obtained from a standard logistic regression. It is also worth noting that the ensemble classifier does not appear to outperform the discriminatory performance of the best classifier individual learner within its ensemble, the random forest, suggesting that it is not improving performance over and above its individual learners.

Overall, the ensemble classifier did improve discriminatory performance over logistic regression but failed to substantially improve discriminatory performance above that of the random forest and the risk estimates were less reliable. Previous research (Fumera and Roli, 2003) shows that, for ensemble classifiers to improve performance over and above the performance of individual learners, they must be accurate and diverse. Ensemble diversity, when the classifiers output posterior probabilities, can be measured by individual learner correlations (Brown et al., 2005), with lower correlations indicating higher diversity.

Exact specifications on how diverse or how accurate the constituent learners should be is unclear in literature. Here, the average correlation between individual learners' posterior probabilities across the 999 bootstraps is:

1. 0.794 for random forest and support vector machine
2. 0.921 for random forest and neural network

3. 0.731 for support vector machine and neural network

These are high correlation coefficients, and it is likely that trying to increase the diversity of the ensemble, by using either the random subspace method on the features or bootstrapping the dataset further such that each learner was trained with different datasets and features, may improve performance (Dutta, 2009). It also may be that the performance of the ensemble was hampered by insufficient sample size, insufficient hyperparameter optimisation or insufficient feature count that are likely to have limited the performance of machine learning in **section 5**.

7. Sample Size

Another potential reason why the performance improvement from supervised machine learning is not clear cut is insufficient sample size. It is plain from previous studies (Couronné et al., 2018) that performance differences are potentially a result of a datasets meta-features, the characteristics of an individual dataset. Such meta-features include the sample size, feature count and sample-feature ratio. It is plausible that sample size is insufficient to demonstrate the gains in discriminatory performance for machine learning which are widely known to be "data hungry" (Van Der Ploeg et al., 2014), requiring ten times more events per variable than logistic regression to achieve stable discrimination performance. Van Der Ploeg et al. (2014) recommend that supervised machine learning should only be used in a medical context with a binary outcome if very large datasets are available, which may explain why support vector machines and neural networks are unable to outperform logistic regression in previous analysis as the amount of data was insufficient to train these algorithms.

7.1. Methods

It is not possible to increase the sample size without using additional waves of the NHANES dataset. Using additional waves of the NHANES dataset is beyond the scope of the project, and using additional waves uses a sample from a different cross-section in time, potentially resulting in changes to the underlying population across waves. However, it is possible to evaluate how the relative performance changes with respect to the sample size and compare the relative performance when the methods receive different levels of training data. This analysis trains the logistic regression, random forest, support vector machine, neural network, and ensemble classifier in precisely the same way as in previous analysis, but with different levels of training data. Logistic regression and each of the machine learning algorithms were fitted with {100, 200, 400, 800, 1600 and 3200} training data points. Within this training dataset, three-fold cross validation was used to optimise hyperparameters using ten random searches in the same way as before.

This method was still applied to different bootstraps to measure uncertainty in discriminatory performance, however, due to computational constraints, the number of bootstraps was reduced from 999 to 199. This reduces the reliability of confidence intervals; however, it will still facilitate a comparison of each method's performance as sample size varies. The data points were randomly selected without replacement per bootstrap, and the data points were kept consistent across algorithms in each bootstrap to facilitate a fair comparison and remove additional variation resulting from models being trained on different data points in each bootstrap. For simplicity, model evaluation is now reduced to only differences in AUC and ICI. For the remainder of the analysis, there will be no further ROC or calibration plots.

7.1.1. Logistic Regression

For logistic regression, the method for each of the 199 bootstrapped datasets entailed:

1. Randomly drawing {100} data points from the dataset as training data.
2. Allocating the rest of the dataset as test data.
3. Fitting the logistic regression to the training data.
4. Reporting final performance on the test set (AUC and ICI)

The above method was then repeated for the differing sample sizes levels {200, 400, 800, 1600 and 3200}. This allowed a comparison of how performance changes when logistic regression is exposed to a greater number of data points for training.

7.1.2. Supervised Machine Learning

The method for each supervised machine learning is again more involved owing to the optimisation of hyperparameters. However, it is broadly in line with previous model development.

Overall, the method for each of the 199 bootstrapped datasets entailed:

1. Randomly drawing {100} data points from the dataset as training data.
2. Allocating the rest of the dataset as test data.
3. For each supervised machine learning algorithm:
    a. Using random search to get candidate hyperparameters.
    b. Fitting the model with these hyperparameters three times to the training set (two-thirds train, one-third validation), each with a different partition of the training set as the validation set.
    c. Repeating steps a and b ten times, performing ten different random searches.
    d. Selecting the optimal hyperparameters that maximised validation AUC.
    e. Fitting the model with optimal hyperparameters on the entire training set
    f. Reporting final performance on the test set (AUC and ICI)
4. Repeat steps 1-3 with differing sample size levels {200, 400, 800, 1600, 3200}

Note that for each bootstrap, the random selection of data points was done prior to the fitting of any of the models. This means that for each bootstrap, the logistic regression and each of the supervised machine learning algorithms are trained on the exact same training data and measured on the exact same test data, removing the possibility that differences in performance are the result of idiosyncrasies generated from the methods being trained on different datasets.

Figure 7-1 highlights the methodology applied to the differing amounts of training data. The methodology for logistic regression is the same, except it is directly fitted to the training data as there is no hyperparameter optimisation.

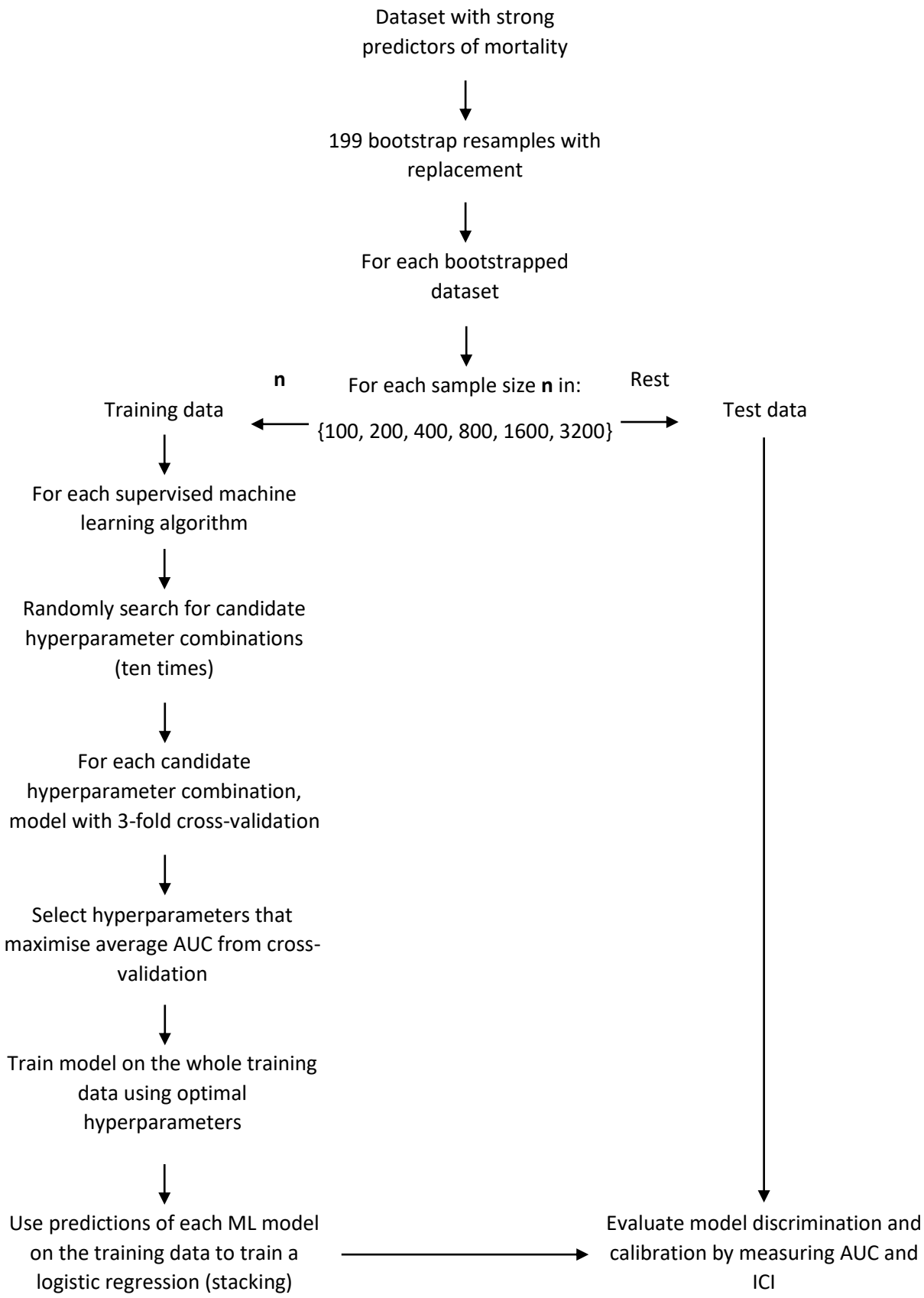

Figure 7-1: Methodology for supervised machine learning applied over different levels of training data

7.2. Results

Figures 7-2: 7-7 compare the difference in the AUC between logistic regression and each supervised machine learning algorithm with different sample sizes.

As before, the difference in AUC measures the difference in discriminatory performance; the difference in a model's ability of a model to differentiate high risk patients from low risk patients, with higher differences in AUC showing greater relative discriminatory performance for machine learning. At lower sample sizes, the performance improvements for machine learning were not apparent, and logistic regression has better discriminatory performance at lower sample sizes, though this result is not statistically significant at the 5% level. However, when the training sample size reaches 800 data points, there begins to be evidence of a greater AUC for both the random forest and ensemble classifier, and this becomes statistically significant when the training sample size reaches 1600 data points. As before, there is no evidence of improved discriminatory performance over logistic regression for either neural networks or support vector machines, although their performance relative to logistic regression does improve as the sample size increases, suggesting that a potential reason for the lack of performance improvement for these methods over logistic regression is insufficient sample size.

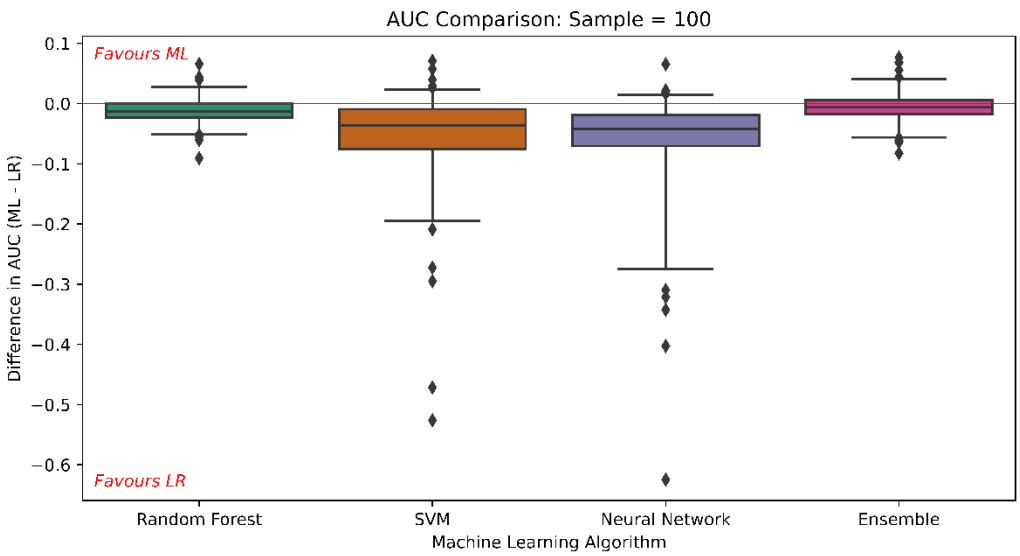

Figure 7-2: Comparing the AUC of each supervised ML algorithm to logistic regression with 100 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

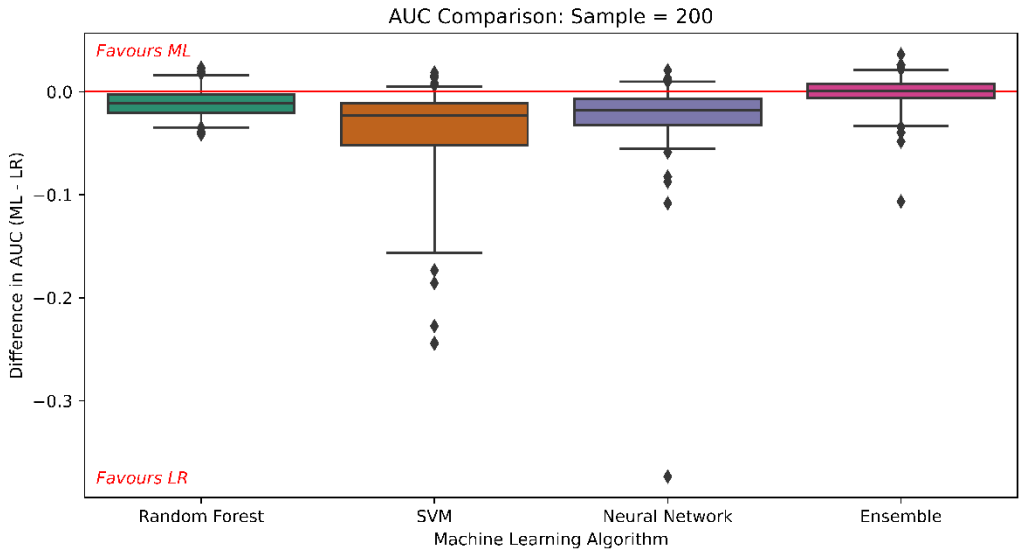

Figure 7-3: Comparing the AUC of each supervised ML algorithm to logistic regression with 200 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

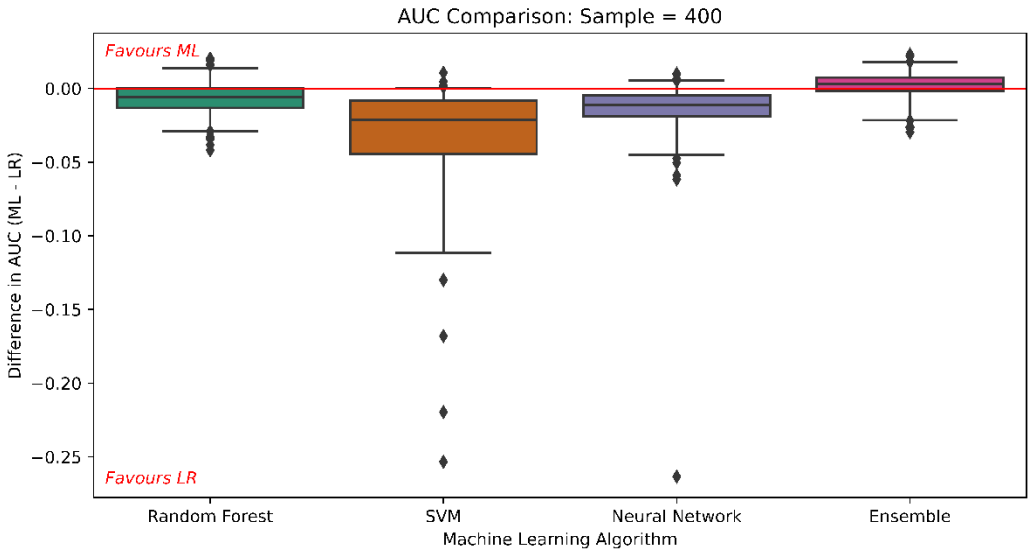

Figure 7-4: Comparing the AUC of each supervised ML algorithm to logistic regression with 400 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

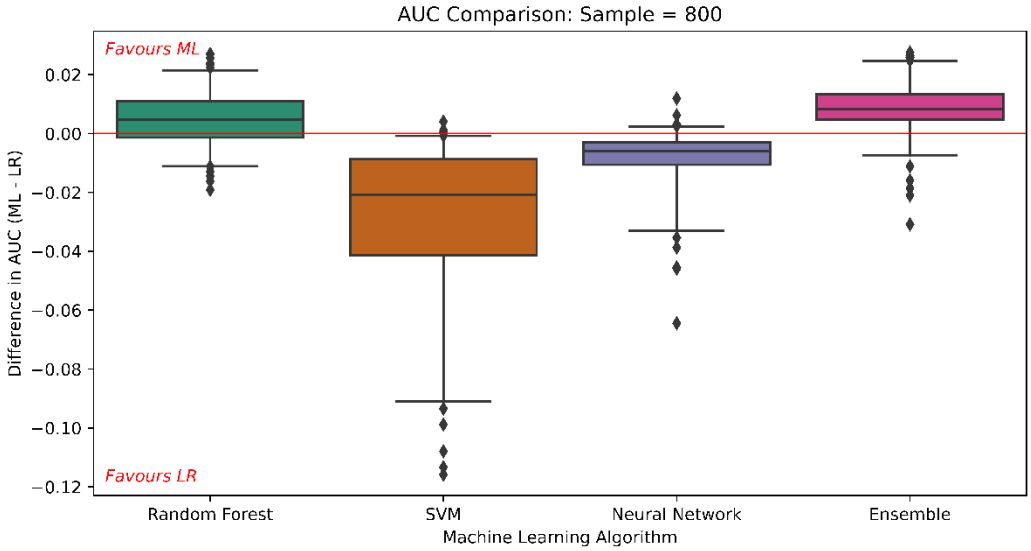

Figure 7-5: Comparing the AUC of each supervised ML algorithm to logistic regression with 800 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

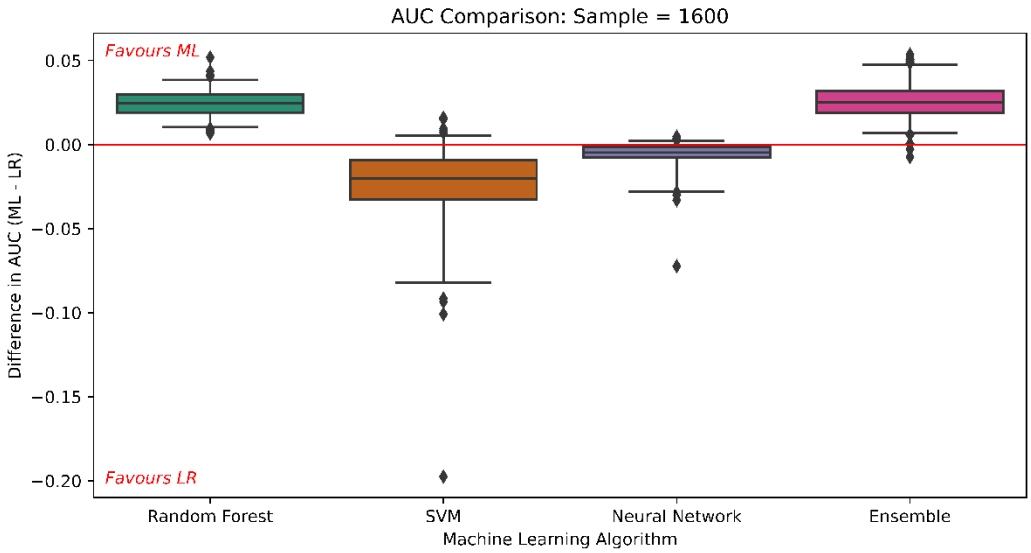

Figure 7-6: Comparing the AUC of each supervised ML algorithm to logistic regression with 1600 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

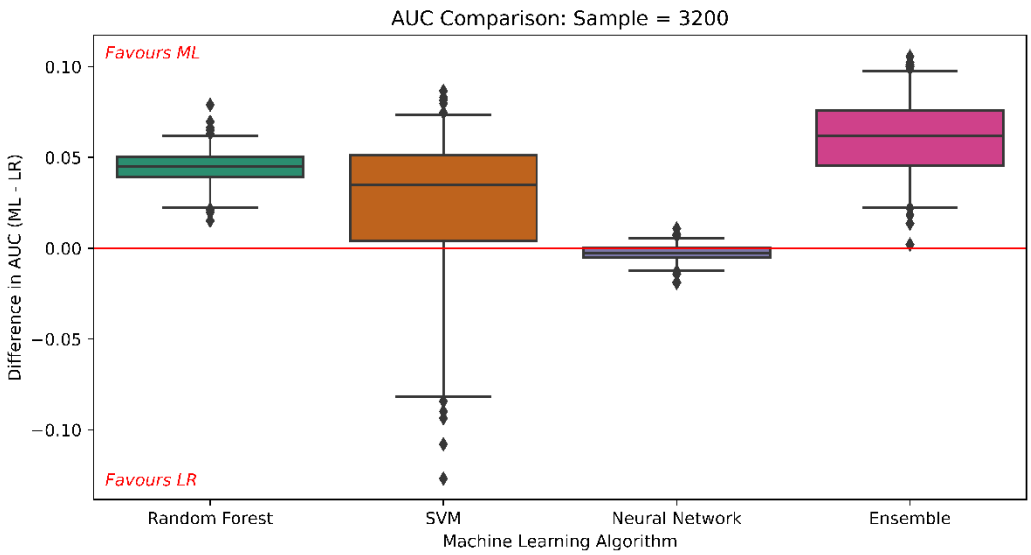

Figure 7-7: Comparing the AUC of each supervised ML algorithm to logistic regression with 3200 training data points. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

The comparison of the ICI across the 199 bootstraps for the different levels of training data can be found in **appendix A**. From these plots, it can be seen that as the size of the training dataset increases, the relative calibration performance of logistic regression improves relative to all supervised machine learning algorithms. This can be seen as the training size increases, the median difference in the ICI between logistic regression and supervised machine learning decreases, and is negative for all machine learning algorithms at all training sample sizes larger than 100 data points. However, only the calibration of the support vector machine and ensemble classifier have poorer calibration than logistic regression that is significant at the 5% level, and this only arises at higher levels of training samples.

7.3. Discussion

These results demonstrate that when modelling ten-year mortality as a binary outcome with a small number of strong candidate predictors, as the sample size increases, the discriminatory performance of supervised machine learning as measured by the AUC improves relative to logistic regression. This performance gain is most notable for random forests and ensemble classifiers. There is limited evidence of a performance gain for SVM and Neural Networks as the sample size increases, though it is plausible this is due to insufficient training data, insufficient hyperparameter optimisation, and the choice of only including a small number of strong predictors of mortality; all of which are factors that are likely to favour logistic regression. These results reaffirm previous work by Van Der Ploeg et al. (2014) and show that contemporary machine learning methods are more “data-hungry”, requiring more training data in order to achieve greater discriminatory perfromance.

It is also important to note that the relative calibration performance is especially poor for machine learning algorithms at larger levels of sample size, where the calibration performance of supervised machine learning is poorer at the 5% level for some algorithms. This suggests logistic regression has an advantage with respect to calibration perfromance when the gains in discrimination performance for machine learning are largest, highlighting a trade-off in performance at larger sample sizes, whereby logistic regression shows better calibration performance, but machine learning shows better discimination performance at larger sample sizes.

8. Features

It has been speculated that one of the advantages of machine learning algorithms, particularly random forest, is that they are able to better manage a larger number of features relative to logistic regression (Couronné et al., 2018). As such, the previous analysis that used only nine features that were anticipated to be strong predictors of mortality may have favoured logistic regression, and more complex modelling using a greater number of features may be able to demonstrate discrimination performance improvements for more than one supervised machine learning algorithm. This work is also interested in exploring what factors influence the differences in performance of logistic regression and supervised machine learning. As the NHANES dataset is rich in the number of features, it is possible to assess how each method fares in the context of a greater number of features than the initial nine analysed in the previous analysis.

8.1. Additional Variables

This analysis leverages data in the demographic, examination, laboratory, and questionnaire data. The dietary data was excluded**,** as this data is panel data and is therefore more difficult to use in combination with the other data sets. Every other variable in the 2007-2008 NHANES wave was considered as a candidate variable. Feature selection was necessary as many of the variables had missing observations. A candidate variable was accepted if less than ten percent of the eligible patients in the linked mortality file had missing data for the variable. As the means of feature selection was purely software controlled, noise variables that likely have no association with mortality are accepted in this analysis. This was deliberate, as it is also important to see how the models can handle irrelevant features. Applying the criteria of less than ten percent missing data, 322 variables from the demographic, examination, laboratory, and questionnaire datasets remained. A variable was deemed categorical if it was encoded as an integer with less than or equal to 20 categories. Categorical variables were then one hot encoded, increasing the number of candidate features to 744. After this, again a complete case restriction was applied to the dataset, resulting in a sample size of 3,246. The complete list of features is given in **appendix B**, and a complete description of each feature can be found on the NHANES website (CDC, 2008).

8.2. Methods

After merging and cleaning the NHANES dataset and dropping variables with excessive missing data, it was possible to investigate the effect of differing feature counts on the performance of each algorithm. Algorithms were fitted to the data with {10, 25, 50, 100, 250 and 500} features. These features were randomly selected without replacement per bootstrap, and kept constant across algorithms, to facilitate a fair comparison and not affect the results as before by predicting features that were anticipated to be strong predictors of mortality. As with all previous analysis, bootstrapping was applied to get a sense of uncertainty in discrimination and calibration performance. Discrimination performance is again modelled by the area under the receiver operator curve (AUC), and calibration is again modelled by the integrated calibration index (ICI). Due to computational constraints, the number of bootstraps was reduced from 999 to 199 for each feature count, again worsening the reliability of the bootstrapped confidence intervals.

8.2.1. Logistic Regression

For logistic regression, the method for each of the 199 bootstrapped datasets entailed:

1. Randomly drawing {10} features from the dataset
2. Splitting the data into 80% train and 20% test.
3. Fitting the logistic regression to the training data.
4. Reporting final performance on the test set (AUC and ICI)

The above method was then repeated for the differing feature levels {25, 50, 100, 250 and 500}. This allowed a comparison of how performance changes when logistic regression is exposed to a greater number of features.

8.2.2. Supervised Machine learning

The method for each supervised machine learning is again more involved owing to the optimisation of hyperparameters. However, it is kept deliberately like the model building previously described.

Overall, the method for each of the 199 bootstrapped datasets entailed:

1. Randomly drawing {10} features from the dataset
2. Splitting the data into 80% train and 20% test.
3. For each supervised machine learning algorithm:
    a. Using random search to get candidate hyperparameters.
    b. Fitting the model with these hyperparameters three times to the training set (two-thirds train, one-third validation), each with a different partition of the training set as the validation set.
    c. Repeating steps a and b ten times, performing ten different random searches.
    d. Selecting the optimal hyperparameters that maximised validation AUC.
    e. Fitting the model with optimal hyperparameters on the training set
    f. Reporting final performance on the test set (AUC and ICI)
4. Repeat steps 1-3 with differing feature levels {25, 50, 100, 250 and 500}

Note that for each bootstrap, the random selection of features was done prior to the fitting of any of the models. This means that for each bootstrap, the features that logistic regression and each of the supervised machine learning algorithms are using is the same, removing the possibility that differences in performance are generated from one algorithm being fitted with stronger predictors of mortality. The algorithms are also fitted to the same bootstrapped datasets to prevent any differences in performance across algorithms resulting from using different datasets.

The hyperparameter optimisation was done using the same hyperparameter ranges as in the previous analysis, except for random forests. As the default maximum number of features used to split at each node is $\sqrt{features}$, the search range for this hyperparameter was modified to $[0.5\sqrt{features}, 2\sqrt{features}]$ (rounded to nearest integer), again randomly sampled from a discrete uniform distribution. All other hyperparameter optimisation used the same ranges.

The development of each of the machine learning models with various levels of features is shown in **Figure 8-1.** The development of the logistic regression was done using the same bootstrapped datasets and in the same way, just without the hyperparameter optimisation.

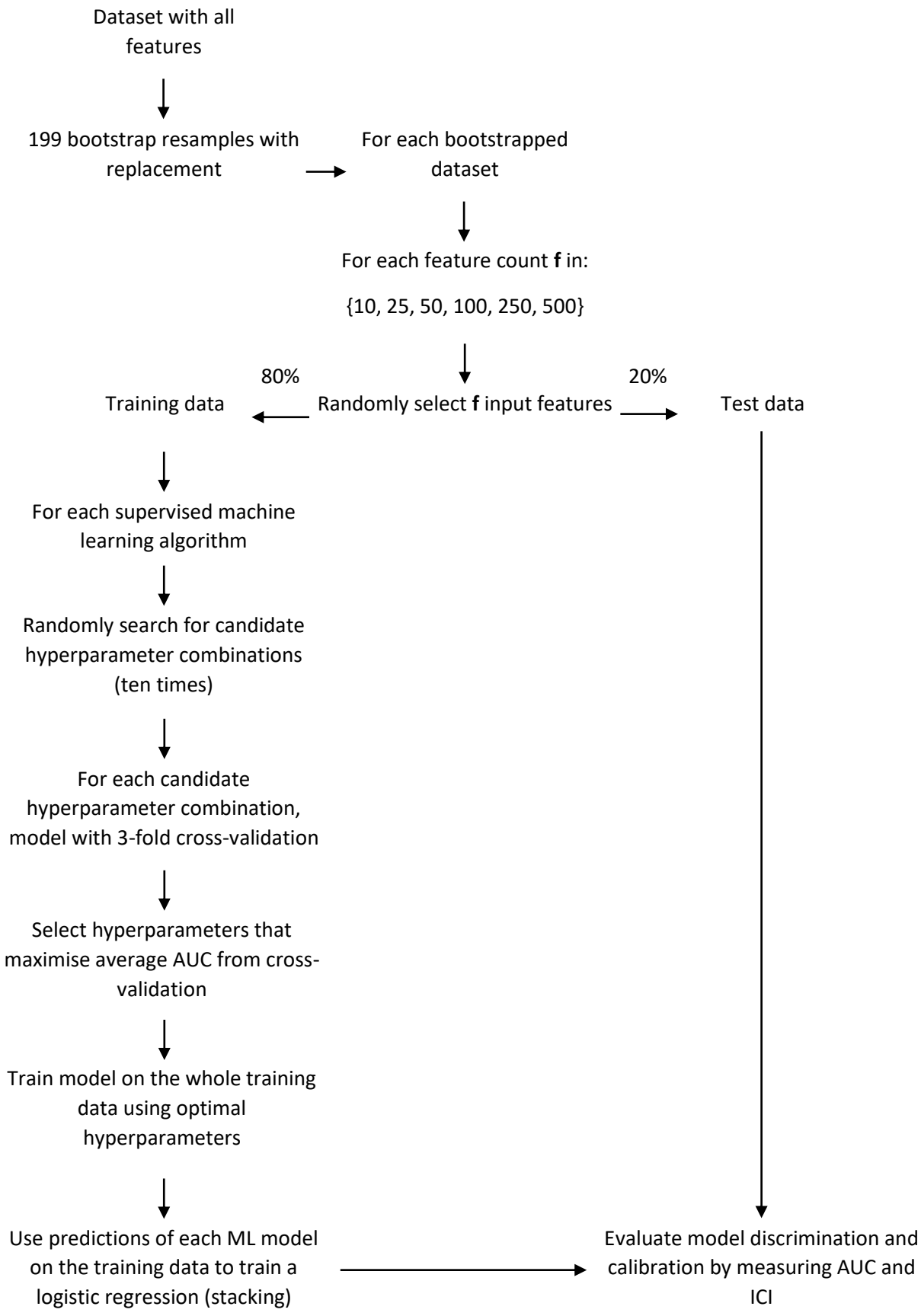

Figure 8-1: Methodology for evaluating the effect of feature count

8.3. Results

Figures 8-2: 8-7 compare the difference in the AUC between logistic regression and each supervised machine learning algorithm with different levels of features. These figures demonstrate that as the number of features increases, the discriminatory performance of each supervised machine learning algorithm improves relative to logistic regression. Once the number of randomly selected features exceeds 100, the difference in AUC is statistically significant at the 5% level for all machine learning algorithms, indicating improved discriminatory performance for machine learning over logistic regression.

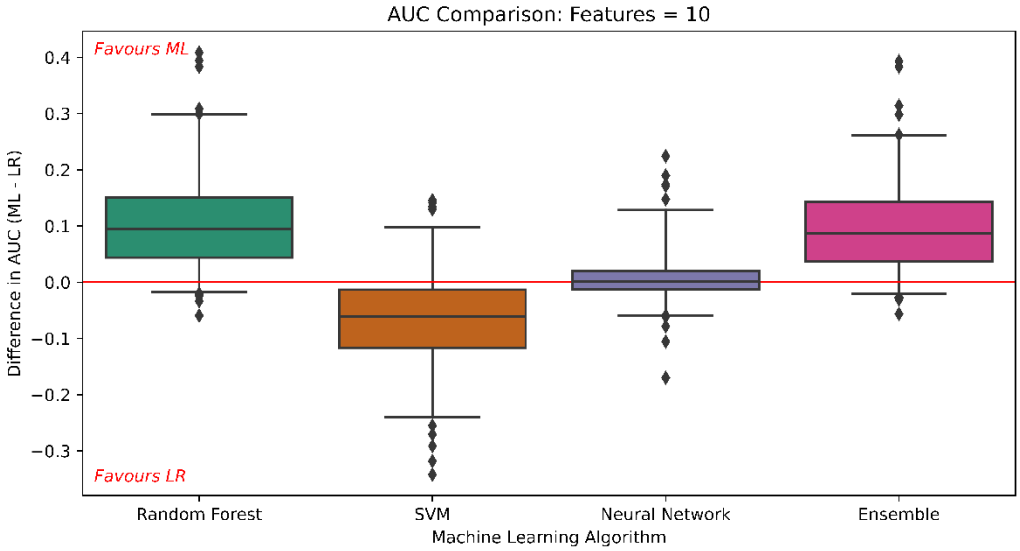

Figure 8-2: Comparing the AUC of each supervised ML algorithm to logistic regression with 10 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

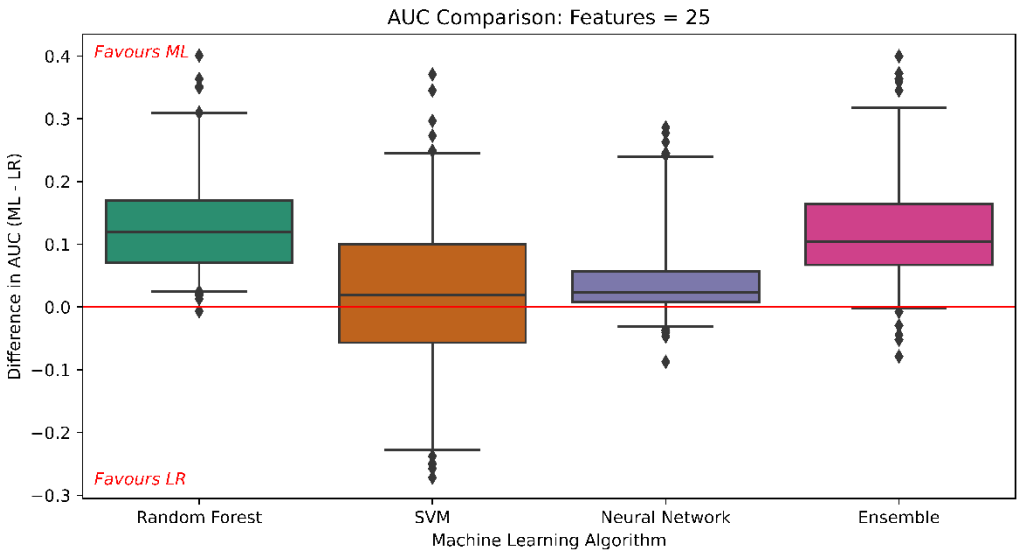

Figure 8-3: Comparing the AUC of each supervised ML algorithm to logistic regression with 25 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

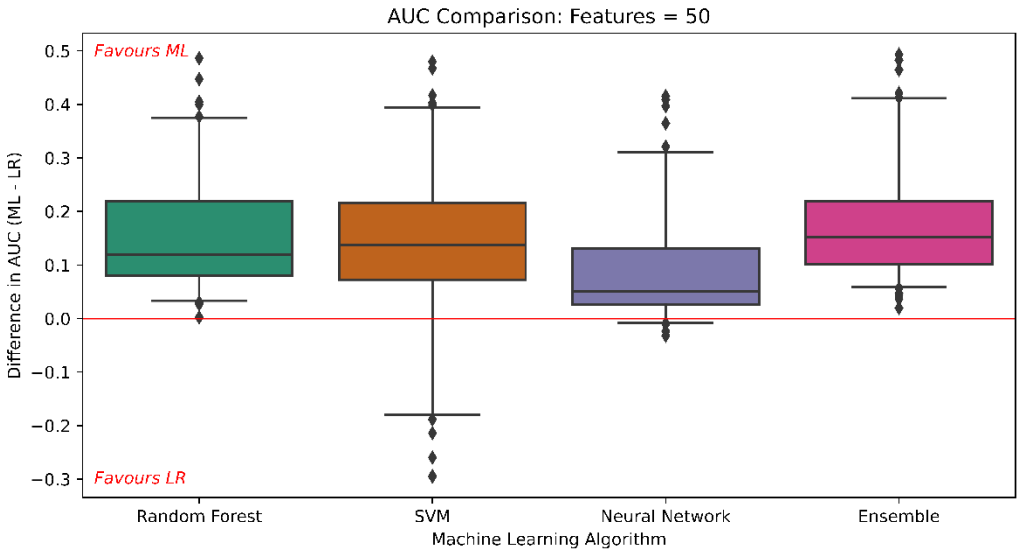

Figure 8-4: Comparing the AUC of each supervised ML algorithm to logistic regression with 50 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

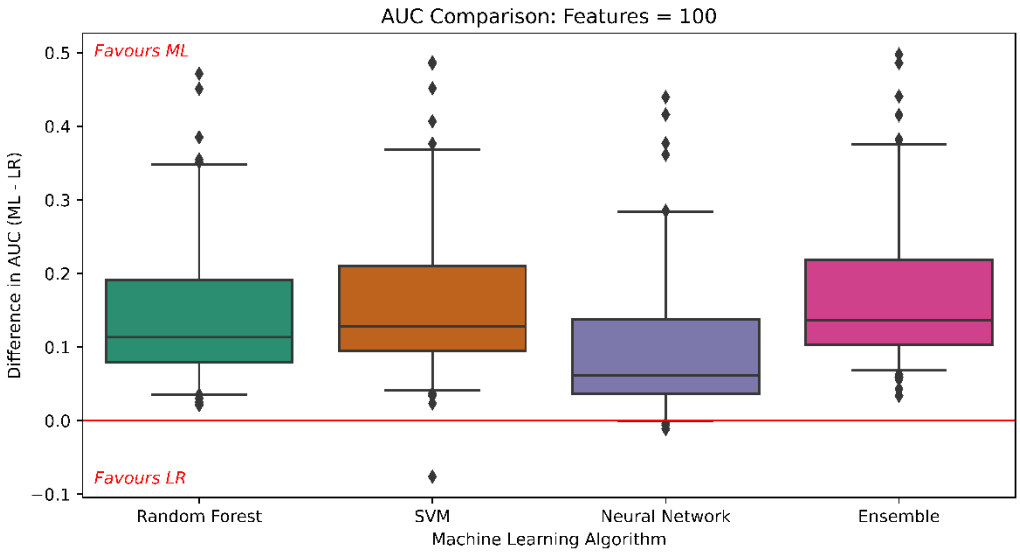

Figure 8-5: Comparing the AUC of each supervised ML algorithm to logistic regression with 100 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

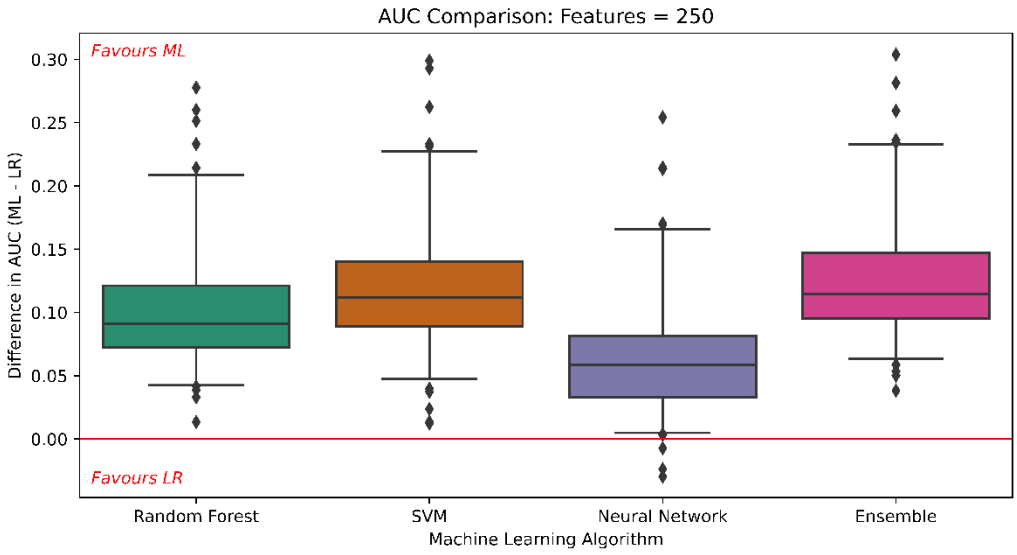

Figure 8-6: Comparing the AUC of each supervised ML algorithm to logistic regression with 250 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

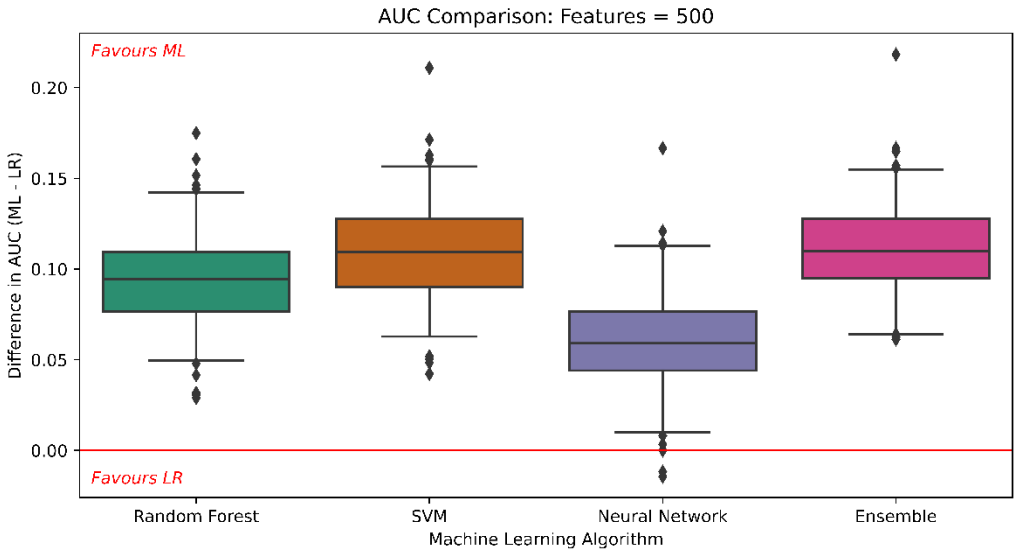

Figure 8-6: Comparing the AUC of each supervised ML algorithm to logistic regression with 500 randomly selected features. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

Appendix C contains a complete set of comparisons of the ICI across the different feature levels. It is apparent from these plots that logistic regression still confers an advantage with respect to the reliability of its risk estimates at all feature counts, as the median difference in ICI between logistic regression and supervised machine learning is less than zero for all algorithms for all feature counts, indicating logistic regression is better calibrated overall. However, the effect of increasing features had differing effects on the different algorithms, with the relative calibration performance of support vector machines improving as the feature count increases and the relative calibration performance of random forests worsening as the feature count increases, having significantly worse calibration performance than logistic regression at the 5% level at higher feature counts.

8.4. Discussion

These results demonstrate that in the clinical risk prediction space with a binary outcome, as the number of features increase the discriminatory performance of supervised machine learning as measured by the AUC improves relative to logistic regression, irrespective of whether it is a random forest, a support vector machine, a neural network, or an ensemble classifier. However, the calibration performance, or the reliability of the risk estimates, remains generally poorer for supervised machine learning than logistic regression, though this result is often not statistically significant.

These results suggest that the performance gain of machine learning is at least partially a consequence of the quantity and quality of the input features, with around 100 input features required to show a substantial increase in discriminatory performance for a supervised machine learning algorithm over logistic regression. This dataset's number of high-quality features is unusual in modern datasets, and it is unclear how often the number of features is this large in clinical practice. For instance, if a doctor requires 100 different features to make an assessment on clinical risk, this is unlikely to be practical or particularly useful at present, though the rise of on body sensors may make this more practical in the future (Appelboom et al, 2014).

These results also have clear limitations. The confidence intervals are no longer reliable, as the number of bootstraps is too small. However, with the exclusion of neural networks, when the models were trained with 500 features the supervised machine learning algorithms had higher AUCs in all the 199 bootstraps, giving confidence that the improvement in discriminatory performance is not a consequence of unreliable confidence intervals.

Problems remain with the specification of the logistic regression, which assumes linearity and additivity and therefore is likely to have worse performance than a more carefully considered specification that includes interactions and non-linearities. This is likely to favour performance for machine learning. There is also likely insufficient hyperparameter optimisation, favouring logistic regression as further hyperparameter tuning would likely improve performance. However, this analysis does suggest that supervised machine learning has an advantage in discrimination performance in clinical risk prediction at handling many features.

9. Feature Selection

A further limitation with the previous analysis is that there was no feature selection or feature engineering. At each bootstrap, a random subset of features were chosen then applied to all algorithms. This is not how machine learning algorithms or statistical models are applied in practice and is likely to result in overfitting for all algorithms when including this many features relative to this number of data points (Steyerberg et al., 1999). Comparing how the different methods work when responding to different feature selection is beneficial as this is likely to mirror real world models more closely and be of use to researchers when deciding which model to use.

9.1. Principal Component Analysis

A more realistic setup than randomly selecting variables uses feature selection or feature engineering. Principal component analysis (PCA) is a dimensionality reducing algorithm that extracts the most important information from a set of variables and compresses the size of the dataset by creating linear combinations of the original variables (Abdi and Williams, 2010).

PCA works by creating new variables, called principal components, that are linear combinations of the original variables. The first principal component is a linear combination of all candidate features that maximises the variance of the principal component, and the subsequent components are linear components that maximise variance conditional on orthogonality to all previous principal components. The result of PCA is a set of orthogonal principal components that reduce the number of features in data whilst maximising the variation present in the dataset (Abdi and Williams, 2010).

9.2. Methods

PCA was applied to all the 744 candidate features outlined in the features section. In each bootstrap, PCA was applied to the training data, and the same linear combination of features were then applied to the test set. PCA was performed in this way as it prevents leakage from the test set in model training, and best demonstrates how model predictions generalise to new observations. The top thirty principal components were used in the subsequent modelling, and the remaining principal components discarded. Scikit-learn was used to implement the PCA (Pedregosa et al., 2011).

The top thirty principal components were then used to fit the logistic regression and each of the supervised machine learning algorithms in the same way as described in the features section. The hyperparameter ranges are unchanged from the features section, and the same principle of using ten random searches and three-fold cross validation to find the optimal hyperparameters and measuring the AUC and ICI for each of the 199 bootstrapped datasets is unchanged. Again, 199 bootstraps were used instead of the recommended 999 due to computational constraints, reducing the reliability of confidence intervals.

Figure 9-1 outlines the methodology of fitting all supervised machine learning algorithms to the data. The model development of logistic regression was the exact same as in **Figure 9-1**, though there was no hyperparameter optimisation.

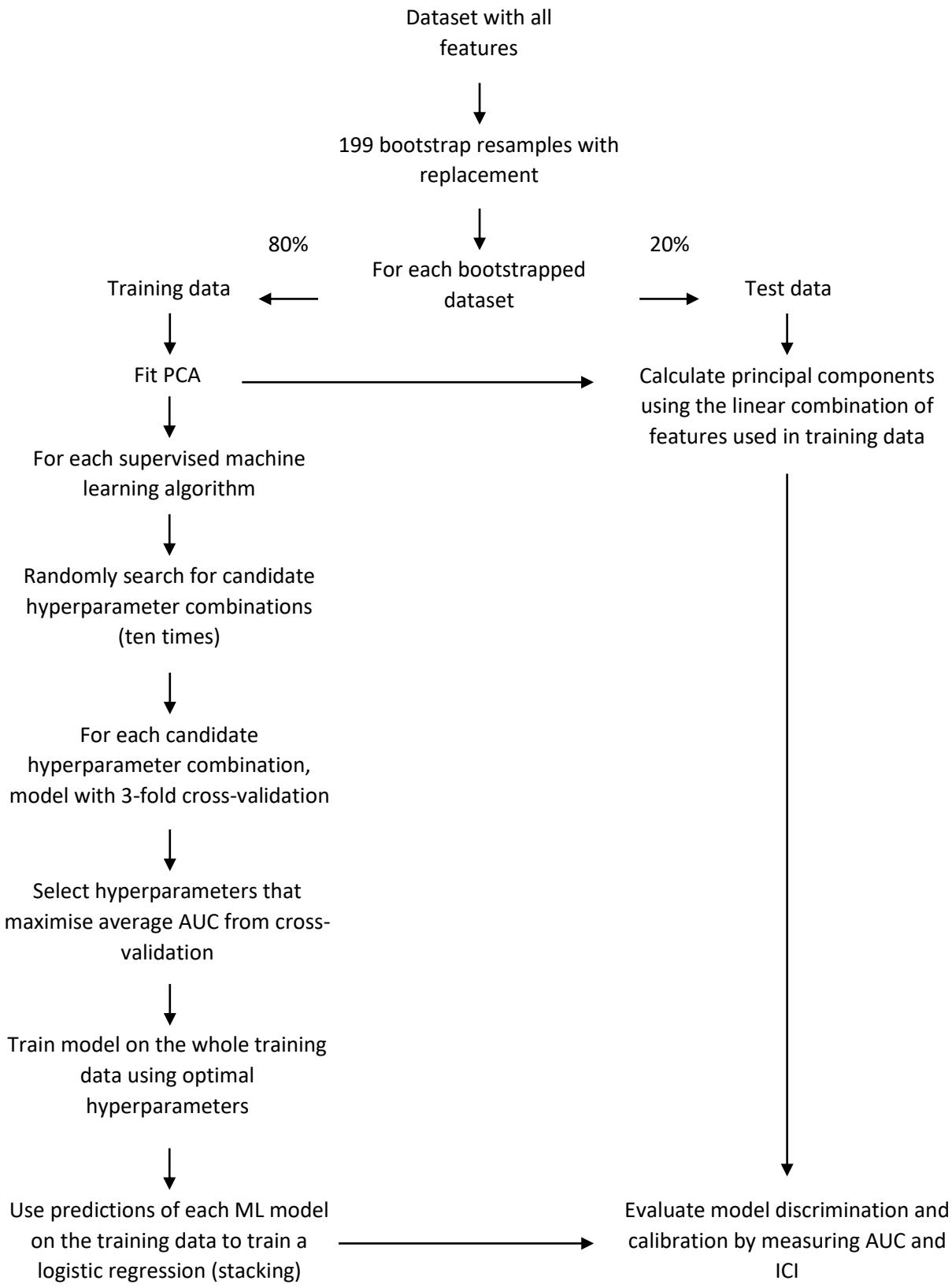

Figure 9-1: Methodology for evaluating the effect of feature selection

Figure 9-2 demonstrates a scree plot that accompanies the PCA, showing the proportion of variation captured in each of the top thirty principal components. This figure demonstrates that on average, most of the variation (76.7%) was not preserved in the top thirty principal components, indicating a large amount of potentially useful information is not being used in the subsequent modelling.

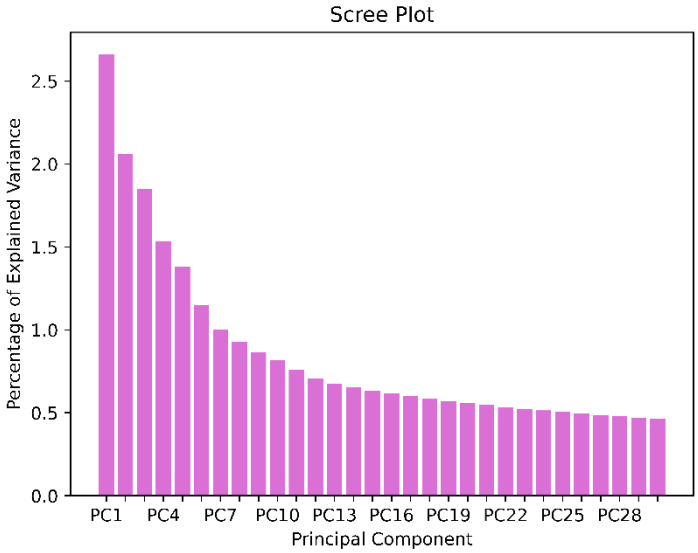

Figure 9-2: Scree Plot (average across 199 bootstraps)

9.3. Results

Figure 9-3 compares the difference in AUC between logistic regression and each of the supervised machine learning algorithms, when these models utilise the top 30 principal components as features as opposed to random subsets of the 744 candidate features. From this plot, discriminatory performance is still superior for machine learning algorithms and this difference in discriminatory performance as measured by AUC differential is statistically significant at the 5% level for all algorithms, though the point estimate of this difference is reduced from when 500 randomly selected features were included in model building. This suggests that machine learning still has an advantage in discriminatory performance in the context of many features, even if the features are condensed or selected by use of a dimensionality reducing algorithm such as PCA. It is worth noting that these confidence intervals are no longer reliable, and that the difference in discriminatory performance at the 5% level for SVMs and Neural Networks is especially questionable as the decision to reject the null hypothesis of no difference in discriminatory performance is marginal.

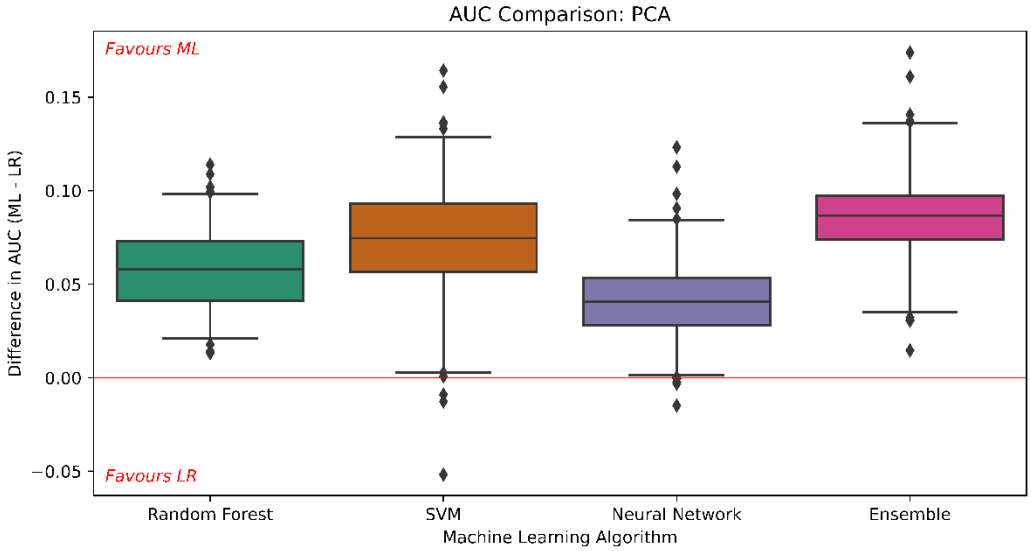

Figure 9-3: Comparing the AUC of each supervised ML algorithm to logistic regression with the top thirty principal components. AUC differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

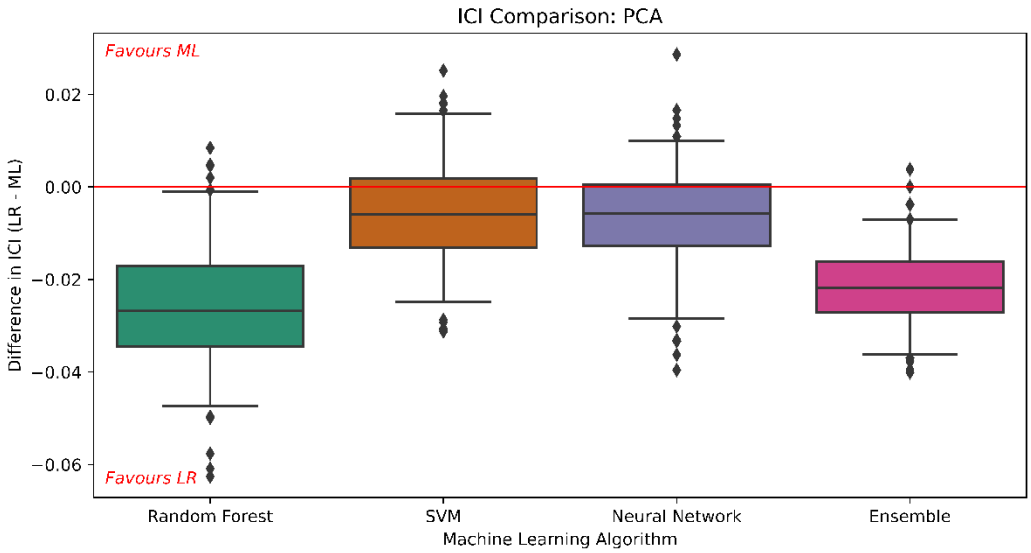

Figure 9-4: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with the top thirty principal components. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

Figure 9-4 compares the calibration of logistic regression and the supervised machine learning algorithms when the models were trained with the top thirty principal components. This figure shows that the median difference in ICI between logistic regression and each supervised machine learning algorithm is negative, suggesting logistic regression is better calibrated. This result is statistically significant for both random forests and the ensemble classifier, and the result is generally similar to the case of 500 randomly selected features, suggesting that PCA has not reduced the relative calibration performance improvement of logistic regression.

9.4. Discussion

These results demonstrate that in the clinical risk prediction space with a binary outcome, as the number of features increase the discriminatory performance of supervised machine learning as measured by the AUC improves relative to logistic regression, even if the features are selected or reduced by means of principal component analysis. However, PCA does not appear to change the calibration performance of supervised machine learning relative to logistic regression, and the calibration performance of logistic regression remains superior to all supervised machine learning algorithms, and this is sometimes significant at the 5% level depending on the algorithm.

These results further highlight that supervised machine learning has more substantial gains in discrimination performance when the feature count is larger and may go some way to explaining the mixed results of comparing these methods in the clinical risk prediction space, as these methods are applied to datasets with different feature counts in recent literature (Couronné et al., 2018). There are still limitations with the above analysis. The confidence intervals are no longer reliable, as the number of bootstraps is insufficient to achieve reliable confidence intervals. This is especially concerning with the SVM and neural network, as the conclusion that these algorithms have greater discriminatory performance at the 5% significance level is very close to the cut-off.

Furthermore, the means of feature selection applied here is arbitrary. The top thirty principal components were chosen, but it could just have easily been the top twenty or top forty, and the number of principal components selected may have affected each model's performance. Similarly, whilst principal component analysis is common, it is by no means the only choice when it comes to feature selection, nor it is not the only dimensionality reduction algorithm (Van Der Maaten et al., 2009), and other choices may affect the improvements in discrimination and calibration performance.

10. Conclusion

10.1. Findings

In this project, logistic regression and a multitude of contemporary supervised machine learning algorithms were applied to the NHANES dataset to model ten-year mortality, with the purpose of evaluating the differences in their performance. Through the analysis, the discrimination performance of each algorithm was measured by the AUC, and the calibration performance was assessed using the ICI.

The analysis showed that in the context of a smaller number of strong predictors, of the three most used supervised machine learning algorithms (random forests, SVM and neural networks) only the random forest was able to achieve superior discriminatory performance (statistically significant at the 5% level) to a logistic regression that assumed linearity and additivity. Generally, the reliability of the risk estimates from supervised machine learning were poorer than those produced by the logistic regression, though this difference in reliability of risk estimates was only significantly poorer for the SVM at the 5% level.

However, state-of-the-art machine learning solutions regularly leverage ensemble learning, as combining multiple learners tends to improve generalisability and reduce the risk of a poor local optimum in any given learner. An ensemble classifier of the random forest, SVM and neural network was also able to achieve statistically significantly superior discriminatory performance, but this improvement was no larger than the improvement achieved by random forest, and this ensemble classifier had poorer calibration performance than the logistic regression that was statistically significant at the 5% level as measured by difference in ICI.

The analysis then turned to different factors that may explain differences in performance in the clinical risk prediction space. Firstly, the effect of different sample sizes was evaluated, and it was plain that at lower training sample sizes, neither the discrimination nor calibration performances were superior for supervised machine learning, reaffirming the conclusion that machine learning methods are “data hungry” and require more data to train. As the sample size grew larger, statistically significant discrimination performance improvements were notable in both the random forest and ensemble classifier, however, the median calibration performance of supervised machine learning was poorer than logistic regression, especially at larger sample sizes.

The analysis finally looked at how performance varies with differing levels of features, and how the different algorithms performed with features engineered using principal component analysis. This analysis concluded in the context of larger feature counts, supervised machine learning has a clear discrimination performance advantage across all four algorithms compared to logistic regression, and this improvement in discrimination performance is maintained irrespective of whether the features are engineered using principal component analysis. As the level of features increased, the median calibration performance of supervised machine learning relative to logistic regression worsened for most algorithms, and the calibration performance of both the random forest and the ensemble classifier was poorer relative to logistic regression at the 5% significance level.

10.2. Limitations and Scope for Further Work

This project did not evaluate the effect of differing levels of hyperparameter optimisation, as was initially planned. This was because evaluating the effect of differing levels of hyperparameter optimisation was deemed too computationally expensive, however, this is a significant limitation as the performance of the supervised machine learning algorithms depends to some extent on the choice of hyperparameters. It is therefore plausible that the lack of performance improvement for

support vector machines and neural networks is simply a consequence of insufficient hyperparameter optimisation. An interesting extension of this work would evaluate how these algorithms fare with different levels of hyperparameter tuning, or more advanced hyperparameter tuning methods, particularly Bayesian optimisation. This limitation of this work reflects a broader inherent limitation in machine learning methods, in that their complexity and computational expense makes the methods, whilst theoretically better, much more difficult to implement in practice.

This project similarly did not investigate different model specifications for the logistic regression, remaining with the assumption of linearity and additivity that is the default of many researchers. It is almost certainly the case that this assumption artificially worsened the performance of logistic regression relative to these algorithms, however, there is no standard way of incorporating non-linearities and interactions when modelling a logistic regression. An interesting extension of this analysis would remove the assumption of linearity and additivity, exploring different means of incorporating non-linearities such as cubic splines and fractional polynomials to see if supervised machine learning offers any performance benefit over more advanced specifications of logistic regression.

This work also did not consider calibration models. These are models that are applied to the risk predictions in order to improve their calibration (Huang et al., 2020). The lack of calibration models is likely to have worsened the calibration performance of all models. An interesting avenue for further research would entail applying these calibration models to the risk estimates for each model to see how they affect differentials in calibration performance.

Additionally, this analysis concerned itself with only one dataset, and thus it is unclear if the results may be a result of features such as the signal to noise ratio. A further extension of this work could apply these methods to many datasets to give confidence that these results are not the result of idiosyncrasies of any one dataset. Such a study could also give more practical guidance to researchers on when each algorithm may be appropriate, as this project has only demonstrated that machine learning has more significant performance improvement in the context of larger sample sizes and features but did not specify how large or how many features are required to obtain benefits in discriminatory performance.

10.3. Recommendations

Based on this analysis, random forests appear to be the best model with respect to discrimination performance for datasets of this size. They have the best discrimination performance when evaluating a small number of strong candidate predictors, and the discrimination performance of random forests is comparable to that of other supervised machine learning algorithms even as the number of features becomes very large.

However, based on the analysis of this dataset, the calibration performance of logistic regression is at least as good for logistic regression, and often the calibration performance of logistic regression is superior, though this result was rarely significant at the 5% level. Clinical risk prediction is a field where the calibration of risk estimates is paramount as these models are used in clinical practice to affect the allocation of treatment.

The choice of which model has trade-offs, with random forests generally outperforming with respect to discriminating higher risk patients from lower risk patients, but logistic regression performing at least as well with respect to the reliability of risk estimates. I would advise developers of clinical risk prediction models that value improved discrimination performance to opt for random forests, and

those that value the best calibration performance to stay with logistic regression. With respect to those models that are applied in clinical practice, where the consequences of unreliable predictions can affect how healthcare is allocated, it is likely logistic regression should be favoured due to its better calibration performance.

Due to their prevalence in clinical practice, clinical risk prediction models are clearly an important area of research for society, as accurately measuring clinical risk has benefits in diagnosis and prognosis. The analysis above gives some advice on when each approach is appropriate and gives recommendations for further research in the field.

11. Appendices

11.1. Appendix A: Comparison of ICI by Sample Size

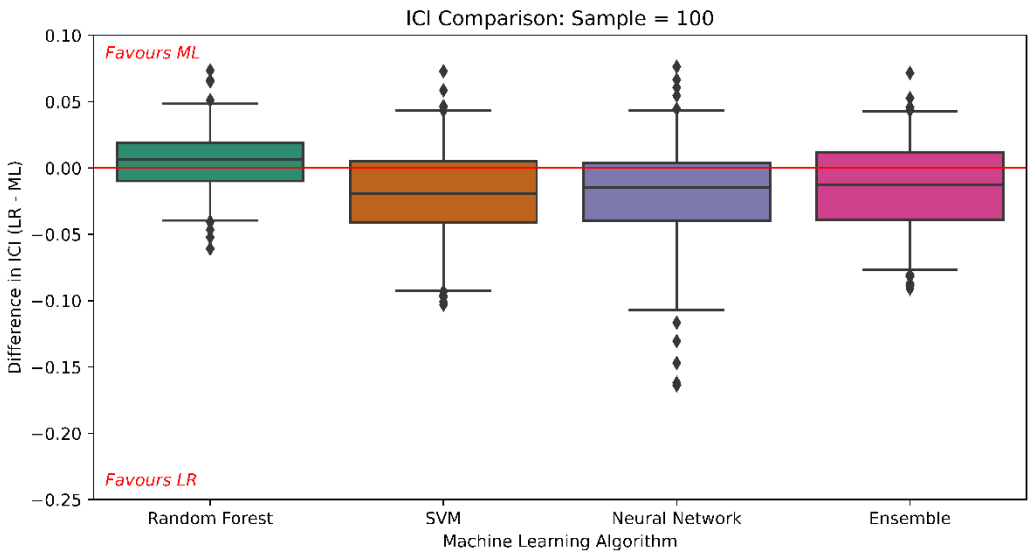

Figure 11A-1: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 100 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

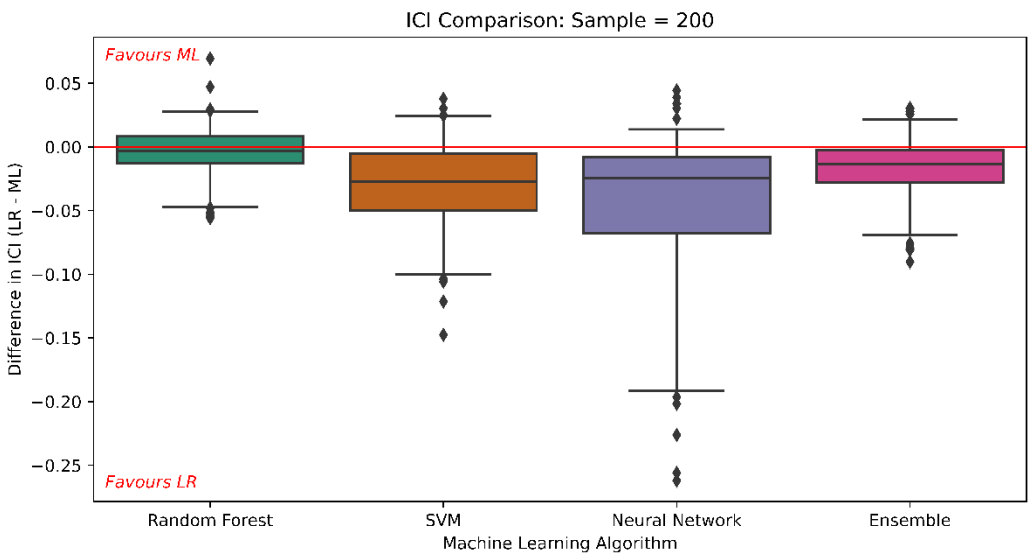

Figure 11A-2: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 200 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

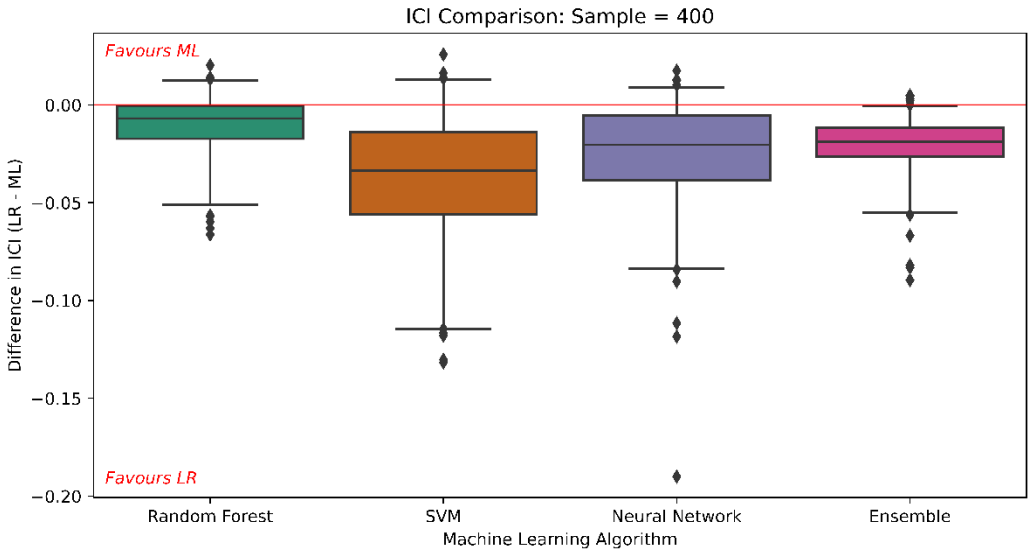

Figure 11A-3: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 400 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

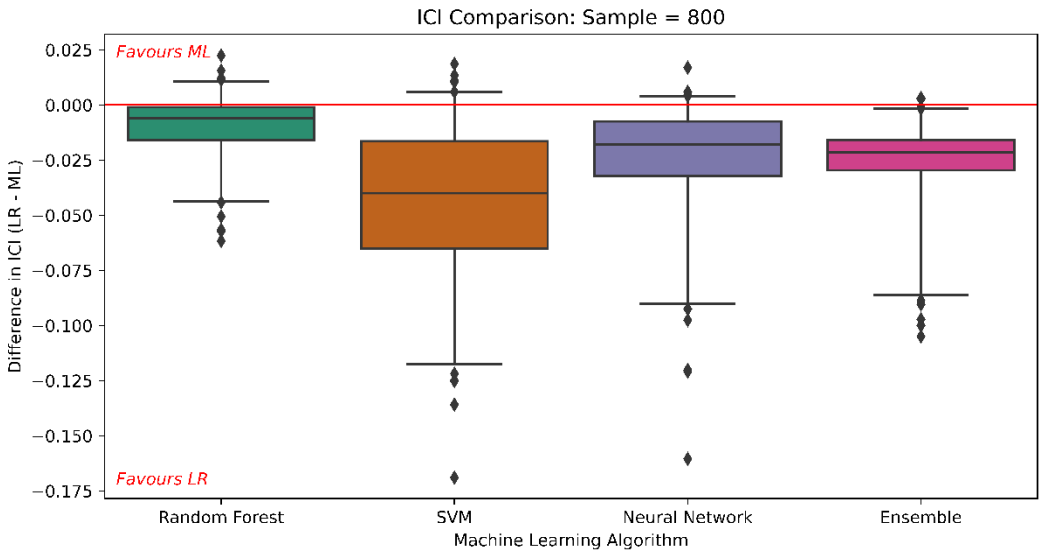

Figure 11A-4: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 800 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

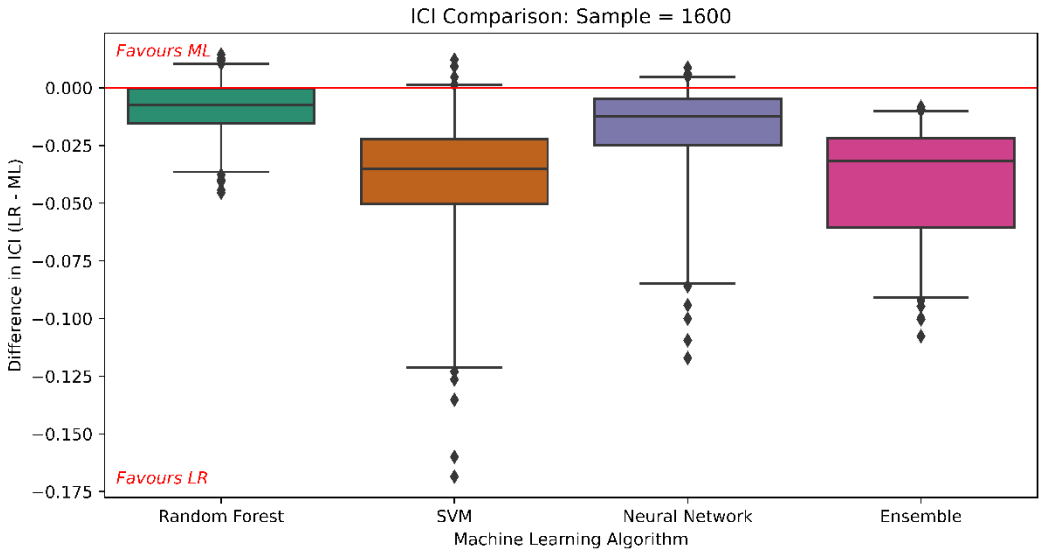

Figure 11A-5: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 1600 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

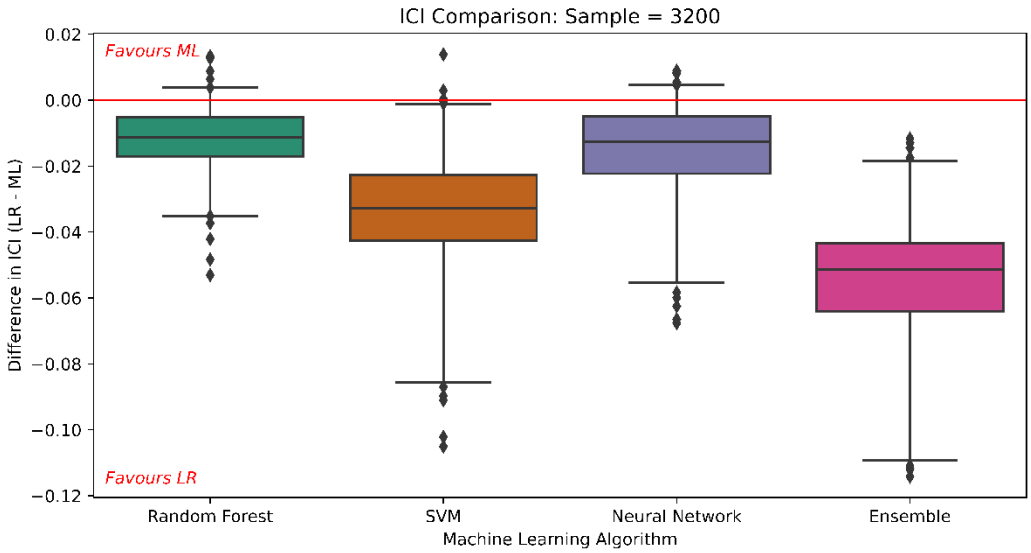

Figure 11A-6: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 3200 training data points. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

11.2. Appendix B: Complete Feature List

Note that variables can be searched for a complete description using the NHANES website:

https://wwwn.cdc.gov/nchs/nhanes/search/default.aspx

Note that the encoding of these variables and choice of which variables to include was entirely software controlled, such that some of these variables are likely to be noise variables that have no association with mortality. It is plausible some of these variables have been encoded as categorical when they are continuous in nature.

Variable Code	Categorical	Number of Categories
AGQ030	Categorical	3
AIALANG	Categorical	2
AUQ131	Categorical	6
BMDSTATS	Categorical	2
BMXARMC	Continuous	NA
BMXARML	Continuous	NA
BMXBMI	Continuous	NA
BMXHT	Continuous	NA
BMXLEG	Continuous	NA
BMXWAIST	Continuous	NA
BMXWT	Continuous	NA
BPAARM	Categorical	2
BPACSZ	Categorical	4
BPAEN3	Categorical	2
BPQ020	Categorical	3
BPQ052	Categorical	2
BPQ057	Categorical	3
BPQ060	Categorical	4
BPQ150D	Categorical	2
BPXML1	Categorical	14
BPXPLS	Continuous	NA
BPXPTY	Categorical	2
BPXPULS	Categorical	2
CBD010	Categorical	2
CBD070	Continuous	NA
CBD090	Continuous	NA
CBD110	Continuous	NA
CBD120	Continuous	NA
CBD130	Continuous	NA
CBD150	Continuous	NA
CBD160	Continuous	NA
CBD170	Continuous	NA
CBD180	Continuous	NA
CBQ020	Categorical	5
CBQ030	Categorical	6

CBQ040	Categorical	5
CBQ050	Categorical	6
CBQ060	Categorical	5
CBQ140	Categorical	7
DBD895	Continuous	NA
DBD905	Continuous	NA
DBD910	Continuous	NA
DBQ197	Categorical	5
DBQ229	Categorical	3
DBQ700	Categorical	6
DBQ915	Categorical	2
DBQ920	Categorical	3
DBQ930	Categorical	2
DBQ935	Categorical	2
DBQ940	Categorical	3
DBQ945	Categorical	3
DIQ010	Categorical	4
DIQ050	Categorical	2
DIQ190A	Categorical	3
DIQ190B	Categorical	3
DIQ190C	Categorical	3
DIQ200A	Categorical	2
DIQ200B	Categorical	3
DIQ200C	Categorical	2
DMDBORN2	Categorical	4
DMDCITZN	Categorical	3
DMDEDUC2	Categorical	5
DMDFMSIZ	Categorical	7
DMDHHSIZ	Categorical	7
DMDHRAGE	Continuous	NA
DMDHRBR2	Categorical	4
DMDHREDU	Categorical	6
DMDHRGND	Categorical	2
DMDHRMAR	Categorical	6
DMDMARTL	Categorical	6
DMQMILIT	Categorical	2
DXAFMRST	Categorical	5
DXASPNST	Categorical	5
ENXSTAT	Categorical	5
FIALANG	Categorical	2
FSD032A	Categorical	4
FSD032B	Categorical	3
FSD032C	Categorical	3
FSD151	Categorical	2

FSDAD	Categorical	4
FSDHH	Categorical	4
FSQ165	Categorical	2
HIQ011	Categorical	3
HOD050	Categorical	15
HOD060	Categorical	5
HOQ040	Categorical	7
HOQ065	Categorical	3
HOQ070	Categorical	4
HOQ080	Categorical	3
HUQ020	Categorical	3
HUQ030	Categorical	3
HUQ050	Categorical	6
HUQ071	Categorical	3
HUQ090	Categorical	2
IMQ011	Categorical	4
IMQ020	Categorical	5
IND235	Categorical	14
INDFMIN2	Categorical	12
INDFMMPC	Categorical	5
INDFMPIR	Continuous	NA
INDHHIN2	Categorical	12
INQ012	Categorical	2
INQ020	Categorical	3
INQ030	Categorical	2
INQ060	Categorical	3
INQ080	Categorical	3
INQ090	Categorical	3
INQ132	Categorical	3
INQ140	Categorical	4
INQ150	Categorical	3
KIQ022	Categorical	3
KIQ026	Categorical	3
LBDBANO	Continuous	NA
LBDBCDSI	Continuous	NA
LBDBPBSI	Continuous	NA
LBDEONO	Continuous	NA
LBDFOL	Continuous	NA
LBDHBG	Categorical	2
LBDHCV	Categorical	3
LBDIHGLC	Categorical	2
LBDIHGSI	Continuous	NA
LBDLYMNO	Continuous	NA
LBDMONO	Continuous	NA

LBDNENO	Continuous	NA
LBDRBF	Continuous	NA
LBDTHGLC	Categorical	2
LBDTHGSI	Continuous	NA
LBXBAPCT	Continuous	NA
LBXBCD	Continuous	NA
LBXBPB	Continuous	NA
LBXEOPCT	Continuous	NA
LBXFOLSI	Continuous	NA
LBXGH	Continuous	NA
LBXHBC	Categorical	2
LBXHCT	Continuous	NA
LBXHGB	Continuous	NA
LBXIHG	Continuous	NA
LBXLYPCT	Continuous	NA
LBXMC	Continuous	NA
LBXMCHSI	Continuous	NA
LBXMCVSI	Continuous	NA
LBXMOPCT	Continuous	NA
LBXMPSI	Continuous	NA
LBXNEPCT	Continuous	NA
LBXPLTSI	Continuous	NA
LBXRBCSI	Continuous	NA
LBXRBFSI	Continuous	NA
LBXRDW	Continuous	NA
LBXTHG	Continuous	NA
LBXWBCSI	Continuous	NA
MCQ010	Categorical	3
MCQ053	Categorical	3
MCQ080	Categorical	3
MCQ092	Categorical	3
MCQ140	Categorical	3
MCQ160A	Categorical	3
MCQ160B	Categorical	3
MCQ160C	Categorical	3
MCQ160D	Categorical	3
MCQ160E	Categorical	3
MCQ160F	Categorical	3
MCQ160G	Categorical	3
MCQ160K	Categorical	3
MCQ160L	Categorical	3
MCQ160M	Categorical	3
MCQ160N	Categorical	3
MCQ220	Categorical	3

MCQ245A	Categorical	2
MCQ300A	Categorical	4
MCQ300B	Categorical	4
MCQ300C	Categorical	4
OCD150	Categorical	4
OCD390G	Categorical	5
OHAEXSTS	Categorical	3
OHQ011	Categorical	7
OHQ620	Categorical	6
OHQ630	Categorical	6
OHQ640	Categorical	6
OHQ650	Categorical	6
OHQ660	Categorical	6
OHQ670	Categorical	5
OHQ680	Categorical	6
OSQ010A	Categorical	3
OSQ010B	Categorical	3
OSQ010C	Categorical	3
OSQ060	Categorical	3
OSQ080	Categorical	3
OSQ130	Categorical	3
OSQ150	Categorical	4
OSQ170	Categorical	4
OSQ200	Categorical	4
PAD680	Continuous	NA
PAQ605	Categorical	2
PAQ620	Categorical	3
PAQ635	Categorical	2
PAQ650	Categorical	3
PAQ665	Categorical	2
PAQ685	Categorical	4
PEASCTM1	Continuous	NA
PFQ051	Categorical	2
PFQ054	Categorical	2
PFQ057	Categorical	2
PFQ090	Categorical	3
PHAFSTHR	Continuous	NA
PHAFSTMN	Continuous	NA
PHDSESN	Categorical	3
PHQ020	Categorical	2
PHQ030	Categorical	2
PHQ040	Categorical	2
PHQ050	Categorical	2
PHQ060	Categorical	2

RDQ070	Categorical	3
RDQ140	Categorical	3
RIAGENDR	Categorical	2
RIDAGEEX	Continuous	NA
RIDAGEMN	Continuous	NA
RIDAGEYR	Continuous	NA
RIDEXMON	Categorical	2
RIDRETH1	Categorical	5
SDMVPSU	Categorical	2
SDMVSTRA	Categorical	16
SIAINTRP	Categorical	2
SIALANG	Categorical	2
SIAPROXY	Categorical	2
SLD010H	Categorical	13
SLD020M	Continuous	NA
SLQ030	Categorical	5
SLQ040	Categorical	5
SLQ050	Categorical	3
SLQ060	Categorical	3
SLQ080	Categorical	6
SLQ090	Categorical	5
SLQ100	Categorical	6
SLQ110	Categorical	6
SLQ120	Categorical	6
SLQ130	Categorical	6
SLQ140	Categorical	6
SLQ150	Categorical	6
SLQ160	Categorical	5
SLQ170	Categorical	6
SLQ180	Categorical	6
SLQ190	Categorical	5
SLQ200	Categorical	6
SLQ210	Categorical	5
SLQ220	Categorical	5
SLQ230	Categorical	5
SLQ240	Categorical	5
SMD410	Categorical	2
SPXNQEFF	Categorical	4
SPXNQFV1	Categorical	6
SPXNQFVC	Categorical	6
URDSCNLC	Categorical	2
URXCRS	Continuous	NA
URXNO3	Continuous	NA
URXSCN	Continuous	NA

URXUCR	Continuous	NA
URXUCRA	Continuous	NA
URXUCRB	Continuous	NA
URXUCRC	Continuous	NA
URXUMA	Continuous	NA
URXUMS	Continuous	NA
URXUP8	Continuous	NA
VIQ041	Categorical	5
VIQ051A	Categorical	7
VIQ051B	Categorical	6
VIQ051C	Categorical	7
VIQ051D	Categorical	6
VIQ051E	Categorical	7
VIQ056	Categorical	7
VIQ061	Categorical	5
VIQ071	Categorical	3
VIQ180	Categorical	3
VIQ200	Categorical	3
VIQ220	Categorical	3
VIXKLCD	Continuous	NA
VIXKLCG	Continuous	NA
VIXKLD1	Continuous	NA
VIXKLD2	Continuous	NA
VIXKLDM	Continuous	NA
VIXKLG1	Continuous	NA
VIXKLG2	Continuous	NA
VIXKLM1	Continuous	NA
VIXKLM2	Continuous	NA
VIXKLMM	Continuous	NA
VIXKRCD	Continuous	NA
VIXKRCG	Continuous	NA
VIXKRD1	Continuous	NA
VIXKRD2	Continuous	NA
VIXKRDM	Continuous	NA
VIXKRG1	Continuous	NA
VIXKRG2	Continuous	NA
VIXKRM1	Continuous	NA
VIXKRM2	Continuous	NA
VIXKRMM	Continuous	NA
VIXOLAM	Continuous	NA
VIXOLCM	Continuous	NA
VIXOLSM	Continuous	NA
VIXORAM	Continuous	NA
VIXORCM	Continuous	NA

VIXORSM	Continuous	NA
WHD010	Continuous	NA
WHD020	Continuous	NA
WHD050	Continuous	NA
WHD140	Continuous	NA
WHQ030	Categorical	4
WHQ040	Categorical	3
WHQ150	Continuous	NA
WTDRD1	Continuous	NA
WTINT2YR	Continuous	NA
WTMEC2YR	Continuous	NA

11.3. Appendix C: Comparison of ICI by Feature Count

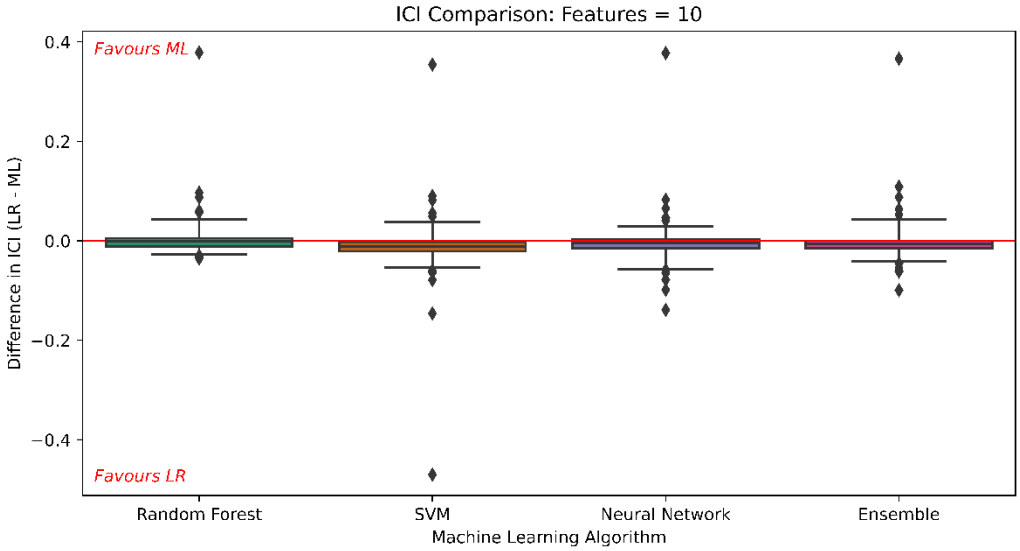

Figure 11C-1: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 10 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

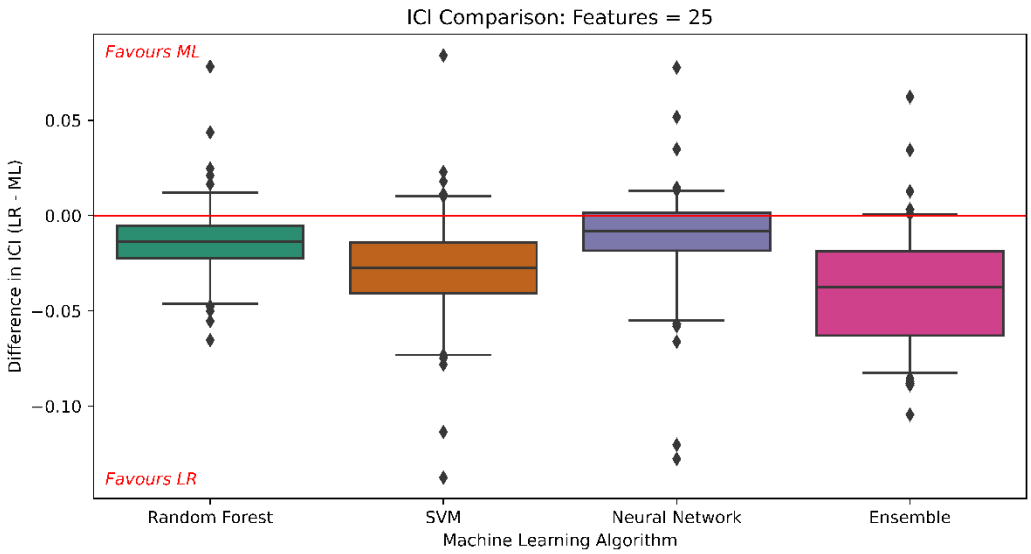

Figure 11C-2: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 25 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

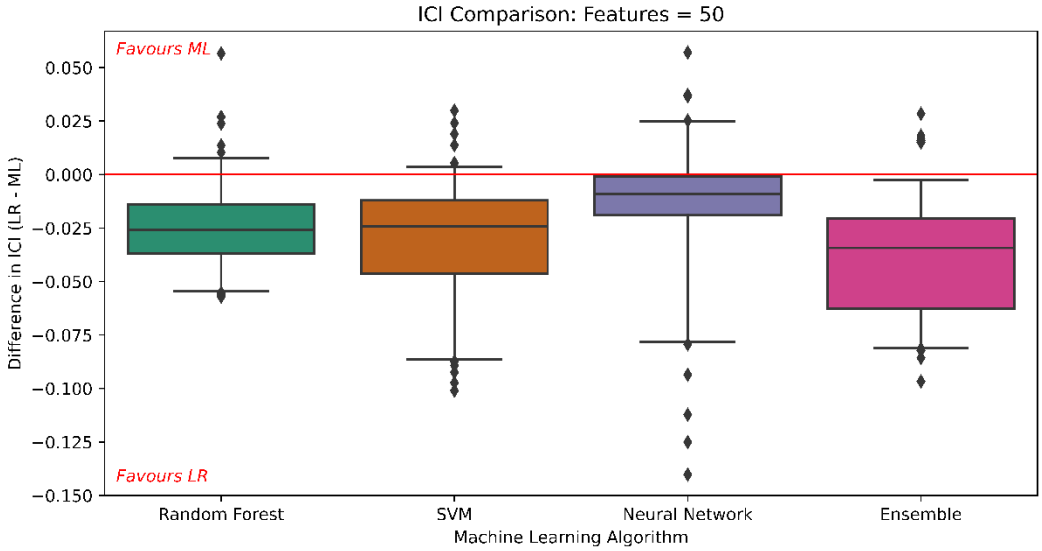

Figure 11C-3: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 50 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

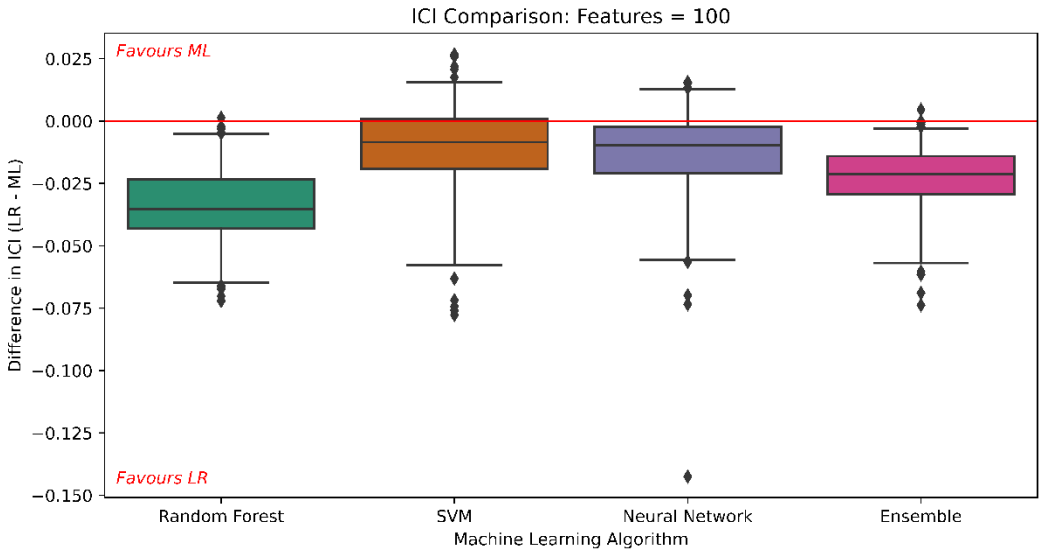

Figure 11C-4: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 100 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

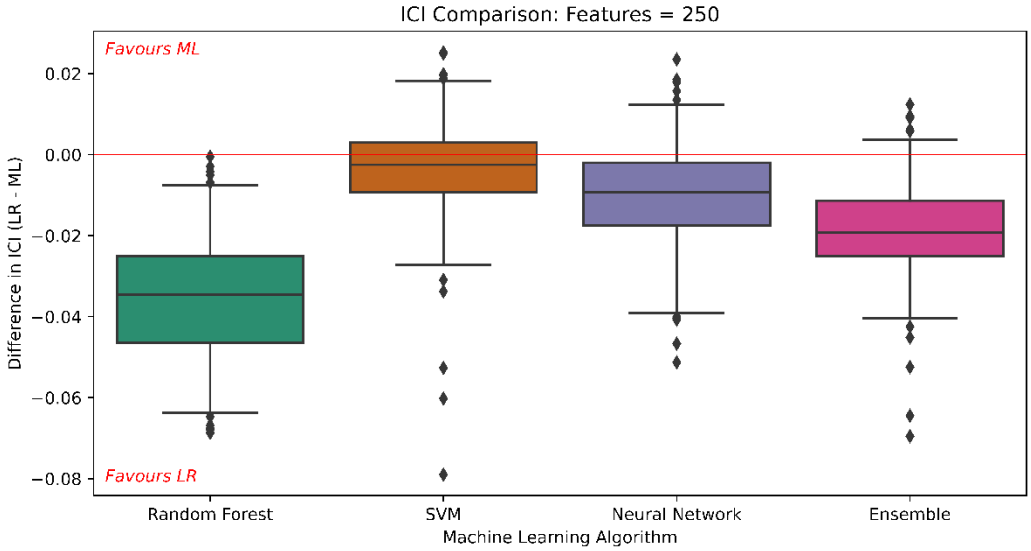

Figure 11C-5: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 250 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

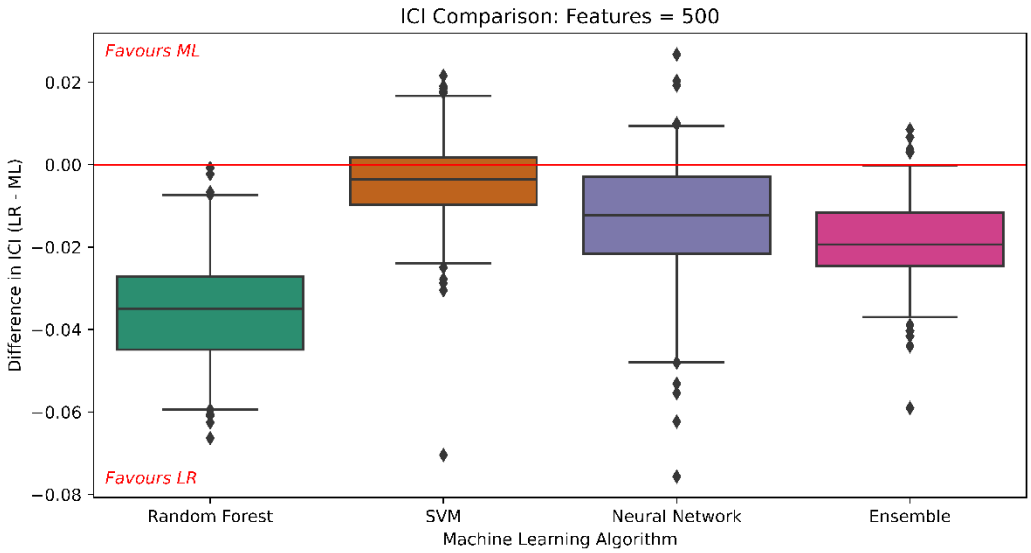

Figure 11C-6: Comparing the difference in the ICI of logistic regression to each supervised ML algorithm with 500 randomly selected features. ICI differences were calculated across 199 bootstraps, with whiskers showing 95% percentile confidence interval.

12. References

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